Associative Memory in Transformers: Topology, Iterative Retrieval, and Controlled Activation with Progressive Retrieval Attention

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Abstract

Global self-attention can contextualize distant evidence while making the resulting memory geometry less explicitly traversable. We use Progressive Retrieval Attention (PRA) to ask whether transformer states expose an associative topology, whether iterative retrieval can traverse it, and where useful retrieval fails to influence computation. A controlled family of 25 matched causal decoders varies only the training attention window $W \in \{16, 32, 64, 128, \text{global}\}$ across five seeds. Each checkpoint is converted without learning to native-K/V PRA. We complement this causal study with frozen Qwen, Gemma, and Llama-family diagnostics on multi-hop language tasks.

At the controlled model's final layer, $W = 16$ attains native edge R@4 $.428 \pm .076$, versus $.187 \pm .065$ under global attention. Under the same 20 layer-token K/V budget, evolving-state iterative PRA improves complete-path recovery over one-shot retrieval in 59/400 (14.8%) model–example units. Within this path-improved subset, correct-label margin increases by 2.164, with a paired-bootstrap 95% interval $[1.336, 2.971]$, while aggregate answer-accuracy improvement remains statistically unresolved. Ordinary executable memory is distractor dominated, assigning .147 final-query attention mass to evidence and .521 to memory distractors. A matched oracle intervention raises frozen accuracy from .140 to .398 and correct-label margin from -3.103 to $-.703$.

These results identify controlled activation, not absolute frozen-consumer incapacity, as the dominant limitation. Transformer memory contains topology that depends causally on receptive field; iterative PRA can traverse that topology; and the frozen computation can use correctly activated memory. The unresolved problem is to make useful evidence dominate sparse external-memory influence and to preserve that influence through subsequent layers.

1 Introduction

If attention provides associative access to contextual memory, does a transformer preserve a traversable memory topology, and can sparse iterative activation exploit it? Consider a query that directly identifies memory A, while A supplies the relation needed to find memory B. One-shot retrieval may rank A but miss B because the initial query does not yet contain the intermediate relation. Raising top-k can recover B accidentally, at the cost of activating more unrelated memory. Iterative retrieval instead asks whether A itself can become a new search state.

This question has two parts that are often conflated. First, the model must expose useful memory-to-memory geometry. Second, selected memories must influence the residual stream in the right direction. A recovered path can fail at either boundary: routing can select distractors, the shared softmax can assign them most of the memory mass, or later layers can erase an initially useful change. Conversely, a weak aggregate output effect does not prove that associative traversal or frozen memory consumption is absent.

PRA separates compact routing state from layer-native K/V payloads [16, 15, 17]. That separation lets us observe root activation, associative traversal, physical activation, attention allocation, residual change, and final output as distinct events. It also enables a strict comparison: one-shot and evolving-state iterative PRA receive the same layer-token K/V budget, and selected URI identities cannot be replayed merely to consume another intervention.

The paper has four principal findings. First, local receptive fields preserve stronger explicit native Q/K topology than global attention. Second, matched iterative PRA exploits that topology to recover more complete paths. Third, better traversal is functionally useful when it changes selected evidence: correct-answer margin rises substantially in the path-improved subgroup. Fourth, ordinary activation remains noisy, whereas matched oracle memory produces a large frozen-model output gain. Table 1 states the resulting causal decomposition.

Table 1: Headline questions, results, and consequences. Controlled values use five receptive-field settings and five seeds unless explicitly described as model–example diagnostics.

Scientific question	Result	Consequence
Does transformer memory expose associative topology?	Yes: native Q/K contains recoverable evidence edges	Compact states support explicit traversal.
Does receptive field affect topology?	Yes: final-layer R@4 is .428 at $W = 16$ and .187 globally	Locality preserves stronger explicit structure.
Can iterative PRA traverse it?	Yes: path recovery usually rises at the same 20-state budget	Evolving-state access is an operational primitive.
Does improved traversal help computation?	Yes when useful evidence changes: margin +2.164	Traversal is useful, but improved cases are sparse.
Why is aggregate answer gain weak?	Selected memory gives .147 mass to evidence and .521 to distractors	Controlled activation is the dominant bottleneck.
Can the frozen model consume correct PRA memory?	Yes: oracle accuracy .140→.398	Frozen-consumer incapacity is rejected as an absolute explanation.
Do layers matter?	Yes: search, consumption, and preservation profiles differ	There is no universal best PRA layer.

Our contributions are: (1) controlled causal evidence that receptive field shapes associative topology; (2) matched-budget evidence that evolving-state PRA traverses that topology; (3) a causal decomposition of retrieval, attention allocation, residual change, and preservation; (4) oracle evidence that frozen transformers can use correctly activated external memory; and (5) pretrained validation of root-entry, granularity, layer, and distractor-competition effects. The chronological mechanism search, including negative results, remains intact in the appendices.

2 PRA as a Probe of Associative Memory

Let $G = \{g_1, \ldots, g_N\}$ be compact routing gists at one transformer layer and let q be the current query state. One-shot PRA returns

$$N_B(q) = \mathrm{TopB}_{g \in G}\, s(q, g). \tag{1}$$

This is sufficient when each useful memory is independently similar to q. It is incomplete when

$$s(q, g_A) \gg 0, \qquad s(q, g_B) \approx 0, \qquad s(g_A, g_B) \gg 0. \tag{2}$$

Here A is the *root activation*: the first query-to-memory transition. The transition $A \to B$ is *associative traversal*. PRA performs this search over compact state and activates full layer-native K/V only for the final selected identities.

Five terms remain distinct throughout the paper. *Associative topology* is the native Q/K geometry linking memory states. *Controlled activation* selects a working set in which useful evidence can dominate external-memory influence. *Consumption* converts attended memory into a useful residual or output change. *Preservation* or *assimilation* asks whether that change survives later transformer computation. Selection is therefore not equivalent to attention, and attention is not equivalent to useful output.

At a PRA layer, memory and direct-context K/V share one softmax. For the final query token we measure

$$A_E = \sum_{j \in \text{evidence}} \alpha_j, \quad A_J = \sum_{j \in \text{memory distractors}} \alpha_j, \quad A_N = \sum_{j \in \text{native context}} \alpha_j, quad A_E + A_J + A_N = 1. \tag{3}$$

This decomposition exposes whether a correct identity merely appears in the working set or actually competes successfully for influence.

logical source or long prompt
↓ model-safe encoding blocks ↓
layer-native K/V + contextual chunks and compact gists
↓ root activation: $q \to A$ ↓
associative traversal: $A \to B \to C$
↓ bounded identity selection ↓
selected native K/V → shared-softmax direct + memory attention
↓ residual assimilation ↓
frozen transformer output

Figure 1: PRA as an associative-memory probe. Compact search and full native-K/V activation are separate phases, so topology, traversal, activation, consumption, and preservation can be measured independently.

3 Controlled Model: Receptive Field and Memory Topology

3.1 Controlled task and model family

Each example samples a fresh entity/relation chain of depth $D \in \{1, 2, 3, 4\}$. The terminal entity maps to one of eight balanced labels, so chance accuracy is .125. Facts contain five tokens and the query contains a start entity plus a relation sequence; the standalone PRA query has $D + 4$ tokens including `[BOS]` and never includes the answer. In the mechanistic cohort, distractor interleaving and controlled gaps produce 7–55-token evidence spans and 5–11-token maximum hop distances. The artifacts record these quantities and their ratios to W for every paired example.

Multiple distractors use the same terminal relation with wrong source entities, preventing a unique relation scan from solving the task. Evidence appears only in reverse causal order, while distractors are randomly interleaved. Train, validation, test, topology, and PRA corpora use disjoint deterministic seeds and are shared across every model seed and W. Gold paths are joined only after inference and never enter cache metadata or executable selection.

query: start entity + relation sequence   answer: one of 8 balanced labels

↓

randomly interleaved five-token facts: evidence chain + matched relation distractors

↓

matched 6-layer decoder with $W \in \{16, 32, 64, 128, \text{global}\}$

↓

native Q/K edge ranks, path survival, contextual dependence, and answer prediction

Figure 2: Controlled scientific instrument. Only the native training attention window varies across the five model families; corpus, architecture, optimization, and evaluation remain matched.

Table 2: Controlled model and data configuration.

Property	Fixed value
Decoder	6 pre-norm layers, width 96, 4 heads, FFN width 384, RoPE, no dropout
Parameters / vocabulary	739,968 / 230 controlled tokens
Native attention	$W \in \{16, 32, 64, 128, \text{global}\}$; the current token is included
Train / validation / test	4,096 / 512 / 512 disjoint deterministic examples
Optimization	AdamW, batch 48, 800 updates, gradient norm 1.0, warmup plus cosine decay
Checkpoint rule	Validation accuracy every 100 updates; loss breaks ties; test remains untouched
Initialization seeds	17, 29, 41, 53, 67

A one-layer local mask exposes at most the current token and previous $W-1$ positions. Across L layers, theoretical causal reach can grow to $1+L(W-1)$ tokens. We therefore do not equate a local mask with permanently isolated state. Native edge metrics and restricted-context interventions are measured at every layer and compared with actual evidence span and hop distance.

3.2 Native topology depends on receptive field

For every evidence edge $A \to B$, we score A's native per-head query against mean native keys for all facts. Table 3 reports held-out model behavior and final-layer topology. The output threshold lies near $W = 32$, while explicit topology moves in the opposite direction: R@4 declines from .428 at $W = 16$ to .187 globally and shortcut rate rises from .429 to .584. Complete four-hop survival at R@4 is .203 at $W = 16$ and .119 globally.

Table 3: Held-out quality and final-layer native topology. Intervals are five-seed Student-t 95% intervals.

Window	Test accuracy	Test loss	Edge R@4	Shortcut rate
16	$.415 \pm .030$	1.767	$.428 \pm .076$	$.429 \pm .027$
32	$.511 \pm .012$	1.218	$.294 \pm .072$	$.524 \pm .052$
64	$.506 \pm .020$	1.118	$.277 \pm .039$	$.497 \pm .061$
128	$.514 \pm .010$	1.077	$.196 \pm .062$	$.537 \pm .080$
Global	$.522 \pm .017$	1.085	$.187 \pm .065$	$.584 \pm .069$

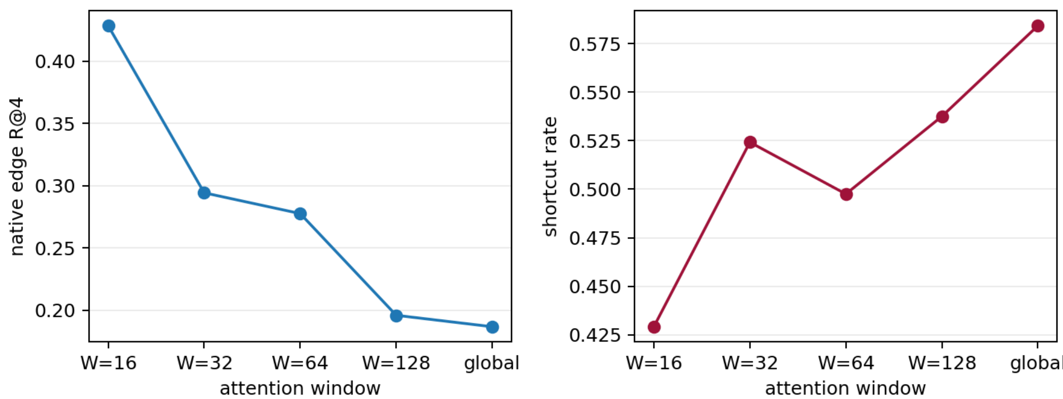

Figure 3: Native edge R@4 and shortcut rate across the matched receptive-field family. The trend is smooth rather than a $W = 16$ endpoint artifact.

3.3 Contextualization is not traversability

For $W = 16$, R@4 rises from .221 at layer 0 to .408 at layer 1 and remains near .4, while dependence on context outside the final 16 tokens grows from approximately zero to .426. Global restricted-context dependence approaches .494 while R@4 remains near .2. Local states therefore accumulate context through depth without converging to the global model's shortcut-rich topology.

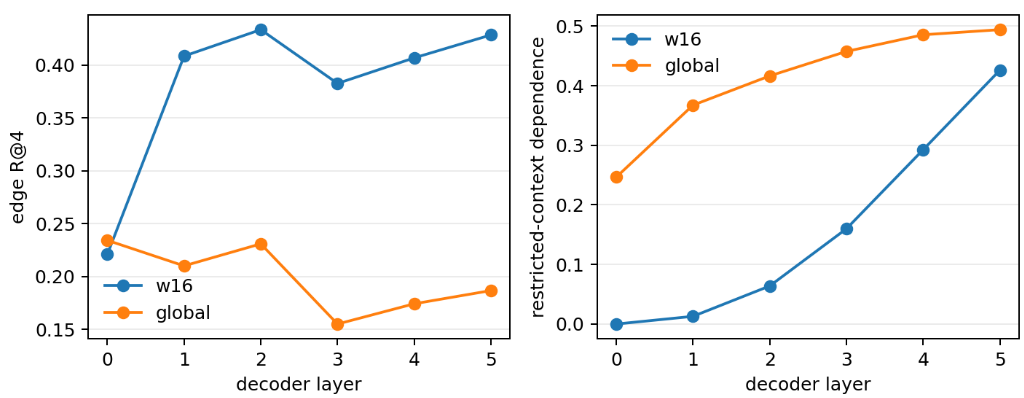

Figure 4: Layerwise edge R@4 and restricted-context dependence. More contextualized does not mean more explicitly traversable.

4 Controlled Iterative Retrieval

4.1 Evolving-state retrieval under a matched physical budget

After training, selected self-attention blocks are replaced with PRA blocks carrying the same layer normalization, Q/K/V/O projections, and FFN weights. Disabled-memory logits match the source model to numerical tolerance. One-shot PRA selects four facts at one layer. Matched iterative PRA selects one unseen fact at layers 0, 2, 4, and 5. Both materialize exactly 20 layer-token K/V states. The later query is the transformed hidden state after previous memory integration and intervening self-attention/FFN computation:

$$h_{\ell_1} \xrightarrow{\text{PRA}(A)} h'_{\ell_1} \xrightarrow{\text{SA/FFN}} h_{\ell_2} \xrightarrow{\text{PRA}(B)} h'_{\ell_2}. \tag{4}$$

Visited URI identities are removed from later caches, so iterative depth cannot be simulated by replaying the first selected fact.

Table 4: Matched one-shot and iterative PRA over in-domain depths 1–4. Acc. is answer accuracy, Rec. is evidence recall, and Path is complete-path recovery.

	One-shot, 20 states			Iterative, 20 states			Δ Acc.
Window	Acc.	Rec.	Path	Acc.	Rec.	Path	Iter.-one
16	.167	.349	.109	.164	.388	.177	-.004
32	.183	.365	.113	.256	.383	.166	+.073
64	.178	.378	.177	.195	.412	.165	+.018
128	.212	.405	.190	.234	.451	.232	+.021
Global	.212	.420	.197	.177	.453	.221	-.036

Iterative retrieval raises path recovery in four of five windows, including +.068 at $W = 16$, but every five-seed accuracy interval crosses zero. The largest mean answer difference is +.073 at $W = 32$, with a 95% interval of $\pm$.151. This does not mean graph traversal is irrelevant. It means the executable policy changes useful evidence too infrequently to establish a stable architecture-level answer gain.

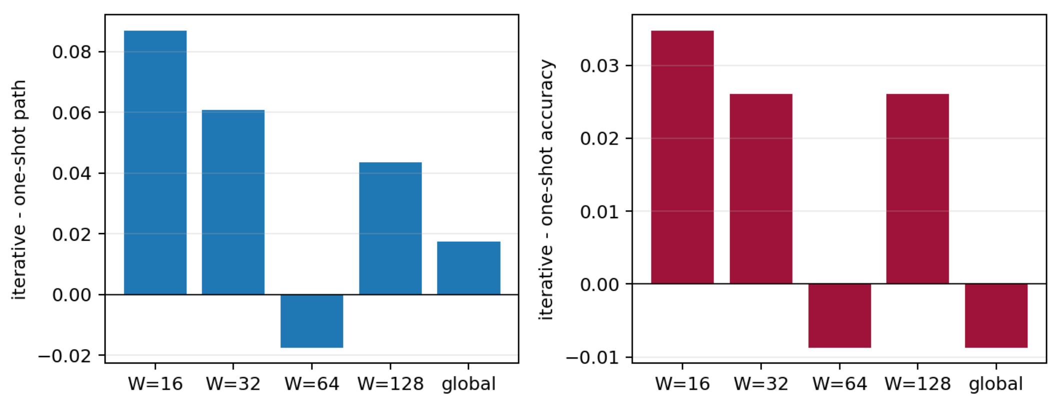

Figure 5: Matched iterative-minus-one-shot path and answer changes by receptive field. Traversal improves more consistently than aggregate answer accuracy.

4.2 Intervention density is a diagnostic frontier

PRA at every layer averages 29.8 layer-token states and raises recovery further, but it is no longer budget matched. Sparser spacing reduces cost to 15, 10, and 5 states and generally reduces path recovery. Answer accuracy is non-monotonic. More interventions make memory more active and alter the residual stream; they do not guarantee more useful computation.

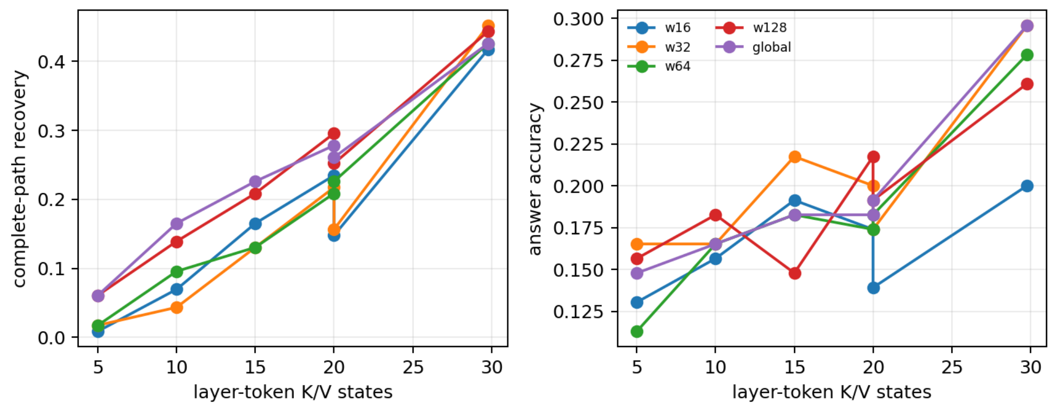

Figure 6: Layer-token K/V states versus complete-path recovery and answer accuracy. Every-layer PRA is retained as a diagnostic frontier, not a deployment policy.

5 Controlled Activation: Why Better Retrieval Sometimes Fails

The matched traversal result identifies a boundary but not its cause. We therefore reuse all 25 validation-selected checkpoints as frozen experimental instruments. The causal cohort contains 16 paired held-out examples per checkpoint, or 400 model–example units. No backbone, router, readout, or memory projection is updated.

For the same model, query, layers, example, and physical budget, we compare no memory E_0; ordinary selected memory E_+; oracle evidence plus enough unique distractors to preserve 20 states E_{oracle}; four non-evidence facts E_{wrong}; and selected URI/gist identities whose native K/V payloads are rotated among facts E_{shuffle}. Oracle identities are available only to the named intervention and never to executable retrieval.

5.1 Ordinary selected memory is distractor dominated

Table 5 is the central causal result. Ordinary iterative selection raises accuracy only from .140 to .195. It assigns .147 attention mass to evidence and .521 to memory distractors. Shuffling leaves accuracy at .193, indicating that some selected-memory benefit is a generic state perturbation or correlated selection effect rather than content-specific evidence use.

Table 5: Frozen-consumer controls averaged equally over windows. Every nonzero row uses exactly 20 layer-token K/V states.

Memory condition	Complete path	Accuracy	Margin	Evidence mass	Distractor mass
E_0	.000	.140	-3.103	.000	.000
E_+	.408	.195	-2.255	.147	.521
E_{oracle}	1.000	.398	$-.703$	.270	.301
E_{wrong}	.000	.058	-3.544	.000	.546
E_{shuffle}	.418	.193	-2.318	.122	.433

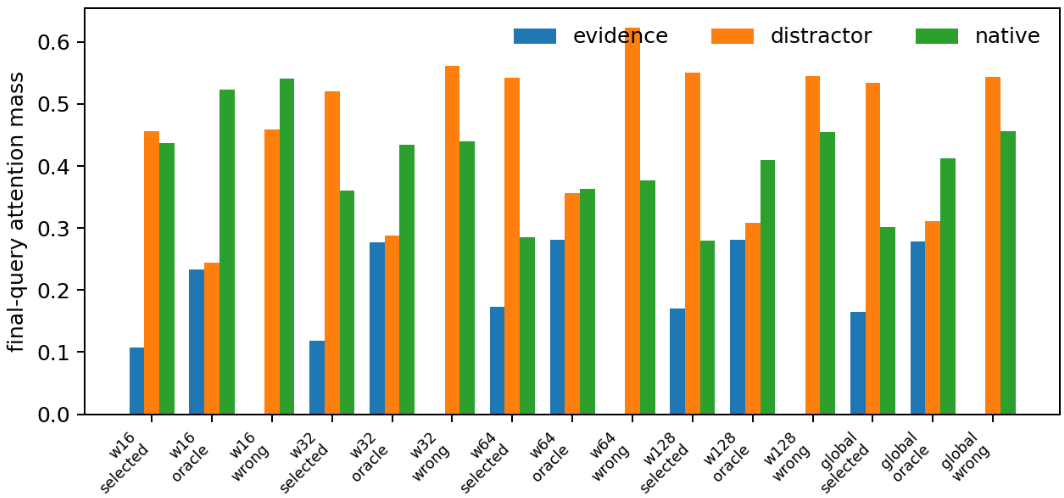

Figure 7: Exact final-query shared-softmax decomposition. Executable selected memory is consistently distractor dominated; matched oracle forcing increases evidence mass while preserving budget.

5.2 Oracle memory establishes consumption capacity

Oracle forcing raises complete-path recovery to 1.0, accuracy to .398, and margin from −3.103 to −.703. Accuracy improves in 23 of 25 window–seed pairs and margin in 24 of 25. With five seeds per window, the strongest exact two-sided sign result is still $p = .0625$; the result is a large, directionally consistent causal ceiling, not a fully powered deployment comparison. Wrong memory lowers accuracy to .058. Memory content therefore matters, and the frozen transformer has substantial latent capacity to consume correctly activated PRA memory.

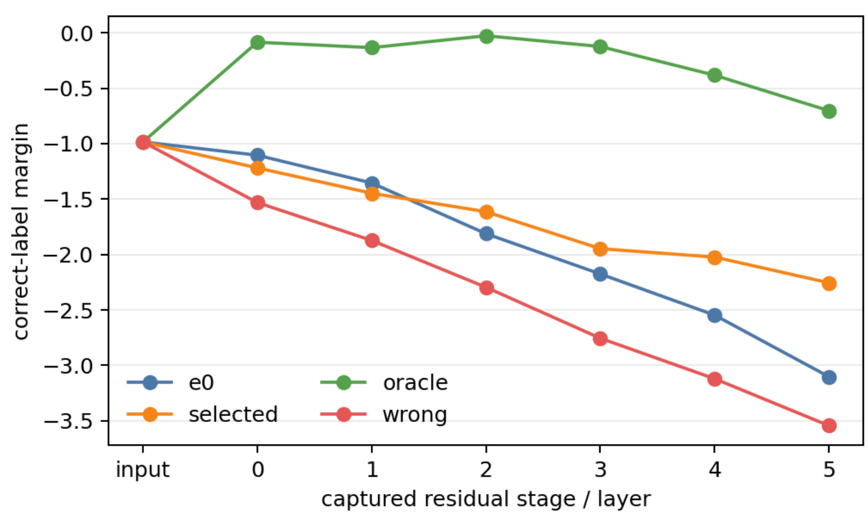

Figure 8: Correct-label margin through the frozen residual stack for no memory, ordinary selected, oracle, and wrong memory. Oracle evidence separates early; wrong memory moves in the opposite direction. Curves average paired checkpoints and examples for diagnosis rather than calibration.

5.3 When does iteration help?

Iterative PRA improves complete-path recovery over one-shot in 59 of 400 model–example units, leaves it unchanged in 293, and worsens it in 48. Within the improved subset, correct-label margin rises by 2.164 with paired bootstrap 95% interval $[1.336, 2.971]$; accuracy rises by .102 with interval $[-.017, .220]$. When path recovery worsens, margin falls by 1.585 and accuracy by .292. Across all rows, path gain correlates .279 with margin gain and .207 with answer gain. Better traversal is therefore functionally useful when it changes evidence, but the tested policy does not produce that event reliably enough for a stable aggregate accuracy improvement.

Let $G_+ = \{\Delta\text{path} > 0\}$ denote the 59 improved units and $G_0 = \{\Delta\text{path} \leq 0\}$ the other 341. Table 6 compares their existing traces. The first block contains measurements available after one-shot retrieval but before choosing a retry; the second contains consequences observed only after iteration. The distinction matters: margin gain, iterative evidence mass, and erasure can diagnose a completed retry but cannot serve as deployable retry triggers.

Table 6: Path-improved versus other model–example units. Entries are means; d is the pooled standardized mean difference except for answer gain, reported as a mean change. Intervals and medians are preserved in the public derived CSV.

Timing	Feature	G_+ (59)	G_0 (341)	Effect
Pre-decision	One-shot reference recall	.085	.603	$d = -1.393$
Pre-decision	One-shot evidence mass	.037	.230	$d = -1.094$
Pre-decision	One-shot distractor mass	.871	.657	$d = 1.119$
Pre-decision	Chain depth	1.17	1.92	$d = -.770$
Pre-decision	One-shot margin	-4.137	-2.330	$d = -.534$
Pre-decision	Evidence span (tokens)	30.83	26.92	$d = .307$
Post-treatment	Margin gain	2.164	.026	$d = .629$
Post-treatment	Answer gain	.102	$-.065$	$+.166$
Post-treatment	Iterative evidence mass	.202	.137	$d = .635$
Post-treatment	Iterative distractor mass	.476	.528	$d = -.387$
Post-treatment	Erased by final layer	.203	.185	$p = .720$

One-shot complete-path recovery is zero in every G_+ unit by construction: a binary complete path cannot improve from one. That fact identifies retry eligibility, not a learned signature. Among the 248 one-shot misses, 59 improve and 189 do not. Relative to those 189 misses, improved retries have shorter chains (1.17 versus 2.50, $d = -1.43$), longer evidence spans (30.83 versus 20.94 tokens, $d = .89$), less one-shot evidence mass (.037 versus .156, $d = -.779$), and more distractor mass (.871 versus .720, $d = .854$). Their baseline margin is also lower (-4.137 versus -3.233, $d = -.319$). Thus the minority has a recognizable controlled-task profile: shallow recoverable paths whose one-shot working set is especially evidence poor. Query length is omitted from the interpretation because this synthetic generator couples it to chain depth. Root-score gaps, routing entropy, facets, per-unit native R@K/MRR, contraction, and frontier competition were not recorded for these paired units.

We do not fit a retry classifier. The cohort contains only 16 repeated task identities per checkpoint, and its strongest features are either structural eligibility or generator-coupled. Treating a random row split across seeds as independent prediction would overstate generalization. The descriptive signature instead motivates the independent adaptive-control question in Paper 3.5.

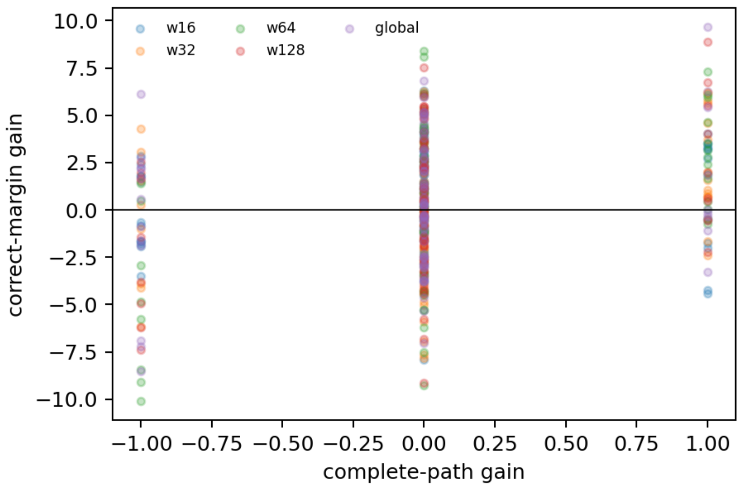

Figure 9: Paired complete-path and correct-margin changes. Positive path changes usually move the frozen decision in the correct direction, but most rows do not improve the path.

Table 7: Causal diagnosis from associative topology to preserved output.

Stage	Finding	Interpretation
Topology	Local W preserves stronger R@K	Associative structure exists and depends on receptive field.
Traversal	Iterative PRA improves path recovery	The structure is operationally traversable.
Ordinary activation	Evidence mass .147, distractor mass .521	Executable selection is noisy.
Oracle activation	Accuracy .140→.398; margin −3.103 → −.703	Frozen computation can use correct memory.
Improved-path subset	Margin +2.164	Better traversal is useful when selected evidence changes.
Later computation	21.6% of initially positive oracle traces are erased	Scheduling and assimilation are secondary limits.

6 Layer Dynamics: Search, Consumption, and Preservation

The best search layer L_{search} is where native Q/K ranks useful successors well. The best consumer layer L_{consume} is where fixed oracle memory most improves the answer state. A preservation profile asks whether the resulting gain survives later layers. These are different measurements and need not peak together.

Oracle usefulness is strongest early in this controlled family. Averaged across windows, injection at layer 0 gives .358 accuracy and −1.239 margin, compared with .138 and −2.565 at layer 5. The native graph-search optimum follows a different layer ordering. Among oracle traces with a positive immediate margin effect, 21.6% lose that advantage by the final layer. Mean local answer-direction alignment is positive for oracle memory (.054), near zero for selected memory (.017), and negative for wrong memory (−.061). Evidence value-contribution norm rises from 1.283 under ordinary selection to 2.105 under oracle selection.

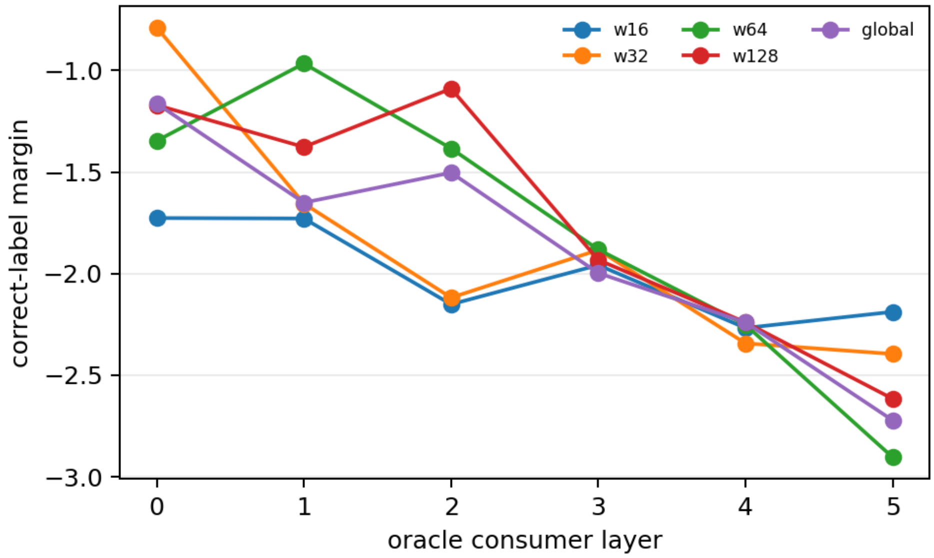

Figure 10: Oracle consumer-layer profile. Early layers use the same correct memory more effectively than late layers, while native graph quality has a separate, non-monotonic profile.

QUERY / CURRENT STATE
$\downarrow$
ROOT ACTIVATION
$\downarrow$
ASSOCIATIVE TRAVERSAL ($W = 16$ path +.068 at matched budget)
$\downarrow$
SELECTED MEMORY
$\searrow$ distractors dominate (.521 vs .147 evidence) $\rightarrow$ poor activation
$\downarrow$
EVIDENCE-DOMINATED ATTENTION (oracle evidence mass .270)
$\downarrow$
USEFUL RESIDUAL CHANGE (path-improved margin +2.164)
$\searrow$ later erasure (21.6%) $\rightarrow$ lost gain
$\downarrow$
OUTPUT (oracle accuracy .140$\rightarrow$.398)

Figure 11: Final causal pipeline. The dominant measured loss is controlled activation between selected memory and evidence-dominated attention; consumer placement and preservation are secondary.

7 Pretrained Validation

The pretrained studies test whether the controlled decomposition remains useful in natural language. They do not repeat the controlled causal intervention. Qwen3-0.6B supplies the main frozen feature and output artifacts; Gemma 3 1B and SmolLM2-135M provide architectural contrasts. The latter is a compact Llama-family model, not a Llama 3.x checkpoint. Backbones, router checkpoints, post-RoPE native K/V, and attention semantics remain fixed within each comparison.

Synthetic chains test exact transitive reachability. QASPER is primarily a direct-evidence/root-entry control; HotpotQA supplies bridge questions; 2Wiki supplies annotated evidence transitions; and MuSiQue supplies deeper task graphs that reveal contraction. These datasets have different roles and are not pooled into one leaderboard.

7.1 Root activation and query facets

The all-offset audit shows that useful root-query views often exist but are hard to select. Some contextual span ranks a correct root in Top-4 for .878 MuSiQue and .950 2Wiki runs. Fixed executable facets achieve .422/.608; the best inexpensive held-out selectors reach .622/.692. Position, hidden norm, margin, entropy, and validation-fitted scale selectors do not close the gap. Root activation is therefore a competition problem, not simply the absence of a useful query representation.

HotpotQA makes that competition concrete. A short bridge question can mix a source entity, first relation, bridge object, second relation, and target property. The correct first passage can be necessary without being answer-like, so $\mathrm{sim}(Q, B)$ can exceed $\mathrm{sim}(Q, A)$ even when A is the correct root and $A \to B$ is a strong native transition. Distractors can match one sub-intent, and the query-to-memory comparison is asymmetric: Q is a question state while A and B are contextual memory states. The inherited seven-identity Hotpot test is not a failure case for candidate availability—global semantic root R@4 is .971—but at a 20% selection budget the correct root is included only .343 of the time. Four-token contextual facets raise inclusion to .457 by preserving relational subphrases, while comparisons rise from 5.9 to 67.6. More facets create opportunities and competitors; selecting the facet that expresses the current retrieval intention remains the hard part.

Table 8: Held-out root Top-4 recall: oracle query view versus executable selection.

Entry rule	MuSiQue	2Wiki
Deployed fixed facets	.422	.608
Bounded all-facet union	.411	.475
Best cheap selector	.622	.692
All-offset oracle ceiling	.878	.950

Table 9: Dataset routing geometry from frozen artifacts. Values are diagnostic and cohorts differ in role; blank oracle entries were not measured.

Dataset	Query → root	Evidence dispersion	Memory → memory	Main bottleneck
QASPER	Low in direct control: R@4 .967; facets add no 20% inclusion gain	Low/moderate: .227 source selected	No annotated transition graph	Direct retrieval and materialization
HotpotQA	Bridge-ambiguous; R@4 .971 but inclusion .343→.457 with facets	Moderate: .501 source selected	Useful after correct root; propagation adds little here	Robust multi-intent root activation and concentration
2Wiki	Executable R@4 .608 versus .950 oracle-facet ceiling	Moderate/high: .770 source selected	Strong: edge R@6 .880	Root/facet selection before successor traversal
MuSiQue	Executable R@4 .422 versus .878 oracle-facet ceiling	Distributed: .513 source selected	Useful but contracted: edge R@6 .833	Root selection, dispersion, and contraction

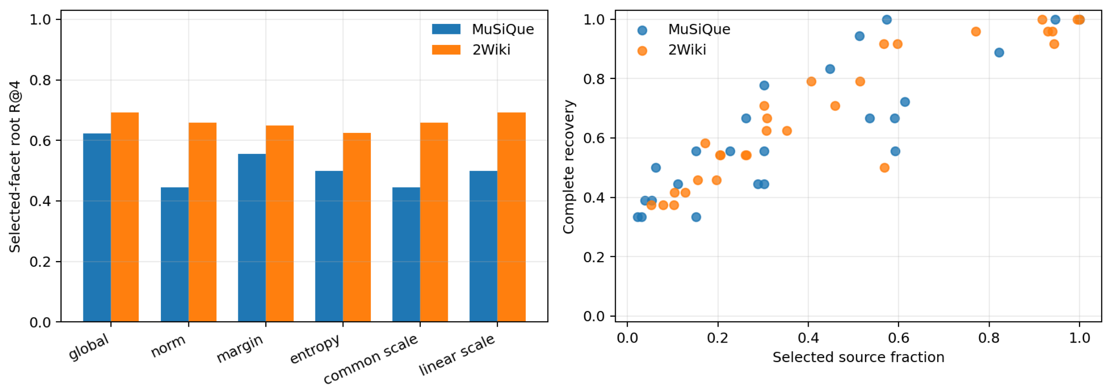

Figure 12: Executable root selectors remain below the oracle query-view ceiling; high complete recovery also requires broad conceptual activation.

7.2 Memory-to-memory topology and granularity

Once 2Wiki search enters a correct 128-token node, direct successor R@4/R@6/R@8 is .720/.880/1.000 and complete-path survival is .588/.824/1.000. This establishes a strong local associative neighborhood after entry. It does not establish executable root routing or faithful reasoning order. On MuSiQue, evidence recovery often saturates by native search depth two even for annotated depth-four questions, exposing shortcuts or contracted neighborhoods.

Table 10: Held-out 2Wiki native-edge and path fidelity at 128-token nodes.

K	Transition R@K	Complete path	Broad evidence recovery
4	.720	.588	.917
6	.880	.824	.958
8	1.000	1.000	1.000

Chunking changes the graph itself. Fine nodes improve payload precision but remove useful edges; coarse nodes contract evidence and distractors into the same conceptual unit. At 16 tokens, MuSiQue/2Wiki select only .063/.172 of source but recover complete evidence in .500/.583 cases. Broad 128-token conditions reach complete recovery while selecting .573/.918 of source. Deeper search cannot recover an edge absent from the chosen graph scale.

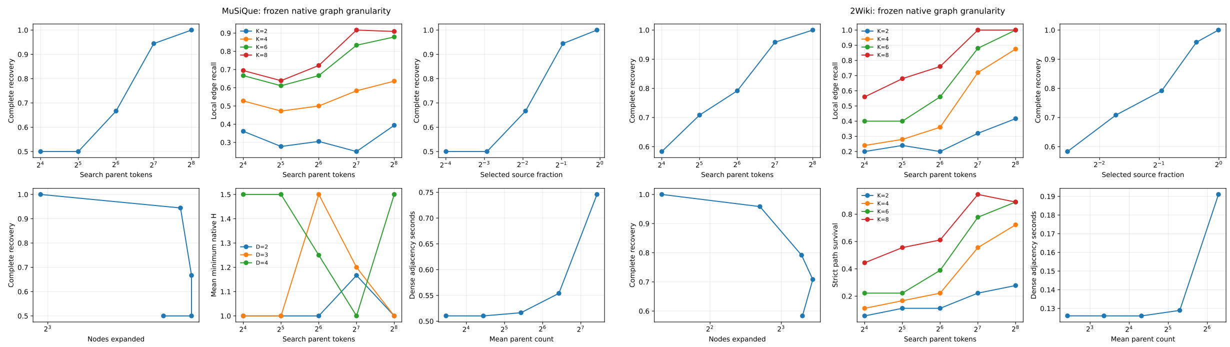

Figure 13: Granularity changes both conceptual payload and complete evidence recovery. Fine memory is sparse but topologically fragile; coarse memory is reachable but distractor rich.

7.3 Layer topology

Layer and granularity interact non-monotonically. For 2Wiki at 32 tokens, layer-0 R@6 is .680 and layer-27 R@6 is .400; at 128 tokens, layer 12 leads with .920. Restricted-context dependence rises strongly with layer but has weak, unstable relationships with edge R@6, MRR, contraction, and shortcut rate. The pretrained model therefore reproduces the controlled distinction between contextualization and traversable topology.

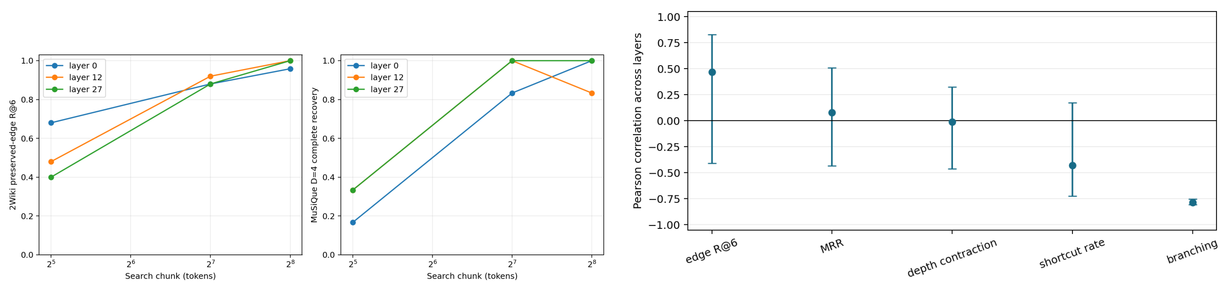

Figure 14: Pretrained layer-by-granularity topology and matched contextualization correlations. No single late or highly contextual layer dominates every graph scale.

7.4 Output validation

Frozen generation reproduces the discovery/activation asymmetry. On held-out 2Wiki, one-shot, balanced, broad, and oracle F1 are .354, .313, .271, and .362. Broad discovery recovers all evidence but activates .934 of source; final-token attention assigns .044 to annotated evidence, .754 to selected non-evidence, and .202 to the local prompt. Oracle activation is denser in evidence (.110/.461/.430) but still yields only a small, uncertain F1 gain. All-layer consumption collapses 2Wiki output relative to the selected sparse band. MuSiQue normalized accuracy is zero even under direct full context, so this backbone cannot identify a useful output policy for that dataset.

Table 11: Compact held-out 2Wiki output summary. Selected source is conceptual activation; active K/V is physical peak native state in thousands.

Condition	Evidence recall	Selected source	Active K/V	F1
Native, no source	.000	.000	1.34	.353
One-shot PRA	.719	.589	4.25	.354
Graph balanced	.760	.586	4.27	.313
Graph broad	1.000	.934	6.22	.271
Oracle evidence	1.000	.320	2.88	.362

A control-qualified GPT-5.6 Sol judge rates 2Wiki balanced versus one-shot at 90.0 semantic equivalence and +2.9 relative quality, but favors the selected sparse band over all-layer output by 37.4 points. Claude Sonnet 5 covers only 51.7% of pairs and fails both calibration anchors; it is retained as a failed-instrument diagnostic rather than efficacy evidence. Pretrained experiments thus do not establish dramatic QA gains. They show that richer search improves access more consistently than output, while layer placement and distractor competition materially affect use.

8 Related Work

Longformer, BigBird, and Routing Transformer sparsify token attention [2, 21, 14]; Transformer-XL, Compressive Transformer, and feedback-memory models carry state across segments [4, 12, 7]. These systems primarily modify access within an extended sequence or recurrent state. PRA instead probes an externally backed logical memory whose compact identities and native K/V payloads remain separately addressable.

REALM, RAG, RETRO, and Atlas retrieve text or representations before or during generation [5, 10, 3, 8]. IRCoT and Self-RAG interleave retrieval with generated reasoning or self-reflection [19, 1]. Iterative PRA does not generate an intermediate text query or rerun the transformer during compact search. This makes associative transitions cheap and auditable, but less expressive than an agent that can reformulate the task.

H2O, SnapKV, and QUEST select or retain K/V entries for an available context [22, 11, 18]. PRA separates root activation, associative identity search, and later native-K/V disclosure. The distinction motivates separate metrics for topology, conceptual activation, physical K/V, attention allocation, residual change, and output. At a computational level, this resembles bounded associative memory: ongoing local processing is occasionally influenced by a limited activated working set. We use this analogy only to organize resource and control questions, not to claim biological equivalence.

9 Discussion

The experiments support a progression from *potential memory* to *activated memory*, *attended memory*, *useful state change*, and *preserved computation*. Receptive field changes potential associative topology. Evolving-state PRA can traverse that topology. Ordinary selection, however, exposes too many plausible distractors, so evidence does not dominate the shared softmax. Oracle forcing demonstrates that when evidence does dominate, the frozen transformer can convert it into substantially better margins and predictions.

Controlled activation is therefore the primary unresolved mechanism. It is not equivalent to choosing the largest K, deepest search, or broadest source coverage. Each plausible memory given another opportunity to enter attention also competes with evidence. The Qwen root-facet ceiling and the controlled oracle intervention show that useful representations already exist; selecting a small, evidence-dense working set remains difficult.

9.1 Parameter directionality and adaptive effort

One fixed PRA configuration is unlikely to be optimal because each control moves opportunity, distractor pressure, physical state, and compute differently (Table 12). More facets F preserve more query intents but create more candidate competition; this helped Hotpot root inclusion while leaving QASPER unchanged at the same 20% budget. More roots R improve entry coverage but broaden the selected working set. More neighbors K improve successor availability at superlinear comparison cost: the held-out Hotpot oracle-root control recovers all oracle parents at $K = 4, B = 6$, whereas faithful 2Wiki transition and path recovery require $K = 8$. Neither operating point is universal. Greater hop depth H can recover long paths, but global contextualization can contract them into shallow shortcuts before routing begins. A larger final budget B tolerates ranking error while increasing active K/V and softmax competition. Raising an acceptance threshold θ reverses that tradeoff, improving precision while risking weak bridge evidence; earlier threshold-calibration gates did not identify a reliable closure rule.

Consumer placement is a separate axis. More PRA layers offer repeated assimilation but consume more layer-token K/V and can perturb or erase useful residual changes. The best search layer need not be the best consumer layer or preservation schedule. Finally, chunk granularity G changes the graph itself: coarse nodes improve containment at low node count but disclose irrelevant payload, whereas fine nodes improve payload precision while increasing graph size and sensitivity to missing edges, K, and H.

Table 12: Dominant parameter directions, not universal monotonic laws. Effects depend on dataset, layer, granularity, and routing quality.

Parameter ↑	Root	Path	Distractors	Active K/V	Search	Typical benefit
F facets	↑	indirect ↑	↑	indirect ↑	↑	Ambiguous, multi-intent queries
R roots	↑	↑	↑↑	↑	↑	Uncertain root rank
K neighbors	—	↑	↑	↑	↑↑	Noisy successor rank
H hops	—	deep paths ↑	↑↑	↑	↑↑	Distributed evidence
B budget	↑/—	↑	↑↑	↑↑	—/↑	Incomplete coverage
θ threshold	precision ↑, recall ↓	varies	↓	↓	↓	Confidence filtering
PRA layers	—	potential ↑	interference	↑↑	↑	Repeated assimilation
Finer G	varies	edges can ↓	payload ↓	↓/node	nodes ↑	Precise disclosure

These coupled directions motivate adaptive rather than globally maximal effort: query/state → $(F, R, K, H, B, \theta, L, G)$. A QASPER-like direct query may use a narrow policy; a Hotpot bridge query may warrant additional facets or roots; distributed MuSiQue evidence may warrant more search and coverage budget only when confidence justifies the added distractor load. Paper 3.5 asks whether a controller can make that decision; this paper does not train or evaluate one.

Search, consumption, and preservation operate on different computational timescales. A layer can have useful native Q/K topology while being a poor location for injecting value vectors. An early consumer can create a useful change that later layers erase. These observations favor architectures that treat local ongoing computation, sparse nonlocal activation, and assimilation time as separate controls. They do not establish that one layer schedule generalizes across model families.

The bounded-processing analogy is deliberately restrained. A limited working set can be useful because not every associatively plausible memory should dominate current processing. Iterative retrieval permits one activated memory to redirect the next search, while finite budgets force a choice about what becomes influential. Adaptive control is a natural next question, but it is not evaluated here.

10 Limitations

The controlled models are 740K-parameter scientific instruments with a synthetic vocabulary, not language-model scaling evidence. Five seeds leave wide Student-t intervals; the minimum exact two-sided result with five same-direction nonzero effects is $p = .0625$. The primary PRA corpus has 32 examples per seed and the mechanistic cohort has 16. Model–example bootstrap analyses do not turn repeated examples across seeds into independent model replicates. Oracle forcing is a causal ceiling, not an executable router.

The local mask uses a dense kernel, so controlled elapsed time does not estimate optimized LocalSA. Exact SA-to-PRA conversion is intentionally frozen. The experiment tests inherited topology and memory use, not PRA-aware training. Content shuffling also shows that some ordinary selected-memory gain is not evidence specific.

Natural cohorts are diagnostic: 18 held-out MuSiQue identities, 24 2Wiki identities, smaller inherited HotpotQA/QASPER sets, and one principal Qwen3-0.6B family. Oracle-root analyses isolate successor geometry rather than end-to-end routing. Annotation graphs need not be the model's preferred reasoning paths. MuSiQue generation is backbone limited even with full context. Python graph search and trace-oriented materialization are not production-serving kernels. One external judge passes calibration; the second is partial and fails it.

The causal $W = 16$ versus global topology result is established in the controlled 25-model family. Pretrained Qwen layer/chunk diagnostics are directionally consistent with architecture-dependent topology but do not constitute a causal pretrained LocalSA-versus-GlobalSA comparison.

11 Scope Boundaries and Follow-up Work

Paper 3: physical disclosure. Given selected conceptual memory, how much native K/V detail should become active? Evidence-centered materialization, overlap deduplication, and quality at a smaller physical payload belong there. Paper 2.5 measures selected identities and uses matched causal payloads; it does not optimize disclosure.

Paper 3.5: adaptive control. How should query difficulty or confidence adapt facets, roots, K, hops, budgets, retries, and runtime effort? The distractor-dominated result directly motivates a controller that spends additional search only when it improves evidence concentration. No such controller is trained or evaluated here.

Paper 4: learned robustness. Can frozen-to-LoRA-to-full-weight-to-native PRA training make memory representation, consumption, and assimilation robust to imperfect sparse memory? The motivation is not that frozen consumers cannot use memory: oracle evidence proves they can. Training may instead improve robustness, portability, and preservation under noisy executable activation. Because oracle evidence substantially improves frozen-model output, PRA-aware training should evaluate robustness as memory quality degrades from oracle to partially correct to ordinary noisy retrieval, not merely oracle performance.

12 Conclusion

Transformer memory exposes an associative topology whose explicit traversability depends on native receptive field. In the controlled family, final-layer edge R@4 is .428 at $W = 16$ and .187 under global attention. Evolving-state PRA can exploit that structure to recover more paths under the same physical K/V budget.

Better traversal is useful when it changes evidence. It does so in 59/400 (14.8%) model–example units, which gain 2.164 correct-label margin, even though the current policy produces too few such cases for a stable aggregate accuracy gain. The causal intervention explains the operational gap: ordinary memory is distractor dominated, while matched oracle evidence raises frozen accuracy from .140 to .398 and margin from -3.103 to $-.703$. The frozen transformer can consume correct PRA memory.

Paper 2.5 therefore ends at controlled activation and preservation. The immediate challenge is not another graph-search variant or a claim of absolute consumer incapacity. It is to activate a sparse working set whose influence is concentrated on useful evidence, place that activation where the residual computation can use it, and preserve the resulting change. Physical K/V disclosure, adaptive effort, and learned robustness remain deliberately separate research programs.

A Chronological Gate Audit: Projection and Locality

The preceding results freeze the original chunk-level closure as Level 1. We next change one assumption at a time. All comparisons reuse the same five checkpoints, held-out examples, depth, root anchor, path score, and final unique-parent/K/V budget. Neither gate trains a new parameter.

A.1 Gate 1: projection-correct frontiers

Table 13 compares the historical $W_m h_A$ frontier with the contract-correct $W_q h_A$ frontier at the primary 10% chunk budget. Correction materially reduces the QASPER failure: any/chain recall rises by 0.075 and coverage by 0.013 relative to historical closure. It nevertheless remains 0.013 below one-shot any/chain recall. On HotpotQA, correction raises any recall by 0.025 and restores exact identity by 0.013 relative to historical closure, but chain completion falls by 0.013 and exactly matches one-shot. Comparison counts are unchanged because this gate changes only the projection used by each activated frontier.

Table 13: Gate 1 at a matched 10% final chunk budget. Q-frontier is the corrected $W_q h_A \to W_m h_B$ traversal; M-frontier is the historical direct reuse of $W_m h_A$.

Dataset	Method	Any	Chain	Exact	Coverage	Comparisons
HotpotQA	One-shot	.700	.138	.013	.207	50.1
HotpotQA	M-frontier	.663	.150	.000	.202	219.6
HotpotQA	Q-frontier	.688	.138	.013	.201	219.6
QASPER	One-shot	.963	.963	.213	.477	142.6
QASPER	M-frontier	.875	.875	.200	.466	1395.9
QASPER	Q-frontier	.950	.950	.225	.479	1395.9

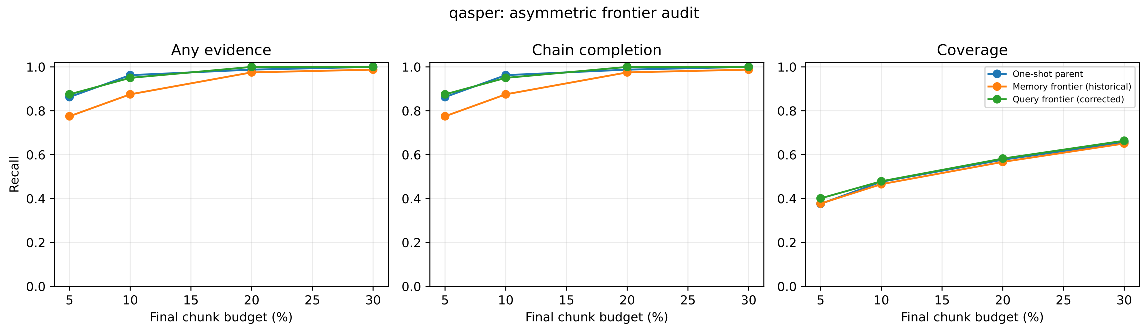

Figure 15: Gate-1 QASPER curves. Respecting the asymmetric projection contract recovers most of the historical closure loss, but one-shot remains the sparse-retrieval reference.

Projection asymmetry therefore explains a meaningful part of the original degradation. It does not explain why the corrected traversal fails to improve chain completion. That result passes the predeclared checkpoint for testing a finer propagation scale.

A.2 Gate 2: contextual local gists

Gate 2 changes the index to 256-token contextual parents with eight 32-token local gists. The one-shot and parent-closure baselines use the same parent features; local closure alone traverses subchunk states. Parent deduplication holds final materialization to the same parent count.

The designed local-bridge control separates the methods cleanly. At two selected parents, one-shot and projection-correct parent closure retrieve only the entry parent: exact identity, chain completion, and coverage are 0, 0, and 0.5 over 320 examples. Local closure retrieves both evidence groups in every case, with exact identity, chain completion, and coverage all 1.0. Its mean Δ_{bridge} is 1.0. The implementation can therefore expose a bridge erased by a parent mean without re-encoding the local window.

Natural data is more restrained (Table 14). At 20%, local closure improves Hotpot any recall, exact identity, and coverage by 0.013 each over one-shot, but chain completion falls by 0.013 and parent closure is stronger. On QASPER, local closure improves exact identity by 0.025 and coverage by 0.016 over one-shot while any/chain recall falls by 0.013. Local accepted edges are stronger than corresponding parent edges: mean Δ_{bridge} is 0.203 on Hotpot and 0.205 on QASPER. That measurable locality does not yet identify the correct cross-parent chain.

Table 14: Gate 2 at a matched 20% final parent/KV budget. Comparisons count semantic gist dot products; no native QK comparisons occur in this gate.

Dataset	Method	Any	Chain	Exact	Coverage	Gist cmp.	K/V tokens
HotpotQA	One-shot parent	.475	.138	.000	.217	6.5	405.8
HotpotQA	Parent closure	.513	.175	.000	.235	11.9	405.0
HotpotQA	Local closure	.488	.125	.013	.229	49.5	403.4
QASPER	One-shot parent	.938	.938	.738	.813	18.2	1036.0
QASPER	Parent closure	.950	.950	.738	.819	66.6	1037.3
QASPER	Local closure	.925	.925	.763	.829	405.2	1031.0

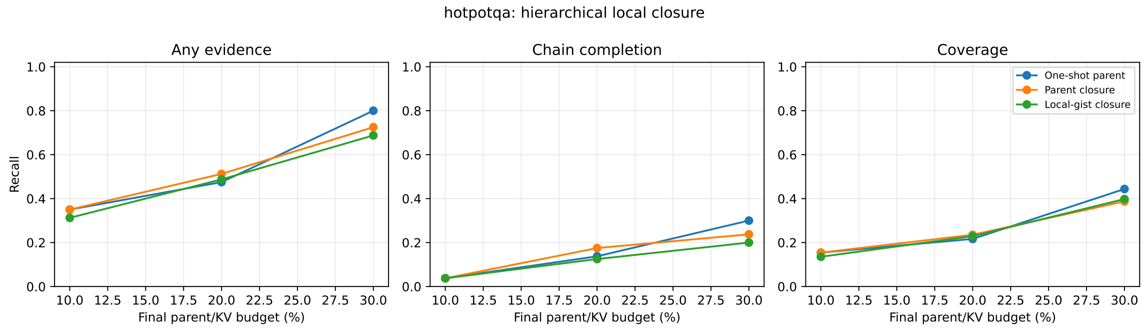

Figure 16: Gate-2 HotpotQA curves under matched final parent budgets. Local routing exposes stronger subchunk edges but does not improve evidence-group chain completion.

The cost is substantial. At 20%, local closure uses 49.5 versus 6.5 comparisons on Hotpot (7.6×) and 405.2 versus 18.2 on QASPER (22.3×). Materialized K/V differs by less than 1.3% within either dataset, confirming that the quality comparison is not purchased with a larger payload. The graph audit validates all 960 saved Gate-1/2 traces against schema v2 and finds no parent-budget violation.

A.3 Gate 3: native local QK propagation

The native-bridge control first verifies the mechanism without relying on natural annotations. The root-semantic rank retrieves entry parent A and a stronger semantic distractor, while a token in A has a model-compatible native QK match only to evidence parent B. Across five seeds and 64 cases per seed, one-shot, parent closure, and local semantic closure retrieve half the evidence and complete no chains. Native QK retrieves both groups in all 320 cases: exact identity, chain completion, and coverage are all 1.0. A distractor with an incompatible native key remains unselected. The vectorized scorer can therefore express the proposed transition.

Natural data does not reproduce that separation (Table 15). At the primary 20% final budget and 10% semantic candidate pool, max native QK matches local semantic chain completion on HotpotQA (0.125) and remains below one-shot (0.138) and parent closure (0.175). Its any-evidence recall and coverage are slightly higher than local semantic, but exact identity falls from 0.013 to zero. Expanding the pool to 20% lowers chain completion to 0.113. On QASPER, max native QK reaches 0.900 with the 10% pool and 0.863 with the 20% pool, below local semantic (0.925) and one-shot (0.938). More candidates amplify rather than repair propagation error.

Table 15: Gate 3 at a matched 20% final parent/KV budget. “QK-10” and “QK-20” are the primary max score with 10% and 20% semantic candidate-parent pools; “QK-Top4” is the sole Top-4 control at 20%. Native dots count token-pair/query-head products.

Dataset	Method	Chain	Coverage	Sem. cmp.	Native dots	K/V tok.	Time (s)
HotpotQA	One-shot	.138	.217	6.5	0	405.8	.0046
HotpotQA	Local semantic	.125	.217	98.3	0	407.0	.0105
HotpotQA	QK-10	.125	.223	114.6	95,712	406.3	.0121
HotpotQA	QK-20	.113	.208	114.6	183,667	407.2	.0132
HotpotQA	QK-Top4	.175	.265	114.6	183,667	406.0	.0143
QASPER	One-shot	.938	.813	18.2	0	1036.0	.0059
QASPER	Local semantic	.925	.829	540.0	0	1042.9	.0463
QASPER	QK-10	.900	.810	540.0	1,198,554	1043.7	.0467
QASPER	QK-20	.863	.794	540.0	1,949,958	1039.8	.0542
QASPER	QK-Top4	.838	.770	540.0	1,949,958	1041.3	.0624

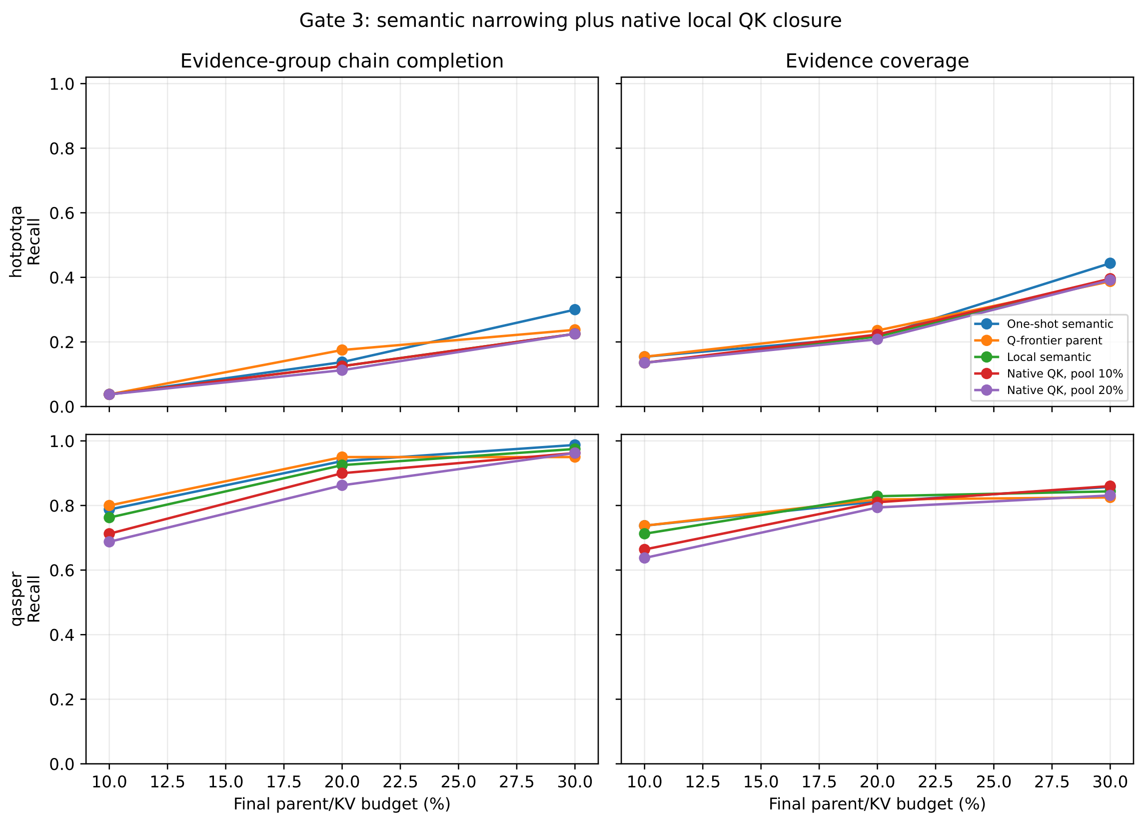

Figure 17: Gate-3 chain completion and evidence coverage across final budgets. Primary native-QK propagation does not dominate local semantic or one-shot routing; the larger candidate pool is usually worse, especially on the direct-retrieval QASPER control.

The Top-4 control is the one localized positive: HotpotQA chain completion at 20% rises to 0.175, matching parent closure and exceeding both one-shot and local semantic. It does not persist as a dominant curve: at 10% all methods tie at 0.038, at 30% Top-4 reaches 0.238 versus one-shot 0.300, and QASPER degrades to 0.838 at 20%. The $\mu + \sigma$ threshold is effectively identical to fixed

Top-k in this hard-cap regime and offers no rescue. These predeclared controls are therefore a mixed point observation, not a positive Gate-3 result. This is specifically a failure of max-reduced Top-1 native closure under the tested competition rule; it does not yet show that useful native associative edges are absent. We test that distinction below.

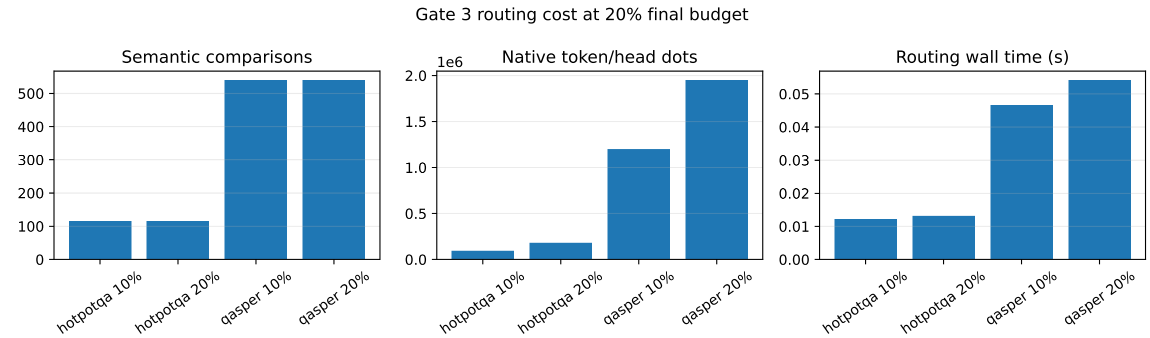

Figure 18: Gate-3 cost at the 20% final parent budget. Semantic and native work are reported separately. Candidate expansion from 10% to 20% nearly doubles native dot products without a quality gain. Wall time measures this batched reference implementation, not an optimized kernel.

At 20%, native refinement compares means of 1.1/1.7 candidate parents and 161/315 candidate tokens on HotpotQA for the 10/20% pools; QASPER compares 3.4/5.6 parents and 874/1,422 tokens. This produces 95,712–183,667 native dots on HotpotQA and 1.20–1.95 million on QASPER per example/seed, while materialized K/V stays within 0.8% of one-shot. Every saved graph reports zero K/V materializations during closure. Gate 3 changes search work and selected identities, not payload quantity or transformer attention semantics.

A.4 Oracle convergence, displacement, and competition

The previous gates score whether a selected set touches each evidence group. Here we ask the stricter question of whether it approaches the exact feasible parent set used by the annotated Paper-2 oracle. We import that oracle constructor unchanged, pass it the cached evidence spans and 256-token parent boundaries, and assert equality with the feature mask for every example. For selected set $S(B)$ and oracle S_O, we report recall, precision, Jaccard overlap, and $\mathbf{1}[S_O \subseteq S(B)]$ at 5, 10, 20, 30, and 40% parent/KV budgets. Figure 19 shows that Hotpot convergence remains slow for every frozen method. At 20%, parent closure has the best non-control recall (0.235); the native Top-4 reduction reaches 0.265, but complete recovery is only 0.013. At 40%, one-shot is strongest at 0.529 recall, while all iterative variants remain between 0.438 and 0.477. Among the few Hotpot seed/example trajectories that ever recover the complete oracle, the mean minimum budget is 31.7–34.0%; only 10–12 of 80 trajectories do so by 40%.

QASPER provides the expected direct-retrieval control. At 20%, one-shot reaches 0.813 oracle recall and 0.738 complete recovery. Parent and local closure reach 0.819/0.738 and 0.829/0.763, respectively, but native max falls to 0.794/0.725 and the Top-4 reduction to 0.770/0.713. At 40%, one-shot and local differ by only 0.002 recall. Iteration therefore does not buy a useful convergence advantage on this mostly one-group task.

That aggregate comparison hides a specific failure. At 20%, one-shot touches at least one oracle parent in 0.475 of Hotpot and 0.938 of QASPER seed/example runs. Of those successful roots, local, native-max, and native-Top-4 closure fail to preserve every touched oracle parent in 0.132/0.052, 0.184/0.117, and 0.158/0.156 of Hotpot/QASPER runs. Parent closure displaces none on Hotpot and 0.013 on QASPER. This is not literal eviction after admission: the frozen implementation reserves approximately half of B for root retrieval, keeps admitted roots, and fills the remaining slots through propagation. A "displaced" one-shot hit therefore falls outside the smaller iterative root allocation and is not recovered. Every replacement observed in this diagnostic is a hop-2 proposal.

Displacement is associated with root uncertainty rather than a larger final payload. For QASPER local closure, displaced cases have mean root Top-1–Top-2 margin 0.037 and entropy 1.842, versus 0.123 and 1.310 for preserved cases. Native-max cases show the same direction (0.089/1.728 versus 0.122/1.286), and native Top-4 cases remain weaker but consistent (0.104/1.618 versus 0.121/1.286). These are descriptive associations over only 4, 9, and 12 displaced runs, not a fitted confidence rule.

We isolate that mechanism with a matched-B protected-root counterfactual. It first preserves one-shot oracle hits, then fills the remaining slots in the frozen iterative order. This uses oracle labels and is an upper bound, not a deployable routing policy. On QASPER it raises oracle recall by 0.006 for parent closure, 0.015 for local closure, 0.044 for native max, and 0.068 for native Top-4; the two native methods both reach 0.838 recall and 0.763 complete recovery. Hotpot gains are smaller (0.000–0.031 recall) because its dominant error remains initial root discovery.

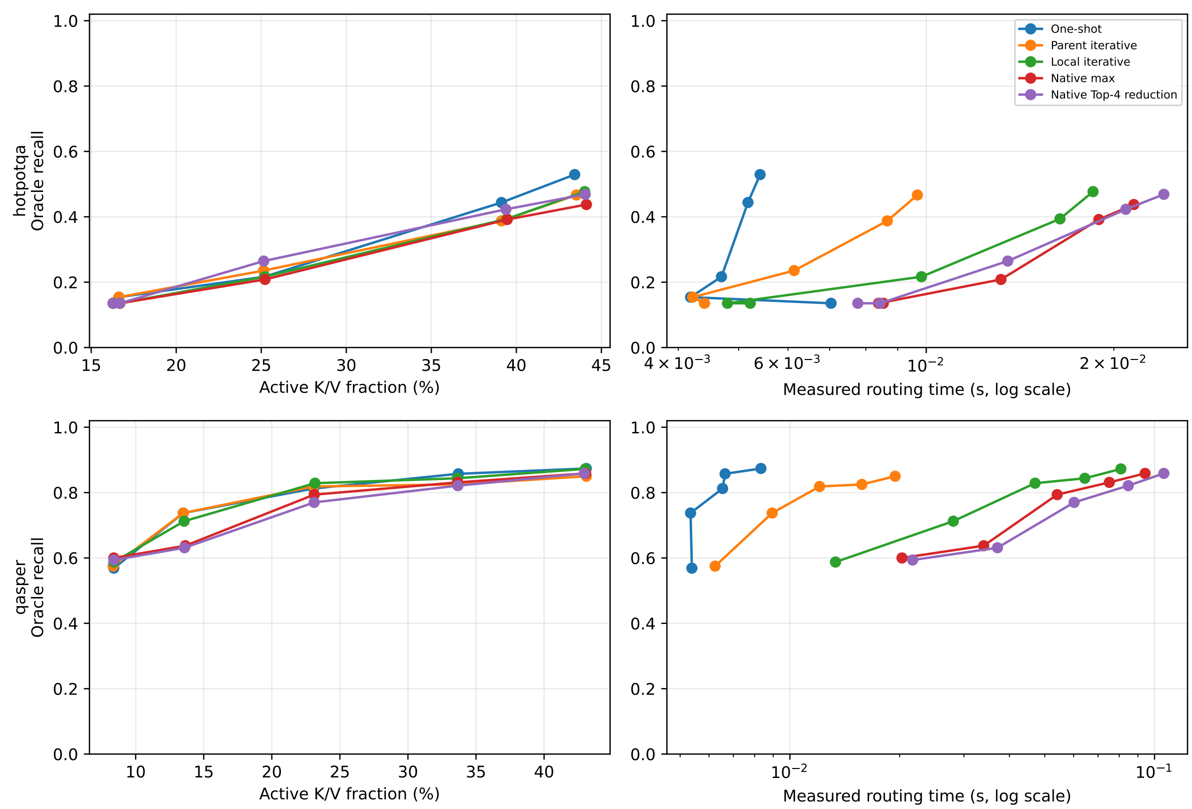

Figure 19: Exact Paper-2 oracle recall against active K/V fraction (left) and measured routing time (right), with HotpotQA above QASPER. The Top-4 label averages four token/head matches; it is not a four-parent beam. Iterative search adds routing work but supplies no cost-adjusted Hotpot advantage. Times describe the batched reference runner, not an optimized kernel.

Table 16: Oracle convergence and root preservation at the primary 20% parent/KV budget. "Displaced" is conditioned on a one-shot oracle hit. Protected recall is an oracle-informed counterfactual at the same B. Semantic comparisons and native dots are different units.

Dataset	Method	Recall	Complete	Displaced	Protected R	Sem. cmp.	Native dots
HotpotQA	One-shot	.217	.000	–	.217	6.5	0
HotpotQA	Parent iterative	.235	.000	.000	.235	11.9	0
HotpotQA	Local iterative	.217	.013	.132	.242	98.3	0
HotpotQA	Native max	.208	.000	.184	.240	114.6	183,667
HotpotQA	Native Top-4 reduction	.265	.013	.158	.290	114.6	183,667
QASPER	One-shot	.813	.738	–	.813	18.2	0
QASPER	Parent iterative	.819	.738	.013	.825	66.6	0
QASPER	Local iterative	.829	.763	.052	.844	540.0	0
QASPER	Native max	.794	.725	.117	.838	540.0	1,949,958
QASPER	Native Top-4 reduction	.770	.713	.156	.838	540.0	1,949,958

Low end-to-end recovery can arise before or after the first evidence group. At 20%, one-shot reaches the first source-ordered Hotpot group in 0.200 of seed/example runs; parent closure reaches 0.213, local closure 0.200, native max 0.188, and native Top-4 0.263. To isolate propagation, we then force only that annotated source group active and rank the next evidence group among all other parents. Any local node in either evidence parent may realize an edge. Parent semantic uses the projection-correct query-to-memory score, local semantic uses the strongest local pair, and native QK uses the exact frozen pre-RoPE max token/head reduction.

The conditioned result is much stronger than end-to-end closure (Figure 20). Median true-edge rank is 3 in all three spaces. Parent semantic, local semantic, and native QK respectively obtain R@1 of 0.141, 0.259, and 0.235; R@4 of 0.765, 0.741, and 0.765; and MRR of 0.424, 0.480, and 0.469. Every semantic transition is within Top-8, as are 16 of 17 native transitions. Mean oracle-minus-best-distractor margins remain negative ($-0.118, -0.143, -0.039$), so the correct edge is usually present but loses hard competition. Across 85 seed-transition comparisons, the best semantic and native ranks are both within Top-4 in 56 cases; semantic alone is near-top in 14, native alone in 9, and both are poor in only 6.

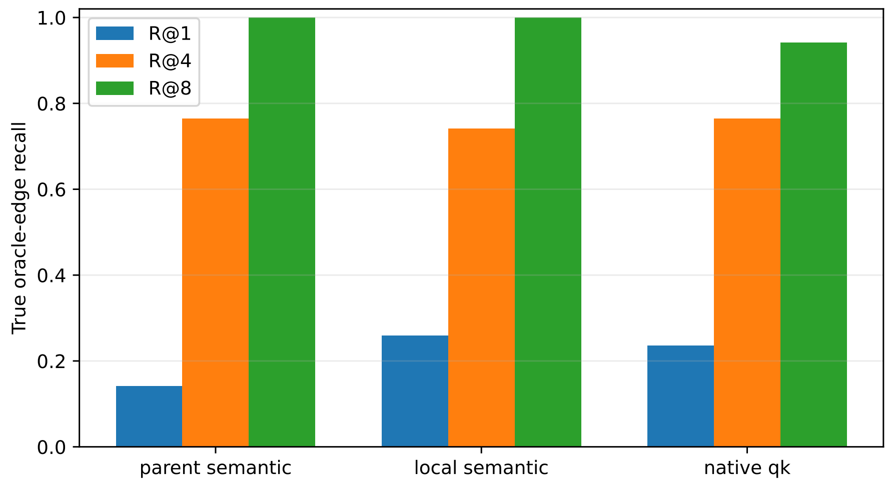

Figure 20: Rank recall for the true next Hotpot evidence group after conditioning on the first group. Most annotated transitions exist within the first four candidates even though they rarely win Top-1. Semantic statistics contain 17 transitions under five projection seeds; native QK has 17 seed-invariant transitions.

The Gate-3 Top-4 signal should not be read as parent-beam R@4. It averages four strong token/head matches before ranking parent transitions. Its four 20% chain wins over max reduction comprise two questions. In one seed-case a near-tied oracle edge moves from rank 2 to rank 1: its margin changes from -0.0038 to $+0.0157$. The other question accounts for three seed-cases, but its source-ordered edge remains rank 6 under both reductions; completion comes from root/reverse-order selection rather than better forward propagation. The cached features contain no entity/topic labels, so we do not assign a semantic relation to those distractors from parent IDs alone.

We also audit whether incomparable score families create the competition error. They never enter a common raw TopK in the frozen router: root and semantic propagation use normalized cosines followed by a 0.25 root anchor and $[0, 1]$ affinity/path products, while native QK uses a sigmoid score followed by its own 0.25 anchor and path product. The audit nevertheless reveals severe native compression. At 20%, maximum and Top-4 native logits average 13.95/12.16 on Hotpot and 14.29/12.47 on QASPER; every sigmoid exceeds 0.999, and 94.8–100% exceed 0.99999. Top-4 does improve the Hotpot oracle-edge score quantile from 0.561 to 0.731, consistent with less sensitivity to one extreme token/head match, but it does not generalize to QASPER.

As an offline control, we fix the candidate identity union already exposed by one-shot and iterative routing, fit per-family z-score or empirical-quantile transforms on a hash-stable validation partition, and evaluate unchanged on held-out examples. The transforms cannot discover a missing parent. They leave QASPER local recall at 0.900 and raise native-max recall from 0.867 to 0.883; quantiles raise native Top-4 from 0.833 to 0.883. They do not improve Hotpot recall: the best normalized native result remains 0.205, and local z-scoring remains 0.190. Calibration can preserve a strong root once it enters the candidate pool, but does not solve Hotpot's missing-root problem.

Finally, we simulate, without changing SDK routing, $k = 1$ when the Top-1/Top-2 gap exceeds a validation-selected threshold and $k = 4$ otherwise. On held-out edges, parent semantic selects $k = 4$ always and reaches 0.686; local semantic averages $k = 3.83$ and reaches 0.657 versus its 0.686 fixed-R@4 ceiling; native averages $k = 3.57$ and reaches 0.714, exactly its fixed R@4. The split is only 10/7 distinct native transitions and 50/35 seed-sensitive semantic transitions, so this is a mechanism diagnostic, not a tuned policy result. The simple margin rule saves little competition and does not explain the Top-4 reduction signal.

These diagnostics motivate the deployable, label-free competition gate evaluated next. A production policy cannot know oracle roots, but it can lock identities using root confidence or cross-seed agreement, then let propagation compete only for remaining slots. This directly tests whether the protected-root upper bound and near-Top-4 Hotpot edges survive without oracle access.

A.5 Monotonic root persistence and adaptive competition

The previous router divides B before seeing the score distribution. We instead compute the ordinary full root ranking $R = (r_1, \ldots, r_B)$, choose a label-free lock set $L \subseteq R$, and enforce

$$L \subseteq S_{\text{final}}, \qquad B_{\text{prop}} = B - |L|, \qquad |S_{\text{final}}| = B. \tag{5}$$

Propagation can compete only for the B_{prop} unlocked slots. Duplicate parent proposals collapse to their strongest path; if propagation supplies too few unique identities, the original root order fills the remainder. The equality above therefore changes identities, not active parent/KV capacity. No evidence label appears in the policy API.

We test fixed prefixes of 1, 2, and 4, first-large-score-drop thresholds, full-vector root z-score thresholds, and the fraction of five router seeds placing a parent within root Top-B. For transitions we retain projection-correct local semantics and the frozen Top-4-mean pre-RoPE native-QK scorer. Native magnitudes are saturated after a sigmoid, so cross-source native proposals use deterministic per-source raw-logit rank. Adaptive breadth chooses $k = 1$ for concentrated semantic/native Top-1 agreement, $k = 2$ for moderate concentration with Top-4 overlap, and $k = 4$ otherwise. Thresholds and policy identities are selected on the hash-stable validation examples only. We freeze two roles per geometry: the unconstrained preservation winner and the best exploratory root policy required to activate propagation on at least 25% of validation runs. This second role prevents a one-shot-equivalent winner from making transition competition vacuous.

Validation selects root $z > -0.5$ for preservation in both geometries. On this sample that rule locks all available Top-B identities, performs no propagation, and becomes an independently implemented one-shot parity check. For exploration, validation selects root $z > 1$ with native rank and seed agreement at 0.6 with semantic propagation. Fixed $k = 1$ wins for native rank; fixed $k = 4$ wins for semantic propagation. None of the nine adaptive threshold pairs wins. Across held-out adaptive-policy traces, native decisions are mostly $k = 1/2$ (1,674/1,014 source decisions, with 30 at $k = 4$); the richer rule does not expose a natural-data gain hidden by fixed breadth.

Table 17: Held-out results at the primary 20% final parent/KV budget. Seven HotpotQA and six QASPER examples are each evaluated under five frozen router seeds. "Prop." is the fraction of seed/example runs that add a propagated identity. The preservation rows lock full root Top-B and therefore exactly match one-shot.

Dataset	Policy	Oracle R	Chain	Root present	Root locked	Prop.	Native dots
HotpotQA	One-shot	.205	.143	.343	.343	.000	0
HotpotQA	Monotonic preserve	.205	.143	.343	.343	.000	0
HotpotQA	Native-rank explore	.233	.143	.343	.314	.314	82,388
HotpotQA	Semantic explore	.176	.086	.343	.257	.314	82,388
QASPER	One-shot	.883	.800	.967	.967	.000	0
QASPER	Monotonic preserve	.883	.800	.967	.967	.000	0
QASPER	Native-rank explore	.883	.800	.967	.967	.767	907,281
QASPER	Semantic explore	.883	.800	.967	.967	.400	621,193

The result separates preservation from discovery. Monotonicity removes the measured QASPER regression exactly, but the selected exploratory policies spend search work without improving 20% chain completion. Native rank gains 0.028 Hotpot oracle recall (0.205 to 0.233) while chain completion remains 0.143. At 30/40%, native chain completion falls from one-shot 0.343/0.371 to 0.257/0.286. Semantic exploration matches one-shot chain completion at 30/40% (0.343/0.371), and reaches QASPER 0.950 recall and 0.900 chain completion at 40%, but it does not produce the required Hotpot advantage. The dual-geometry diagnostic scorer takes about 0.006 seconds per Hotpot and 0.044–0.068 seconds per QASPER run at 20%; these reference timings include the alternate geometry used for confidence measurement and are not an optimized kernel claim.

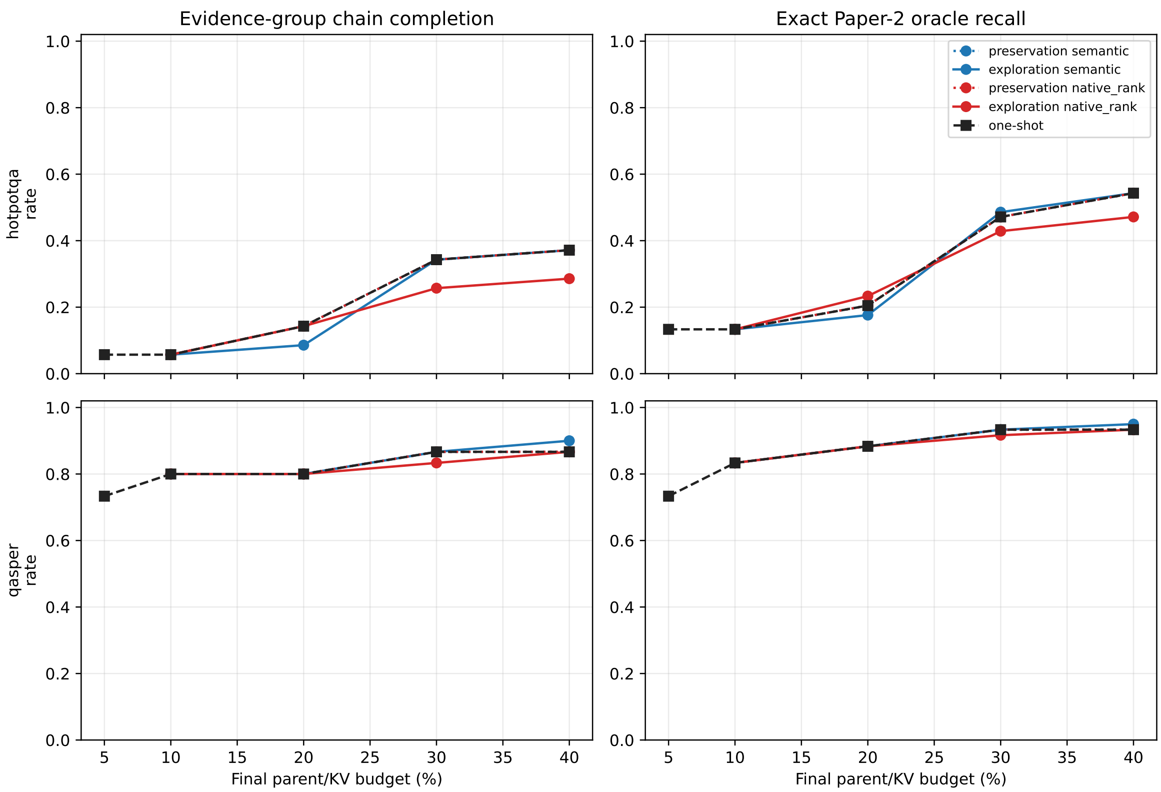

Figure 21: Held-out quality under matched final budgets. Dotted preservation curves reproduce one-shot. Exploratory propagation does not improve Hotpot chain completion; the isolated 20% native oracle-recall gain does not persist as budgets grow.

The failure decomposition is more decisive than the aggregate delta. Across exploratory held-out runs and budgets, the first source-ordered Hotpot evidence group is absent from full root Top-B in 52.9% of cases. It is present but rejected by the lock in 11.1%; the remainder is locked. At 20%, the native policy locks the first group in 31.4% of runs and recovers all later groups in 45.5% of those conditioned runs, yet end-to-end chain completion still cannot exceed one-shot. Competition policy cannot propagate from an entry that was never admitted.

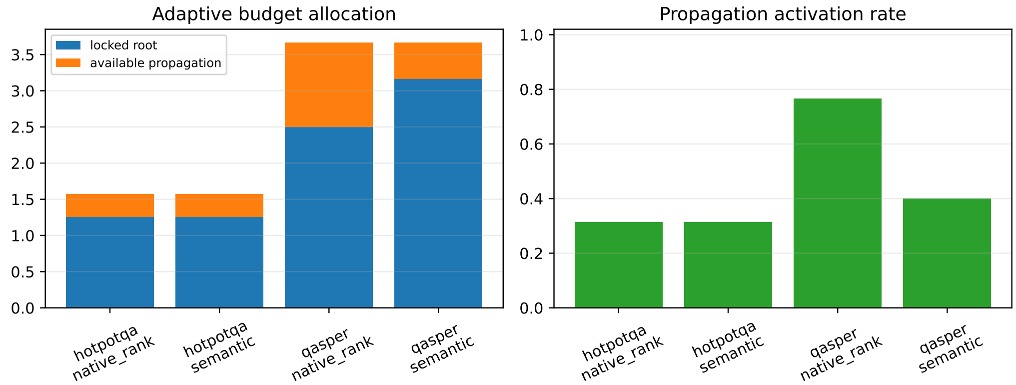

Figure 22: Held-out exploratory budget allocation and propagation activation at 20%. The final parent budget remains matched even when the locally selected lock count differs.

Controlled cases verify the mechanism independently of that natural limitation. Obvious direct retrieval locks the root and stops. With the target outside root Top-B, fixed $k = 1$ obtains 0.5 oracle recall on both an ambiguous rank-2 transition and a distractor-heavy rank-4 transition. Fixed $k = 4$ and adaptive 1/2/4 both recover 1.0 at the same final B; the adaptive rule chooses 2 and 4, respectively. Thus persistence and adaptive breadth are implemented and can regulate designed cases, but validation does not promote them on natural data.

This gate therefore advances a narrower recommendation: refine entry discovery before changing transition geometry. The next experiment below tests whether the pretrained contextual question already contains a useful entry relation that the single final-token query collapses.

A.6 Contextual query facets and native-head entry search

The inherited root is one attention-input hidden state, $h_q = H_q[T-1]$, projected by the frozen query map and cosine-scored against each fixed parent memory gist. We reproduce this operation exactly as condition A. For condition B, the model still sees the complete prompt exactly once. Within the recorded question span, overlapping windows produce

$$g_{q,i} = \frac{1}{b_i - a_i} \sum_{t=a_i}^{b_i - 1} H_q[t], \qquad s_j = \max_{i \in \{\text{global,local}\}} \cos(W_q g_{q,i}, W_m h_j). \tag{6}$$

The global final-token row remains in the set. All facets nominate through one score per parent and one final Top-B; no facet receives a private payload budget. We test 4/8/16-token windows at half-window stride and max versus Top-2-mean facet reduction. Validation selects width 4, stride 2, and max. The query-feature manifest verifies one full prompt forward per example, zero independent window forwards, and bit-exact equality between condition A and the frozen root cache.

The learned 128-D semantic router has no legitimate head axis. We do not invent pseudoheads. Conditions C and D instead audit the model's 16 real layer-27 pre-RoPE query heads against their 8 grouped-query-compatible K heads. Valid local K tokens are vectorized into a parent mean; each query head h compares only with K/V head $\lfloor h/2 \rfloor$. Validation selects independent per-head Top-1 nominations followed by deduplication and one global Top-B rerank. Because this changes semantic projections to native QK, C/D are an operational head audit, not a strict factorial contrast with A/B. Native mean-head controls E/F isolate span effects within that representation.

Table 18: Held-out root-entry quality at the primary 20% parent budget. A/B average seven HotpotQA or six QASPER identities over five frozen router seeds. C/D are deterministic native-QK conditions and are not replicated as fictitious seeds. "Entry" means that the first ordered evidence group occurs in root Top-B.

Dataset	Variant	Entry	Rank	MRR	R@1	R@2	R@4	R@8
HotpotQA	A: global semantic	.343	2.60	.503	.229	.486	.971	1.00
HotpotQA	B: multi-span semantic	**.457**	**2.23**	**.626**	**.429**	**.571**	.943	1.00
HotpotQA	C: global split-head	.429	3.00	.585	.429	.571	.714	1.00
HotpotQA	D: multi-span split-head	.143	3.57	.379	.143	.286	.714	1.00
QASPER	A: global semantic	.967	1.60	.811	.700	.800	.967	1.00
QASPER	B: multi-span semantic	**.967**	**1.50**	**.828**	.700	**.900**	.967	1.00
QASPER	C: global split-head	.667	4.67	.292	.000	.333	.500	.833
QASPER	D: multi-span split-head	.667	5.50	.232	.000	.167	.333	.833

At 20%, multi-span semantic search improves Hotpot entry presence by 0.114, oracle recall from 0.205 to 0.295, mean rank from 2.60 to 2.23, and oracle precision from 0.343 to 0.457. Its mean oracle margin remains negative but narrows from −0.029 to −0.009; Jaccard rises from 0.167 to 0.248 and false-positive parents fall from 1.114 to 0.943. The paired entry changes are six gains, two losses, and 27 ties. At 10/30/40%, entry presence changes from 0.229/0.629/0.686 to 0.429/0.714/0.800. QASPER entry presence, oracle recall, precision, and Jaccard remain exactly 0.967, 0.883, 0.296, and 0.288; MRR rises slightly.

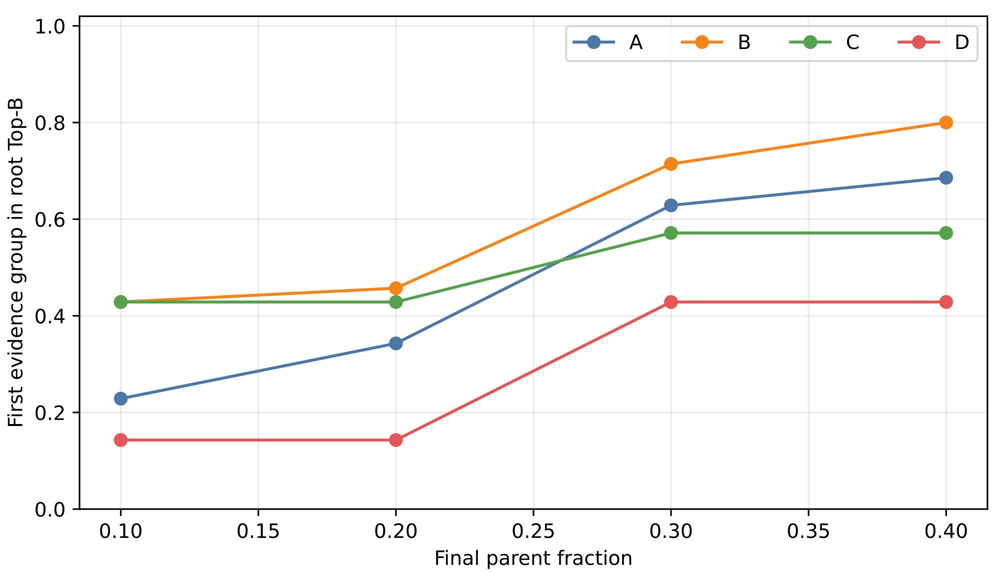

Figure 23: Held-out HotpotQA first-evidence entry across matched parent budgets. Contextual multi-span semantic facets (B) improve the exact global semantic control (A) at every budget. Native head splitting (C/D) does not combine with that gain.

The gain is not free search. Mean Hotpot query-parent comparisons rise from 5.9 in A to 67.6 in B, while C and D require 93.7 and 1,081.1 raw span/head comparisons. QASPER requires 16.3, 82.0, 261.3, and 1,312.0, respectively. Thus B gains about 0.00185 entry probability per extra Hotpot comparison. Batched GPU wall time is approximately 0.0048 seconds for both A and B at this small scale; C/D take 0.017/0.016 seconds on Hotpot and 0.040/0.042 on QASPER. These are diagnostic timings, not optimized-kernel throughput. Native mean-head E/F controls move Hotpot entry from 0.286 to 0.429 but remain far below the semantic QASPER control (0.333 to 0.500 versus 0.967). The native representation change, not merely head aggregation, is therefore consequential.

The provenance audit finds six Hotpot seed/example cases in which A misses and B succeeds. Winning windows include relation-bearing fragments such as "Most Outstanding Defensive Player" and entity fragments containing "Scott". Two paired losses and the still-negative mean margin warn against treating every local maximum as reliable. Four global-head specialization cases exist, but gains are concentrated and do not survive the native split-head aggregate. The span–head interaction audit finds no unique span-only or synergistic win in the deterministic native conditions. Separate span-dilution, head-specialization, and combined synthetic controls all recover their designed parent while conserving the final budget.

The predeclared 0.10 Hotpot gain gate is met without QASPER loss, so we run the small confirmation in Table 19. It freezes B and reconnects only the previously selected $z > 1$ root lock, native-rank transition geometry, and fixed Top-1 transition. No policy is reselected on held-out labels.

Table 19: Held-out confirmation at the same 20% final parent/KV budget. "Exact" requires every canonical oracle parent; "Chain" requires one parent from every ordered evidence group.

Method	HotpotQA				QASPER		
	Entry	Oracle R	Exact	Chain	Entry	Oracle R	Chain
Current one-shot	.343	.205	.000	.143	.967	.883	.800
Current monotonic iterative	.343	.233	.000	.143	.967	.883	.800
New faceted root only	.457	.295	.029	.171	.967	.883	.800
Faceted root + propagation	.457	.295	.057	.171	.967	.883	.800

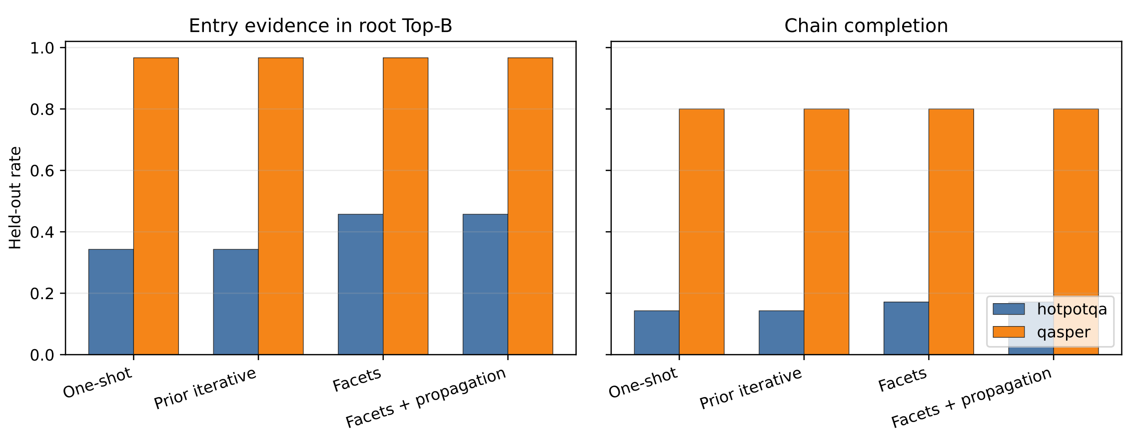

Figure 24: Entry discovery and downstream chain completion for the four confirmation methods. Facets improve Hotpot entry and chain metrics while preserving QASPER; propagation does not add chain completion beyond the new root at this budget.

The root-only facet search averages 68 comparisons on Hotpot and 82 on QASPER. Reconnecting the dual-geometry transition diagnostic raises these to about 112,232 and 1,154,036, respectively, while leaving chain completion unchanged from B. Exact-oracle completion does rise by 0.029 on Hotpot, but the incremental search is not justified as a default. The evidence supports recommendation *A: use multi-span query facets* as an opt-in research selector. Query-span dilution is the dominant tested entry failure; head dilution and span–head synergy are not supported. One-shot remains the SDK default, and learned slots, textual decomposition, new associative training, and partial materialization remain outside this gate.

A.7 Facet types, prompt support, and query-grounded propagation

The preceding experiment established that one fixed window family can improve entry discovery. It did not establish that width four is uniquely useful, that every active prompt token should issue a retrieval query, or that the same facets can choose a successor proposed by associative propagation. We address those questions additively: source memories and five learned routing projections remain frozen, every facet comes from one full contextual prompt pass, and all facets still compete for one final parent budget. Besides the exact global and width-4 controls, we test widths 2/4/8/16 at half-window stride, individual token states, their multiscale union, deterministic punctuation/relation spans, local-only width-4 facets, and one Top-2-mean reduction.

Table 20: Facet-family root entry at the 20% parent budget. Selection uses validation identities; held-out columns are descriptive and never feed selection. "Cmp." is mean held-out Hotpot facet–parent comparisons. All fixed-window rows include the global facet unless marked local.

Facet family	Val. Hotpot	Val. QASPER	Test Hotpot	Test QASPER	Cmp.
Global	.089	.820	.343	.967	5.9
$w = 2, s = 1$	**.222**	.780	.429	.967	134.3
$w = 4, s = 2$	.200	.820	.457	.967	67.6
$w = 8, s = 4$	.178	.820	.429	.967	34.7
$w = 16, s = 8$	.156	.800	.400	.967	19.1
$w = 4$, Top-2 mean	.200	.820	.457	1.000	67.6
$w = 4$, local only	.200	.780	.486	1.000	61.7
Token	.200	.820	.400	.967	140.1
Multiscale 2/4/8/16	**.222**	.780	.429	.967	238.1
Phrase/relation spans	**.222**	.800	.429	.967	**23.3**

The validation rule first preserves QASPER within 0.05, then maximizes Hotpot entry and MRR and breaks ties by false positives and cost. It freezes width 2, stride 1, max reduction, with the global facet retained. This is not the held-out maximum: local-only width 4 happens to reach 0.486 on the seven Hotpot test identities, but selecting it after inspection would invalidate the gate. The exact global and width-4 held-out controls reproduce 0.343 and 0.457, respectively. The result therefore supports fine contextual facets as a family, while the small identity count does not resolve a universal optimal width.

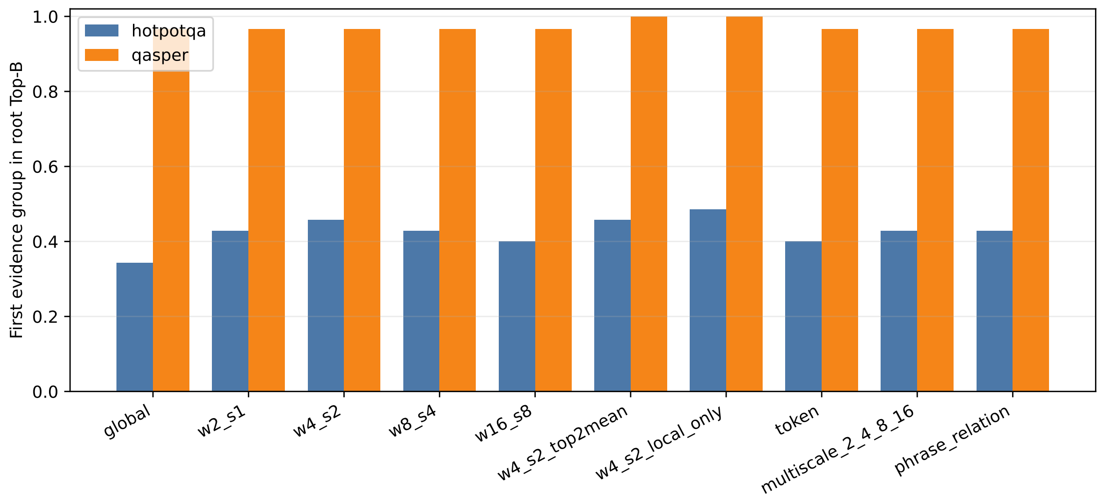

Figure 25: Held-out root entry for every predeclared facet family. Several contextual families preserve QASPER and improve on the global Hotpot control; more facets do not improve monotonically.

For long prompts, active context and query-support context are distinct. Earlier tokens may causally shape a late hidden state without independently nominating memory. We compare the question span, explicit latest-user-message boundary, recent tails of 16/32/64 tokens, and the full prompt. A controlled chat prompt contains an archived user request, a phrase copied from a non-oracle memory parent, tool/history text, and the current question; the stale phrase is varied over 0/16/32/64 target tokens. The validation criterion freezes the latest message. On clean held-out prompts it gives Hotpot/QASPER entry 0.429/0.967 with 134/151 comparisons, versus 0.257/0.933 and 246/462 for full-prompt support. The contamination result is more cautious: over nonzero stale lengths, latest-message stale-parent selection is 0.229 on held-out Hotpot and 0.278 on QASPER, compared with 0.219 and 0.289 for full-prompt support. The boundary reduces search and preserves current-question retrieval; it does not uniformly eliminate stale matches.

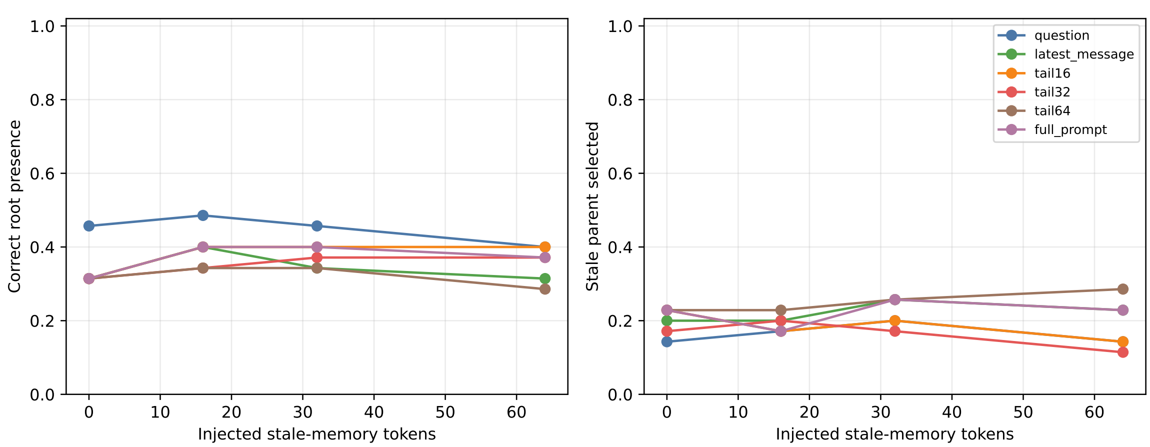

Figure 26: Held-out Hotpot response to injected stale-memory text. Left: current root entry. Right: selection of the explicitly non-oracle stale parent. The validation-frozen latest-message boundary is bounded and cheaper, but stale-context resistance remains mixed on this sample.

We next condition on a correct first evidence group A to isolate successor selection. Existing layer-27 pre-RoPE native QK with Top-4 token/head reduction generates

$$C_A = \mathrm{Top4}_{B \notin A} R_A(A, B). \tag{7}$$

Only those four identities are validated in one batched product by the frozen facets,

$$G_Q(B) = \max_i R_Q(q_i, B), \qquad B \in C_A. \tag{8}$$

We compare query reranking, rank conjunction $r_A(B) + \lambda r_Q(B)$ for $\lambda \in \{.25, .5, 1, 2\}$, and a validation-fitted query threshold. All versus residual facets (excluding the facet that best matches A) are separate controls. Native and semantic raw scores are never added. Oracle successor labels enter only validation selection and metrics, never candidate generation, facet scoring, or ranking.

Table 21: Conditional successor retrieval from a known-correct A. Validation selects all-facet query reranking. There are ten validation and seven held-out transitions, each evaluated with five frozen semantic projections; seed repetitions do not create additional transition identities.

Split	Ordering	R@1	MRR	R@2	R@4	Mean rank
Validation	Association only	.100	.355	.200	.600	3.60
Validation	Query rerank	**.300**	**.508**	**.480**	.600	**3.04**
Held out	Association only	**.286**	**.583**	**.714**	1.000	**2.14**
Held out	Query rerank	.257	.500	.400	1.000	2.69

The validation gain does not transfer. Held-out R@1 changes by −0.029, MRR by −0.083, and R@2 by −0.314, while all true successors remain available within the associative Top-4. Mean held-out query margin between the best oracle candidate and best distractor is −0.113. Residual-facet grounding and rank/threshold conjunctions do not win validation under the R@4 preservation constraint. Exact synthetic controls still behave as designed: query grounding promotes a weaker but relevant proposal, stops an irrelevant continuation, and retains a second-facet bridge. The failure is therefore empirical discrimination on these natural edges, not a missing implementation path.

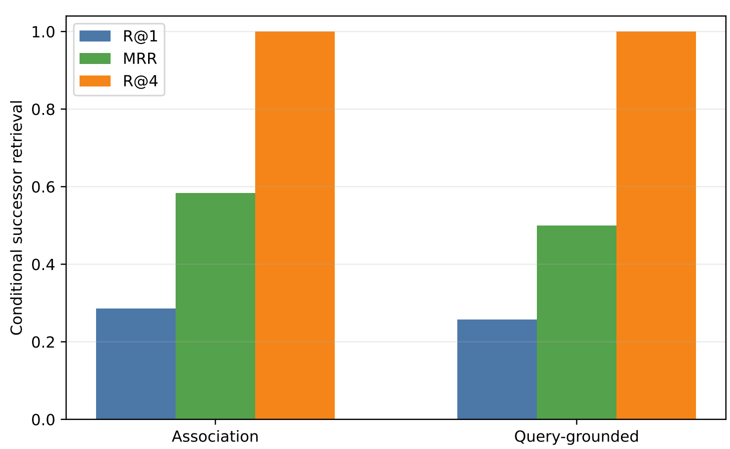

Figure 27: Held-out conditional successor retrieval. Static current-query facets preserve Top-4 availability but worsen the order inside that bounded candidate set.

The exact native proposal averages 6.52 million token/head dot products and about 0.0350 seconds per held-out edge on the reference GPU. Root-facet identification adds 24 comparisons; packed four-candidate validation adds 86 comparisons and about 0.00146 seconds. No K/V is materialized by this diagnostic. Our predeclared success rule required at least +0.10 held-out R@1 with no more than 0.05 R@4 loss. It fails, so we do not run or report a same-budget end-to-end grounded closure. Association proposes useful candidates, but static current state does not yet ground their order reliably; learned or dynamically updated state remains future work rather than a paper result.

A.8 Frozen dynamic query reconstruction

We next test the smallest dynamic-state mechanism without training. Given oracle-conditioned correct A, we decode only its source-group text, combine it with the original question, and run one ordinary frozen transformer pass,

$$H_1 = \text{Transformer}(Q_0 \| A), \qquad Q_1 = \text{Facets}(H_1). \tag{9}$$

No answer is generated. We compare $Q_0\|A$, $Q_0\|$`[Retrieved memory]`$\|A$, and the order control $A\|Q_0$. Width-2, stride-1 facets use either question positions only or question plus A positions; excluded positions can contextualize allowed states but cannot independently nominate a parent. The source and target evidence groups are used to construct the condition and metric, respectively. Target identities or text never enter reconstruction, native candidate generation, facet scoring, or ranking.

Candidate generation remains exactly the preceding layer-27 pre-RoPE native-QK Top-4 token/head mean. We scan $K \in \{1, 2, 3, 4, 5, 6, 8, 11\}$ without changing those scores, then evaluate each candidate in one packed semantic facet product. Exact assertions reproduce the previous static Q0 R@1/MRR and native K=4 candidate recall on both partitions. The predeclared validation rule maximizes K=4 R@1, then MRR, and breaks ties by re-encoding latency and facet count. It selects $A\|Q_0$ with question-only support. This row ties static Q0 on validation but does not transfer: held-out R@1 changes from 0.257 to 0.229 and MRR from 0.500 to 0.479.

Table 22: Frozen dynamic-state variants at native candidate breadth K=4. Selection uses only the validation columns. "Q+A" support permits both regions to issue facets. The final column is the held-out fraction of target-winning facets whose explicit span lies in A; it is descriptive.

Reconstruction	Support	Val. R@1	Val. MRR	Test R@1	Test MRR	A facet
Static Q_0	question	.300	.508	.257	.500	—
$Q_0\|A$	question	.280	.490	.257	.502	.000
$Q_0\|A$	Q+A	.300	.508	**.314**	**.538**	.971
$Q_0\|$`[memory]`$\|A$	question	.260	.477	.257	.502	.000
$Q_0\|$`[memory]`$\|A$	Q+A	.280	.498	.286	.533	.971
$A\|Q_0$	question	**.300**	**.508**	.229	.479	.000
$A\|Q_0$	Q+A	.220	.467	.286	.517	.971

The primary $Q_0\|A$ Q+A control is a useful but non-decisive hint. Its held-out R@1 is +0.057 over static query reranking and +0.029 over association-only ordering; validation ties static, and the held-out gain is below the predeclared +0.10 materiality threshold. In 102 of 105 held-out seed–edge rows across the three Q+A-support orderings, the target's winning facet is an A span. Thus the null gate is not explained by failing to expose A: max-over-facets usually chooses A-derived states, but those states do not discriminate the successor reliably enough.

Table 23: Held-out K ladder for the validation-selected dynamic state. Candidate recall is seed-invariant; R@1 and MRR average five frozen projections over seven transitions. Native search is exhaustive in this diagnostic, so changing K alters only the bounded semantic product.

K	Candidate recall	Static R@1	Dynamic R@1	Dynamic MRR
1	.286	.286	.286	.643
2	.714	.229	.229	.567
3	.857	.257	.229	.507
4	1.000	.257	.229	.479
5	1.000	.200	.171	.436
6	1.000	.200	.171	.431
8	1.000	.200	.171	.431
11	1.000	.200	.171	.431

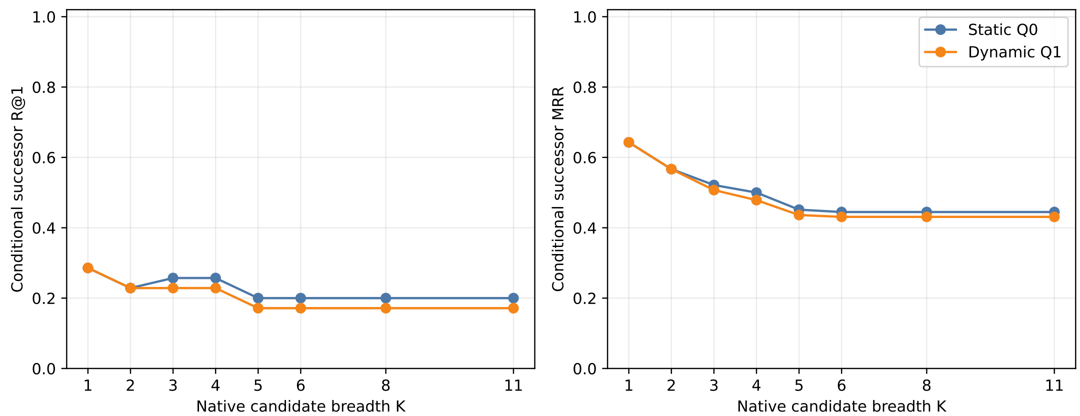

Figure 28: Held-out conditional successor order across native candidate breadth. Reconstructing the validation-selected Q1 does not improve static Q0; once K admits more distractors, both semantic rerankers expose the unresolved discrimination problem.

The selected held-out reconstruction takes 0.723 seconds per edge; native search, projection, and packed scoring bring prefill-to-routing-ready time to 0.765 seconds (p95 1.288 seconds) on the GTX 950M. The frozen model occupies 1.192 GB allocated before capture, and reconstruction peaks at 1.307 GB allocated, an incremental 115.0 MB. Mean per-edge routing structures comprise an 8.82 MB native-search cache, 24.0 KB parent-gist cache, 3.0 KB projected-parent cache, 90.7 KB query facets, and 345 bytes of K=4 candidate scores; measured host-to-device payload is 8.94 MB. The 621 MB source-feature file remains CPU-resident. This routing-only diagnostic conceptually conditions on 2.14 active parents on average but materializes zero native K/V tokens, so its active-KV fraction is zero by construction, not an estimate of later serving cost. It performs no generation; TTFT, TPOT, throughput, concurrency, and dollar cost are therefore not claimed.

The gate required at least +0.10 held-out R@1 and positive MRR. It fails. Following the pre-declared stop rule, we do not run the larger facet-resolution, candidate-width, active-ceiling, or relevance-threshold surface, nor B+threshold dynamic closure. Those deliverables were conditional on evidence that explicit frozen reconstruction repaired successor choice. The next mechanistic alternatives are a learned current-state update, learned query decomposition, or an explicit relational-composition objective; each must retain this candidate, leakage, partition, and memory-accounting protocol.

QASPER does not supply a comparable conditional edge set here: the selected yes/no subset is predominantly one evidence group, so root retrieval is terminal under the available annotation. Its held-out root entry remains 0.967, but this is not evidence for a learned stopping decision. The failed Hotpot conditional gate also means grounded chain completion at a fixed final memory budget is unmeasured rather than zero; we deliberately do not infer it from oracle-conditioned edge rank.

A.9 Semantic graph search with terminal query closure

The preceding rerankers asked the current query to judge every successor. That is too strong for an associative chain: a useful bridge may match its predecessor while remaining weakly related to the original question. We therefore use the query only at the boundaries,

$$q_i \to A \to B \to \cdots \to Z \to q_j. \tag{10}$$

Paper-2 routing supplies one or more entry roots. For a frontier parent X, native layer-27 pre-RoPE local QK scores propose $N_K(X)$. A candidate Y is admitted solely when

$$R_{\text{edge}}(X, Y) \geq \theta_{\text{edge}}, \tag{11}$$

with no query reranking or query threshold on intermediate nodes. Immediately after admission, and only then, the terminal predicate tests

$$G_Q(Y) = \max_{j \leq F} R_{\text{query}}(q_j, Y) \geq \theta_{\text{goal}}. \tag{12}$$

The two thresholds have different jobs: θ_{edge} limits graph branching, whereas θ_{goal} decides that the path has returned to the task. At least one memory edge is required, and the primary search stops on its first terminal. An ablation excludes the root's entry facet from the terminal maximum; it is not the primary condition.

Table 24: The classical-search interpretation. The learned topology is implicit and graded; oracle evidence labels are used only after search for evaluation.

Classical state-space search	PRA semantic graph search
Initial state	Paper-2 routed root A
Successor function	native Top-K local-QK association
Edge predicate	θ_{edge}
Goal predicate	θ_{goal} against any query facet
Frontier / visited set	active candidates / deduplicated parent IDs
Resource bound	parent ceiling B, beam, hop and expansion caps
Path depth	number of memory-to-memory edges

We first grant the first annotated evidence group as the root, keeping entry error out of the traversal test. Five frozen semantic projections provide goal scores; native edges are seed-invariant. The scan covers $K \in \{1, 2, 3, 4, 5, 6, 8, 11\}$, $B \in \{2, 4, 6, 8, 16, \text{None}\}$, depths 1–4, breadth-first, best-first, and beam search, and independent validation quantiles for both thresholds. This diagnostic constructs a dense parent adjacency once from packed local-QK products to make the search surface reproducible. The serving implementation may score successors lazily; no native K/V is materialized here.

The exact conditional candidate curve is reproduced: true successors occur within Top-K in 3/17, 7/17, 11/17, 13/17, 15/17, 16/17, 16/17, and 17/17 transitions, respectively. At $K = 11$, validation $\theta_{\text{edge}} = 12.308$ retains all ten measured successor transitions while changing mean admitted branching only from 5.7 to 5.6. The next quantile, 12.480, lowers branching to 3.9 but successor recall to 0.7. End-to-end validation therefore selects an open edge threshold: the absolute score is not sufficiently calibrated to prune safely.

Terminal calibration is the limiting result. Table 25 reports best-first validation; breadth-first and beam have the same poor closure/false-goal frontier and require more expansions at the closed endpoint. No finite threshold satisfies the predeclared false-goal ceiling of 0.25. The selected policy is consequently "closed," which means no terminal can fire. This is not a successful stopping rule; it is the conservative endpoint of a failed calibration gate.

Table 25: Validation terminal-goal frontier. Correct closure is measured on HotpotQA; false goals pool non-oracle terminals from both datasets. Precision is correct closures divided by all triggered closures.

θ_{goal}	Correct closure	False-goal rate	Precision	Nodes expanded
q50	.244	.853	.120	1.20
q65	.222	.779	.119	1.82
q75	.156	.705	.095	2.49
q85	.111	.589	.082	3.65
q90	.111	.484	.098	4.51
q95	.022	.389	.026	5.37
closed	.000	.000	.000	9.37

With the goal predicate held closed, graph recovery itself is strong. Validation selects best-first $K = 4$, $B = 6$, and depth 4. On seven held-out Hotpot identities, repeated over five projections, it reaches 1.000 oracle recall, 1.000 later-evidence recall, and 1.000 complete-chain coverage. The corresponding six QASPER identities also reach 1.000 oracle recall. Hotpot later recall rises from .143 at $B = 2$, to .857 at $B = 4$, and 1.000 at $B = 6$; at $B = 6$ it rises from .714 at $K = 1$, through .857 at $K = 2$, to 1.000 at $K = 3$. At the current parent granularity, oracle-root HotpotQA collapses to a one-edge evidence-neighborhood problem: depth 1 recovers all held-out annotated evidence. Because HotpotQA is shallow and coarse parent chunks can contract several factual relations into one memory node, this does not test genuinely deeper traversal. Figure 29 preserves the complete held-out $K \times B$ and $B \times \theta_{\text{edge}}$ response surfaces rather than reporting only the winner.

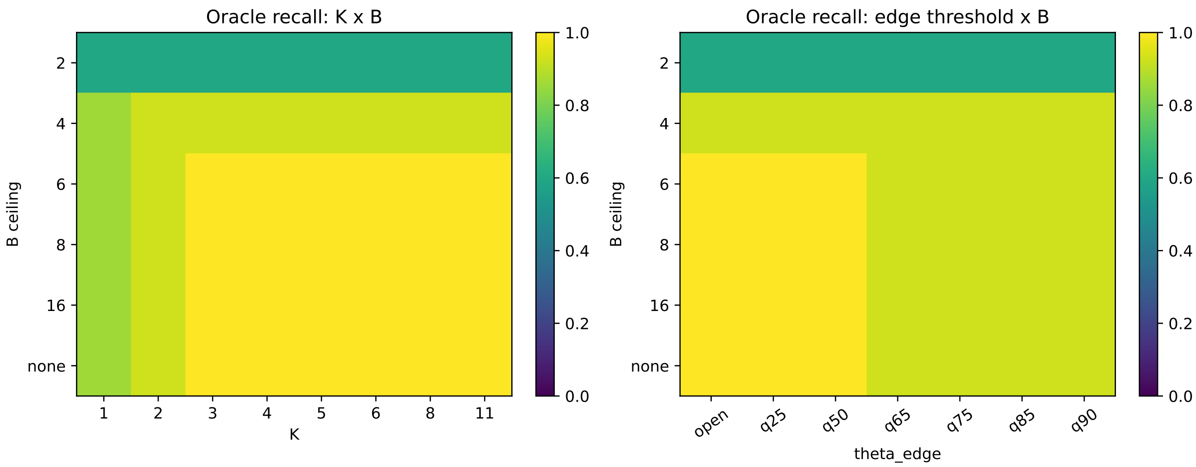

Figure 29: Held-out oracle-root evidence recovery. Candidate breadth and conceptual parent budget are distinct: search may inspect K successors while B caps the shared visited workspace.

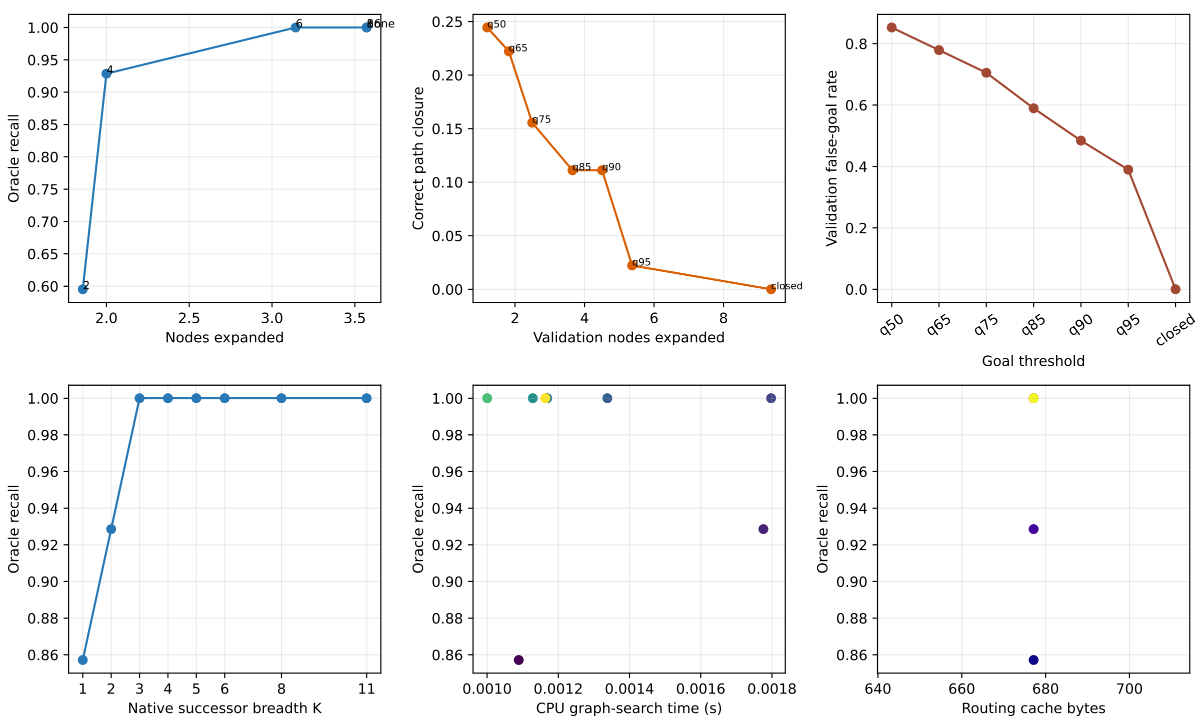

Figure 30: Quality/cost diagnostics. The top-center and top-right panels expose the central trade-off: finite terminal thresholds stop earlier but overwhelmingly stop on non-oracle parents. The bottom timing panel is CPU calibration search only; selected CUDA accounting is reported in text.

We manually inspect the 37 q95 validation false-trigger rows, covering 23 unique example–terminal pairs. Sixteen unique terminals (28 seed-weighted rows) are non-oracle but plausibly relevant to the question; seven (nine rows) are clear semantic or lexical distractors. Examples of the latter include passages about other songs containing “’Em” and unrelated Nashville entities.

Plausibly related passages still do not supply the labeled answer, so this distinction explains some high similarity without converting a false goal into success. The review file retains question, answer, decoded parent text, identity, and classification.

Selected-condition CUDA search averages 13.44 ms on HotpotQA and 12.70 ms on QASPER, including 3.66 and 4.02 ms of terminal facet tests. Mean conceptual work is 5.43/6.00 visited parents, 3.14/2.83 expanded parents, and 12.57/11.33 raw proposals, respectively. The parent adjacency build dominates cold routing at 152 ms for Hotpot and 1.183 s for longer QASPER sources; this dense diagnostic index is not a serving-speed claim. Compact adjacency plus goal caches average 677 and 1,789 bytes, candidate tensors 103 and 145 bytes, and preparation peaks at 86.6 and 218.8 MB CUDA allocated. H2D payloads average 8.94 and 25.31 MB. Materialized parents, native K/V tokens, active-KV fraction, generation, TTFT, and TPOT remain zero or unmeasured by design.

The graph-expansion component therefore passes its oracle-root recovery check at a reasonable K/B, but the terminal-goal component fails: useful closure cannot be separated from false closure under the frozen max-over-facets score. The held-out q95 false-goal rate is still .292; the safe closed policy has zero correct and zero false closures despite 1.000 later-evidence recall. Under the predeclared stop rule we do not restore routed roots or report an $R \times K$ end-to-end result. The next controlled intervention should learn or calibrate evidence completeness at the terminal, while retaining native-only intermediate traversal; more breadth does not address this failure.

A.10 Chunk granularity changes both topology and discovery

The preceding result leaves two explanations confounded. Native association may have found a multi-step chain efficiently, or 256-token parents may have contracted the annotated evidence into a shallow neighborhood. We therefore keep the frozen Qwen revision, layer 27, 256-token contextual encoding blocks, local pre-RoPE QK score, best-first traversal, and open edge threshold unchanged, while repartitioning each Hotpot source into zero-overlap parents of 16, 32, 64, 128, or 256 tokens. The 16 examples contain 25,416 source tokens. Layer-27 token states are captured once and exactly mean-pooled within each new parent. Native edges retain the canonical 32-token atomic QK windows at parent sizes 32–256; the 16-token condition splits those stored tokenwise Q/K tensors exactly rather than re-encoding text.

Each ordered supporting span is remapped to every overlapping parent. The oracle root is the single parent with the largest token overlap with the first supporting span, with lower parent ID breaking ties. Search receives that root but no later oracle IDs. Evidence recall, collisions, and completion are attached only after execution. Terminal query stopping is disabled. The surface contains $5 \times 4 \times 5 \times 3 \times 16 = 4{,}800$ searches over $K \in \{1, 2, 4, 6\}$, $H \in \{0, 1, 2, 3, 4\}$, and $B \in \{6, 16, \text{None}\}$. The five frozen projection seeds are used only for the parallel facet–parent diagnostic because native graph edges are seed invariant.

Table 26 shows direct graph coarse-graining. Mean oracle-parent count falls from 7.25 at 16 tokens to 2.19 at 256. No root contains all annotated evidence, but the mean evidence-token fraction in the root nearly triples, and distinct evidence groups increasingly share a parent: the fraction of roots touching multiple groups rises from zero to .25. The 256-token oracle distribution is 13 examples with two parents and three with three; at 16 tokens the median is 6.5, the 90th percentile 10.5, and the maximum 12.

Table 26: Hotpot evidence topology under deterministic zero-overlap rechunking. Root evidence is the fraction of annotated evidence tokens inside the oracle root; "multi" means that the root touches more than one distinct supporting span.

Tokens	Parents	Oracle parents	Root evidence	Root multi	Later parents
16	99.75	7.25	.201	.000	4.00
32	50.13	4.69	.347	.000	2.56
64	25.31	3.44	.441	.125	1.81
128	12.81	2.44	.508	.188	1.31
256	6.50	2.19	.575	.250	1.13

Discovery does not remain invariant to that refinement. Table 27 holds $K = 4$ and $B = 6$ fixed. At 256 tokens, all-example oracle recall is .969 and 15/16 examples are complete; the seven-example held-out partition exactly reproduces the prior 1.000 oracle and later-evidence recall. At 128 tokens recall is .688 and five examples are complete. At 16 and 32 tokens no example is complete, and recall is only .194/.333. Most gains stop after two hops because the six-parent workspace fills; increasing the hop cap alone cannot add parents.

Table 27: Oracle-root native discovery at the frozen $K = 4, B = 6$ operating point. $H = 0$ contains only the granted root. Complete is measured at $H = 4$.

Tokens	Oracle parents	Root evidence	$H = 0$	$H = 1$	$H = 2$	Complete
16	7.25	.201	.148	.173	.194	.000
32	4.69	.347	.224	.291	.333	.000
64	3.44	.441	.309	.429	.494	.063
128	2.44	.508	.432	.625	.688	.313
256	2.19	.575	.469	.885	.969	.938

The stronger $K = 6, B =$ None diagnostic estimates minimum native-graph evidence-recovery depth without treating it as causal reasoning depth. Fifteen of sixteen 256-token examples are complete at $H = 1$ and one remains unrecovered by $H = 4$. At 128 tokens, five require $H = 1$, two require $H = 2$, and nine remain unrecovered. At 64 tokens the counts are one, three, and twelve; all 16/32-token examples remain unrecovered. Fine chunking therefore creates a harder observed graph, but the current frozen native geometry does not recover enough of it to establish deep natural traversal. The sample has 15 bridge questions and one comparison question; the bridge trend matches the aggregate, while the single comparison item cannot support a subgroup claim.

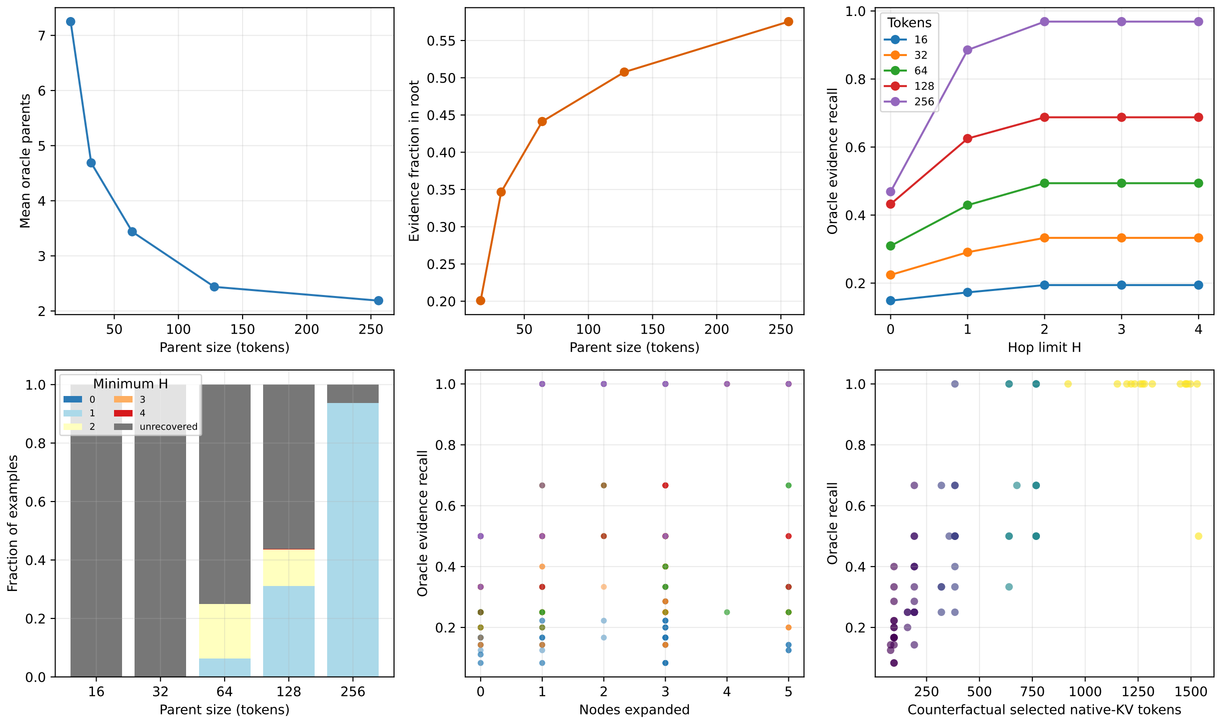

Figure 31: Chunk topology, recovery depth, and the discovery/materialization frontier. The stacked depth panel reports the minimum H under $K = 6, B =$ None and labels failures explicitly as unrecovered. The other quality panels use the frozen $K = 4, B = 6$ operating point. Native K/V is not actually materialized; bottom-right token counts are counterfactual payloads for the visited parents.

Candidate width helps only while the shared workspace can admit useful identities. At 256 tokens, $H = 4, B = 6$ recall rises from .719 at $K = 1$ to .865 at $K = 2$ and .969 at $K = 4$; $K = 6$ falls slightly to .938 because best-first competition under the same six-parent cap is not monotone in proposal width. At 16 tokens, the same ladder moves only from .173 to .194. More successors are thus not a substitute for either a larger active surface or better edge geometry.

The opposite trend appears in payload precision. The selected conceptual set contains 94 tokens at 16-token parents and 1,331 at 256, or .065 and .903 of the logical source. Mean evidence density falls from .211 to .065, while non-evidence selected tokens rise from 74 to 1,245. This is a counterfactual materialization frontier, not a serving measurement: the experiment transfers and materializes zero native K/V tokens. It nevertheless identifies the materialization tradeoff clearly: coarse memory makes discovery easier but activates a polluted payload; fine memory localizes evidence but demands stronger discovery.

Dense adjacency construction averages .326 seconds at 16 tokens and approximately .206 seconds at 32–256 on the GTX 950M. The token/head dot-product count remains 47.8 million because source content and atomic token comparisons are fixed; splitting 32-token windows into twice as many 16-token windows changes scheduling, not the pairwise token total. Parent adjacency storage, however, follows the expected quadratic surface, averaging 47,040 bytes at 16 tokens versus 198 bytes at 256. These are reproducible diagnostic costs, not an indexed-routing speed claim.

Finally, the query-facet matrix does not repair terminal discrimination. For each seed we measure maximum and Top-2 facet score, normalized entropy, winning facet, and incremental min–max path coverage for oracle roots, other oracle parents, discovered non-oracle parents, and manually reviewed 256-token false terminals. Oracle-vs-discovered AUC is .47–.54 across 16–128 tokens. At 256 tokens, incremental coverage gives AUC .253 against plausible false terminals and .342 against clear distractors when oracle evidence is the positive class. Facet entropy is near one for every class. Plausible false terminals often add different query vocabulary precisely because they are related; complementarity is not evidence completeness.

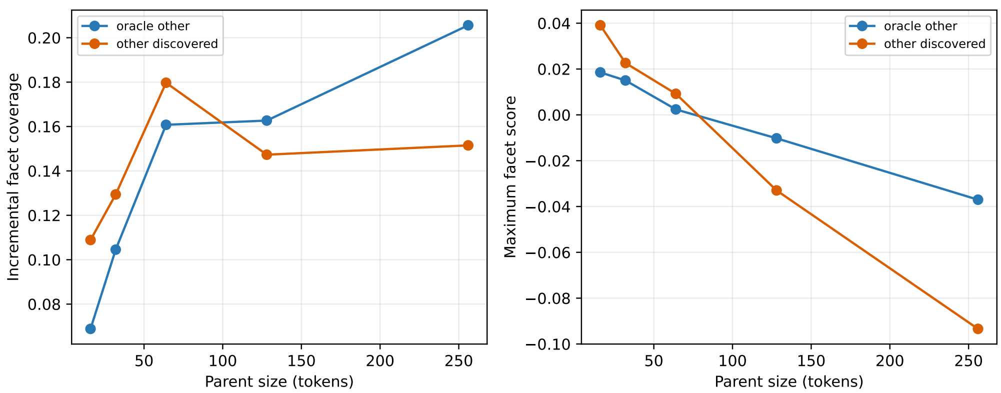

Figure 32: Five-seed facet diagnostics. Oracle evidence is not consistently more query-complementary than other discovered parents, and the direction changes with parent size. These statistics are reported for diagnosis and are not promoted to runtime filtering or stopping.

A.11 MuSiQue and 2Wiki separate task depth from path fidelity

Hotpot supporting spans do not provide a complete ordered reasoning graph. We therefore add two public natural controls without changing the search policy. MuSiQue-Ans v1.0 supplies composed 2/3/4-step questions, a supporting paragraph for each decomposition step, and explicit `#n` dependencies [20]. 2WikiMultiHopQA's April 2021 segmentation-fixed release supplies sentence-indexed supporting facts, relation triples, and entity-ID triples where available [6]. The official test labels are absent in both releases, so we partition official development identities by a fixed SHA-256 rule before model execution. The frozen cohort has 42 validation and 42 held-out examples: 36 MuSiQue examples balanced over $D = 2, 3, 4$ and chain/convergent topology where available, plus 48 2Wiki examples balanced over comparison, compositional, inference, and bridge-comparison types.

The adapter keeps raw annotations apart from PRA mappings. Across the full 2Wiki development split, a deterministic title/sentence rule maps 30,911 of 31,120 evidence triples (.9933); 165 are ambiguous and 44 unmatched. None of those 209 enters the cohort. MuSiQue graph edges come only from explicit `#n` references. For 2Wiki, exact object-to-subject entity-ID joins define edges when IDs exist; normalized lexical joins are retained as *derived*, not promoted to dataset truth. Comparison evidence remains an unordered set. After tokenization, each exact character span maps to every overlapping parent. One maximum-overlap parent represents each oracle root, while all overlaps remain oracle evaluation targets. A transition is preserved only when its target owns at least one parent not already covered by its source.

Hotpot showed strong granularity sensitivity, so 128 tokens is the primary compromise and 64/256 are controls, all with zero overlap. The frozen Qwen revision, independent 256-token encoding blocks, layer-27 local pre-RoPE QK Top-4 token/head reduction, open edge threshold, and best-first policy are unchanged. We scan $K \in \{1, 2, 4, 6\}$, $B \in \{6, 16, \text{None}\}$, and $H \in \{1, 2, 3, 4\}$. Terminal stopping is disabled, query facets are diagnostic/root-only, and labels are attached after search. Of 12 validation operating points at $H = 4$, lexicographic complete recovery/recall/cost selects $K = 6, B = 16$.

The mapping sanity check passes. At 128 tokens, mean MuSiQue oracle parents increase from 4.08 for $D = 2$ to 6.00 for $D = 3$ and 7.67 for $D = 4$, while total parent count stays near 22. Task depth is therefore not erased by chunking. The 48 selected 2Wiki examples contain 52 annotated transitions at this scale: 48 are preserved and four fully contracted; eight preserved transitions also have partial source/target overlap, which is recorded but never counted as a new target. No selected transition is unmappable.

Table 28: Held-out oracle-root recovery at 128-token parents. The first two rows expose the budget tradeoff; the selected broad condition is validation-frozen. Active is the counterfactual selected native-K/V token fraction; no K/V is materialized.

Dataset	K	B	Node recall	Complete	Visited/parents	Active
MuSiQue	4	6	.856	.556	6.00/20.67	.302
MuSiQue	6	6	.815	.444	6.00/20.67	.302
MuSiQue	6	16	.981	.944	10.39/20.67	.513
2Wiki	4	6	.958	.917	5.75/10.92	.567
2Wiki	6	6	.958	.917	6.00/10.92	.597
2Wiki	6	16	.979	.958	7.83/10.92	.770

Identity-bootstrap 95% intervals (2,000 fixed replicates) are [.944,1.000] for MuSiQue node recall and [.833,1.000] for completion over 18 held-out questions; 2Wiki intervals are [.938,1.000] and [.875,1.000] over 24. These strong values need the resource context in Table 28. $B = 16$ is half of MuSiQue's source and more than three quarters of 2Wiki's. Under $B = 6$, increasing K from four to six makes MuSiQue worse because extra high-scoring proposals compete for the same six identities. Breadth is not monotone under a hard global workspace.

Table 29: Held-out recovery versus search-hop cap at the selected $K = 6, B = 16$. MuSiQue improves from $H = 1$ to 2 and then saturates; 2Wiki is already saturated at one expansion.

	$H = 1$		$H = 2$		$H = 3$		$H = 4$	
Dataset	Recall	Complete	Recall	Complete	Recall	Complete	Recall	Complete
MuSiQue	.935	.833	.981	.944	.981	.944	.981	.944
2Wiki	.979	.958	.979	.958	.979	.958	.979	.958

MuSiQue depth strata do not show monotone degradation: held-out completion is 1.000, .833, and 1.000 for $D = 2, 3, 4$, with only six identities per depth. More importantly, the native search does not replay the annotated decomposition. Exact-hop frontier survival for later $D = 3/4$ steps falls to zero even when cumulative evidence recall is one, because many later annotations are reached one or two search rounds early. This is a dense evidence-neighborhood result, not a faithful causal traversal or evidence that four search rounds are useful.

2Wiki gives the sharper path test. Table 30 uses 25 preserved held-out transitions from 17 paths and excludes the one fully contracted transition. Direct transition recall rises smoothly; every transition is present by Top-8. Complete-path survival compounds local misses, reaching only .588 at $K = 4$ and .824 at $K = 6$. MRR is .410. Native paths of at most four steps recover .400/.840/.920/1.000 of transitions at $K = 1/4/6/8$; when reachable, mean path length is 1.80/1.14/1.04/1.00. Detours repair some missing direct edges, but path fidelity still requires broad local successor sets.

Table 30: 2Wiki held-out annotated-transition and complete-path survival at 128-token parents. MRR is computed from the full native rank and is therefore constant across cutoff rows.

K	Transition R@K	MRR	Complete path	Reachable $\leq H4$	Paths
1	.160	.410	.059	.400	17
2	.320	.410	.235	.560	17
3	.600	.410	.412	.720	17
4	.720	.410	.588	.840	17
5	.840	.410	.765	.840	17
6	.880	.410	.824	.920	17
8	1.000	.410	1.000	1.000	17
11	1.000	.410	1.000	1.000	17

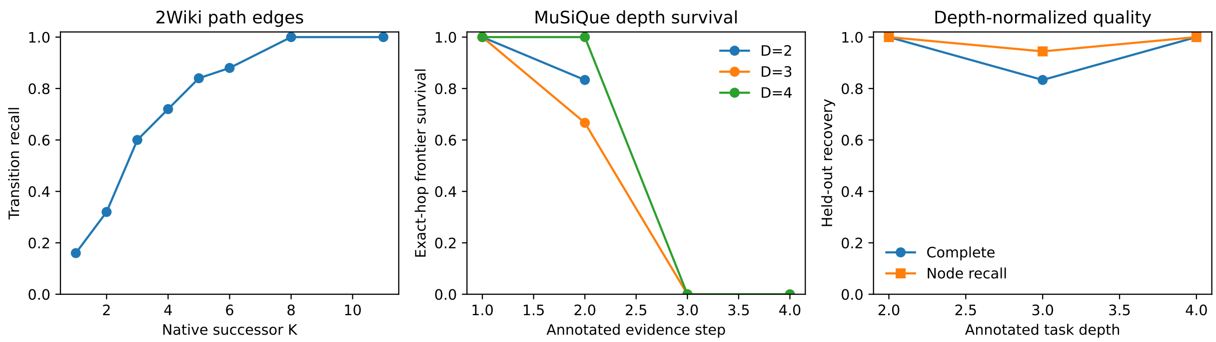

Figure 33: Natural graph diagnostics. Left: 2Wiki direct transition recall. Center: MuSiQue exact annotated-step frontier survival; late steps often arrive earlier than their annotation depth. Right: held-out MuSiQue completion and node recall by task depth.

Because both oracle gates pass, we restore only the previously frozen width-2 latest-message facets for root routing, with $R \in \{1, 2, 4\}$ and one shared $B = 16$. Five projections measure router sensitivity. MuSiQue root recall is .122/.311/.422 and 2Wiki is .250/.383/.608. At $R = 4$, conditional completion is .947/.890, while end-to-end completion is .800/.892. Completion can occur after a root miss because broad native search enters the evidence neighborhood through a non-annotated parent. That is useful robustness, but it cannot be presented as correct root routing. The root bottleneck remains visible even under broad traversal.

Table 31: Cross-dataset oracle-root context. Hotpot uses its fixed $K = 4, B = 6$ 128-token control; MuSiQue/2Wiki use the validation-selected $K = 6, B = 16$. QASPER is the earlier 256-token direct-retrieval control. Protocol differences are shown rather than normalized away.

Dataset	Natural role	Annotation	Parents	Recall	Complete	Visited	Search ms
QASPER	Direct control	mostly one group	256 tok.	1.000	1.000	6.00	12.70
Hotpot	Granularity	ordered spans	128 tok.	.688	.313	5.69	1.70
2Wiki	Path fidelity	relation graph	128 tok.	.979	.958	7.83	2.63
MuSiQue	Task depth	2/3/4-step DAG	128 tok.	.981	.944	10.39	4.61

The systems result is consistent with the earlier dense-adjacency warning. At 128 tokens, adjacency construction averages .517 seconds and 123.5 million native dots on MuSiQue, versus .126

seconds and 28.0 million on 2Wiki. Graph search itself averages only 4.61 and 2.63 ms. One frozen feature capture averages 5.69/2.48 seconds. Compact adjacency occupies only 1,954/469 bytes on average, but exact construction dominates routing latency; this remains a scientific reference implementation, not an indexed serving claim. Facet–evidence matrices are retained as diagnostics (mean held-out path coverage .657/.654 and incremental coverage .250/.282), but never filter intermediate nodes or terminate search.

A.12 Cross-dataset granularity separates missing edges from shallow search

The 128-token result is not the end of the granularity question. We therefore remap every MuSiQue and 2Wiki annotation at 16, 32, 64, 128, and 256 tokens, with zero overlap, and rerun $K \in \{1, 2, 4, 6, 8\}$, $H \in \{0, 1, 2, 3, 4\}$, and $B \in \{6, 16, \text{None}\}$. The source is still encoded in independent 256-token contextual blocks. Thus $G_{\text{encode}} = 256$ while G_{search} varies; the 16-token condition splits stored 32-token Q/K blocks exactly and never independently encodes a context-poor fragment. The earlier $K = 6, B = 16$ operating point remains frozen. $K = 8$ is a ceiling, not a retuned primary condition. Recomputed 128-token 2Wiki R@4/R@6/R@8 reproduces .720/.880/1.000 exactly; conditional preserved-path survival reproduces .588/.824/1.000. The one contracted held-out path is reported separately and counts as failure in the strict all-annotation rates .556/.778/.944.

Table 32: Held-out cross-dataset granularity at the frozen $K = 6, B = 16, H = 4$ condition. Edge is direct recall over preserved annotated transitions. Minimum H averages recovered examples. Source is the counterfactual selected native-K/V token fraction; no K/V is materialized.

Dataset	Tokens	Oracle parents	Root evidence	Edge R@6	Complete	Minimum H	Source	Density
MuSiQue	16	27.72	.066	.667	.500	1.11	.063	.258
	32	15.00	.129	.611	.500	1.11	.131	.255
	64	8.67	.230	.667	.667	1.17	.262	.225
	128	5.72	.314	.833	.944	1.12	.513	.184
	256	4.00	.397	.879	1.000	1.17	.946	.156
2Wiki	16	7.75	.293	.400	.583	.93	.172	.155
	32	5.13	.438	.400	.708	1.06	.302	.148
	64	3.79	.531	.560	.792	.79	.514	.106
	128	2.96	.614	.880	.958	.65	.770	.086
	256	2.33	.679	1.000	1.000	.63	.995	.076

Finer parents sharply improve localization: relative to 128 tokens, the 16-token condition cuts selected source by 8.2× on MuSiQue and 4.5× on 2Wiki, while increasing evidence density by 1.4× and 1.8×. It also changes the graph. MuSiQue oracle-parent count rises almost fivefold and 2Wiki preserved edge R@6 falls from .880 to .400. More search rounds do not repair this loss. Table 33 shows that $D = 4$ recall is unchanged from $H = 2$ through $H = 4$ even at 16/32 tokens. The earlier shallow saturation is therefore not only parent contraction: a native shortcut neighborhood remains. Yet that neighborhood omits enough fine nodes that complete recovery is only .333. The gate is jointly classified B/C/E: fine-edge representation collapses; $H = 2$ saturation survives; and fine nodes offer a useful payload frontier despite lower recovery. Required depth does not rise toward annotated depth.

Table 33: MuSiQue annotated depth by search granularity at $K = 6, B = 16$. Recall at $H = 2$ and $H = 4$ is identical in every fine 16/32-token stratum.

D	Tokens	Oracle parents	H1 recall	H2 recall	H4 recall	Complete	Source
2	16	17.67	1.000	1.000	1.000	1.000	.060
2	32	9.50	1.000	1.000	1.000	1.000	.133
2	64	5.67	1.000	1.000	1.000	1.000	.277
2	128	4.00	.917	1.000	1.000	1.000	.523
2	256	3.17	1.000	1.000	1.000	1.000	.984
3	16	24.67	.667	.667	.667	.167	.067
3	32	13.83	.667	.667	.667	.167	.132
3	64	8.00	.667	.722	.722	.333	.263
3	128	5.50	.889	.944	.944	.833	.505
3	256	3.83	.944	1.000	1.000	1.000	.941
4	16	40.83	.708	.792	.792	.333	.061
4	32	21.67	.708	.792	.792	.333	.128
4	64	12.33	.875	.917	.917	.667	.245
4	128	7.67	1.000	1.000	1.000	1.000	.512
4	256	5.00	.958	.958	1.000	1.000	.913

The edge/product decomposition reaches the same conclusion. At $K = 6$, independent preserved-edge products predict MuSiQue path survival .458/.391/.461/.700/.791 from fine to coarse and 2Wiki .304/.304/.457/.832/1.000. Observed complete graph recovery is never lower: it is .500/.500/.667/.944/1.000 and .583/.708/.792/.958/1.000. Search control therefore introduces no additional aggregate loss on this surface; contractions, alternate native paths, and multiple oracle roots can make observed recovery exceed the direct-edge product. The failing component is local fine-edge availability, not frontier competition after the edge exists.

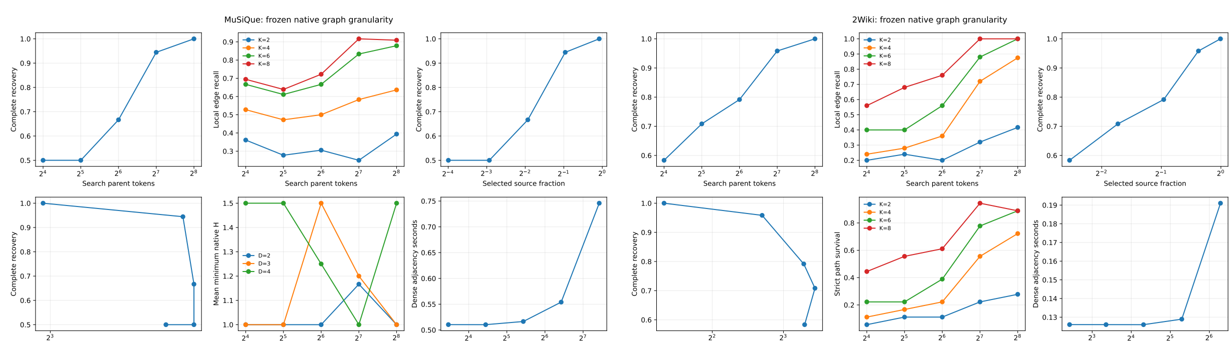

Figure 34: Frozen native graph granularity. Finer nodes reduce selected payload and increase evidence density, but direct edge and complete recovery deteriorate. Dense exact adjacency remains the cold-cost bottleneck; search itself is milliseconds.

Exact dense construction averages .746/.190 seconds per MuSiQue/2Wiki example at 16 tokens and .515/.126 seconds at 128. The underlying tokenwise work remains 123.5/28.0 million native dot products because search-parent grouping changes after contextual capture; splitting 32-token blocks at 16 raises local-pair bookkeeping and temporary memory. Search averages 5.5/4.0 ms at 16 and 4.9/3.2 ms at 128. These are reference-kernel measurements on the same GTX 950M, not serving claims.

A.13 Transformer depth reorganizes the native graph

The preceding experiments inherited layer 27. We now ask whether that choice is neutral. We freeze the same Qwen checkpoint and the complete 84-example split, fix $G_{\text{encode}} = 256$, $G_{\text{search}} = 128$, zero overlap, oracle roots, $K = 6$, $B = 16$, $H = 4$, and the exact pre-RoPE Top-4 token/head reduction, and vary only the zero-based decoder block $\ell \in \{0, 4, 8, 12, 16, 20, 24, 27\}$. Q/K is projected from each block's actual normalized attention input in one shared model pass; no earlier state is reconstructed from layer 27. The final layer reproduces 2Wiki R@4/R@6/R@8 of .720/.880/1.000 and conditional K=6 path survival .824 exactly.

Table 34: Held-out layerwise graph at fixed 128-token search granularity. C_{32} is normalized displacement between the native graph-input state under full and 32-token causal context. $\|\alpha\|/\|h\|$ is the local attention-branch ratio, not contextual information itself. Path is conditional on preserved 2Wiki transitions; D4 depth averages completed MuSiQue cases.

Layer	C_{32}	$\|\alpha\|/\|h\|$	R@4	R@6	R@8	MRR	Path K6	D4 complete	D4 depth
0	.000	6.644	.720	.880	.960	.520	.824	.833	1.80
4	.410	.274	.720	.880	.960	.390	.824	.667	1.50
8	.686	.269	.680	.880	.960	.432	.824	1.000	1.83
12	.816	.213	.760	.920	.960	.438	.882	1.000	1.50
16	.842	.274	.800	.880	.960	.397	.824	1.000	1.67
20	.782	.262	.760	.920	1.000	.471	.882	.833	2.00
24	.996	.178	.800	.880	1.000	.608	.824	1.000	1.67
27	.987	.127	.720	.880	1.000	.410	.824	1.000	1.00

The result rejects both simple hypotheses. Context dependence does grow: the 32-token intervention displacement reaches .410 by layer 4 and approximately one by layers 24–27. Mean native distance between annotated evidence states contracts from 1.23 at layer 0 to 1.06 at layer 27, and two-hop effective branching falls from 3.43 to 2.79. Yet edge quality is not monotonic. R@6 peaks in the middle at .920, MRR peaks later at layer 24, and layer 27 returns to .880/.410. The final matched-example synthesis in Section B replaces the exploratory all-cohort correlation: intervention dependence has uncertain Pearson correlations .470 with R@6 and −.013 with depth contraction, while its stable association is with lower effective branching (−.785). These are descriptive mechanistic relationships over eight layers, not significance claims.

The local branch proxy tells a different story. $\|\alpha_\ell\|/\|h_\ell^{pre}\|$ drops from 6.64 at layer 0 to .127 at layer 27, while intervention dependence rises. Layer 0's ratio is large because the embedding residual has a different scale; its graph input cannot yet depend on prior tokens, so its intervention score is exactly zero. Post-attention displacement numerically equals the branch ratio at every layer, as the residual identity requires. Attention rotation falls from .903 to .007. Thus local update magnitude is a poor proxy for accumulated context. The position-preserving intervention, which changes accessible K/V while retaining token IDs and absolute logical positions, is the stronger measurement.

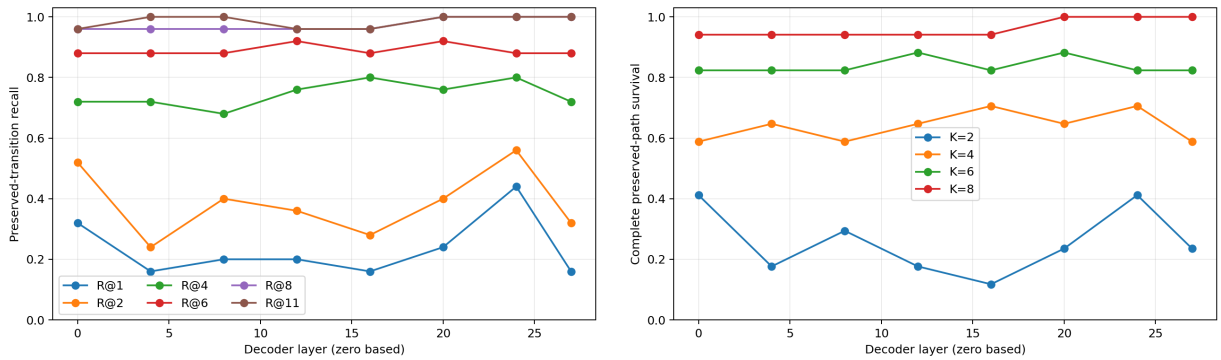

Figure 35: Layerwise local fidelity at fixed 128-token parents. Rank structure changes materially, but neither direct edges nor complete paths improve monotonically with representation depth.

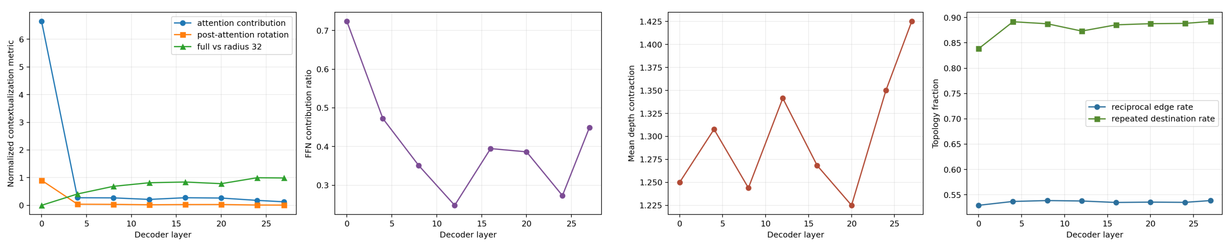

Figure 36: Measured contextualization and topology. Accumulated dependence on context grows while local attention contribution generally shrinks. Depth contraction and neighbor reuse vary, but the graph does not become uniformly more faithful.

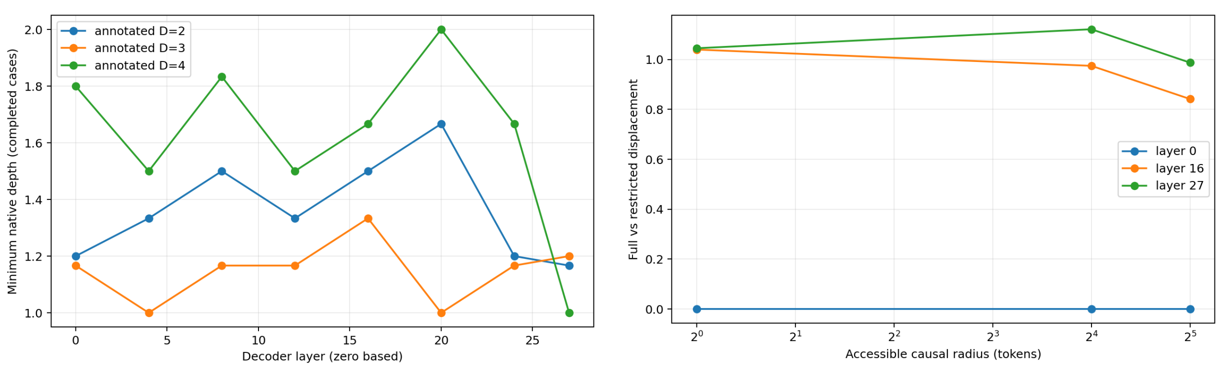

Figure 37: Left: completed MuSiQue cases remain much shallower in the native graph than their annotated task depth at every layer. Right: full-versus-restricted displacement for structural early/middle/late representatives; a wider causal radius approaches the full state, except that the late curve remains strongly dependent on context beyond 32 tokens.

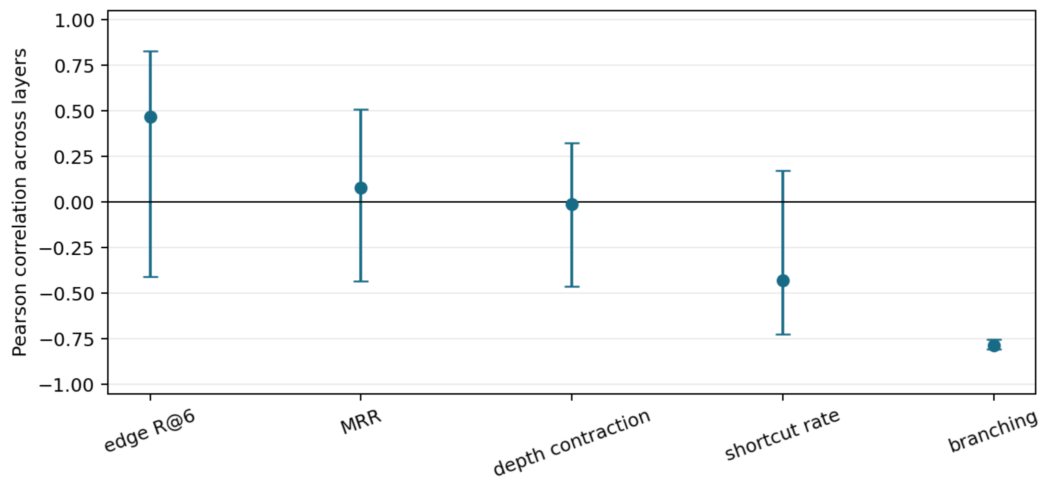

Figure 38: Matched-example layer correlations with identity-bootstrap 95% intervals. Context dependence does not stably predict edge fidelity, MRR, depth contraction, or shortcut rate; its negative relation to effective branching is the only narrow interval.

Because the one-dimensional effect is material, we run the predeclared small cross using a-priori structural representatives 0/12/27 and 32/128/256-token parents. This is not a best-layer search.

Table 35: Held-out layer–granularity interaction at fixed $K = 6, B = 16, H = 4$. Each cell is 2Wiki preserved-edge R@6 / MuSiQue D=4 complete recovery.

Tokens	Layer 0	Layer 12	Layer 27
32	.680 / .167	.480 / .333	.400 / .333
128	.880 / .833	.920 / 1.000	.880 / 1.000
256	.958 / 1.000	1.000 / .833	1.000 / 1.000

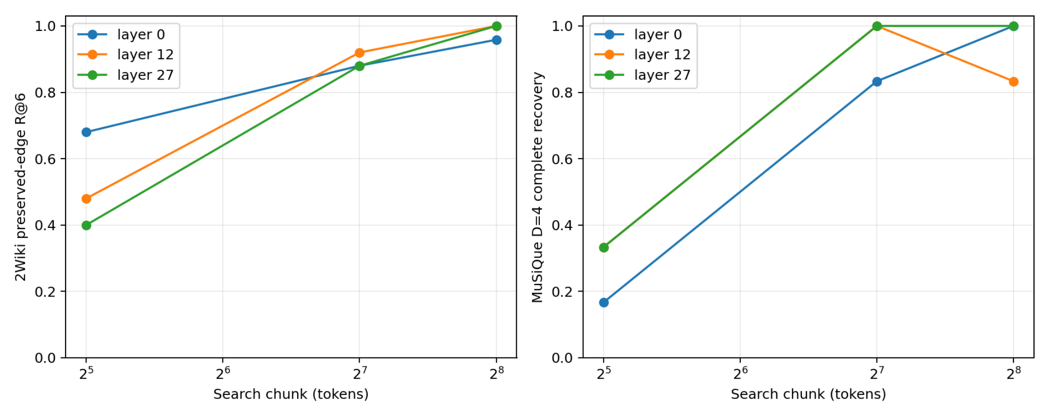

Figure 39: Depth and spatial granularity jointly determine the exposed graph. Early native geometry is strongest at fine 32-token 2Wiki edges; a middle layer leads at 128; coarse 256-token parents largely erase the layer difference through external contraction.

The interaction changes the recommendation. Late-layer topology remains a compatible and often adequate default, but it is not harmless at fine granularity and is not the universal optimum. Future PRA should expose an adaptive multiscale topology—for example early/local and middle/global graphs selected under one budget—rather than merely replacing layer 27 with one new fixed layer. We do not implement layer fusion or learned weighting here. At 256 tokens the near-perfect curve is also not a free win: it inherits the earlier result that coarse parents approach full-source activation. Representation-induced and chunk-induced contraction must be reported separately.

A.14 All-offset query facets reveal a selection ceiling

With graph granularity fixed again at 128 tokens, we ask whether weak routed roots mean that the question lacks a useful view or that the router chooses among views poorly. From one contextual query encoding, we mean-pool every valid stride-one span of width 1, 2, 4, 8, and 16, plus the existing global final state. For every target parent we compute the best rank over all facets *after* scoring. This is an oracle diagnostic: target identity chooses the reported facet but never enters a score, proposal, or graph-search function. We retain token offsets, decoded span, scale, target score, best distractor, and margin. A token-set Jaccard rank is an independent lexical control.

Table 36: Held-out query-visibility ceiling at 128-token parents. Oracle facet means the best existing span is selected with the target label after scoring. Practical R4 unions each facet's Top-1 proposals, deduplicates, normalizes within facet, and applies one shared four-root budget.

Dataset	Current R4	Oracle R1	R2	R4	R8	Terminal R4	Lexical R4	Practical R4
MuSiQue	.422	.711	.833	.878	.967	.711	.889	.411
2Wiki	.608	.592	.800	.950	1.000	.856	.875	.475

The ceiling is high. Existing query geometry contains a Top-4 view of an annotated root in .878/.950 of MuSiQue/2Wiki runs and of ordered terminal evidence in .711/.856. Width one wins 73%/66% of raw root-target rows; global states account for most remaining wins. Lexical overlap is competitive for roots but much weaker for terminals, especially on MuSiQue (.444 terminal R@4 versus .711 semantic), so the semantic result is not merely word matching.

The ceiling is not a router. Searching over many facets also raises a non-oracle parent into Top-4 in .546 MuSiQue and .672 2Wiki rows; even the best target facet has a negative mean margin over its strongest distractor. Validation therefore authorizes one bounded practical control, not unrestricted max aggregation. Each facet nominates Top-1, nominations are deduplicated, and one shared root budget is applied. At $R = 4$ it averages only 4.88/3.52 candidates and costs about .52/.43 ms, yet root recall is .411/.475, below the frozen .422/.608 router, while selected source rises to .587/.798. More facets reveal useful views but do not identify which view to trust. This is a facet-selection and calibration problem rather than evidence that W_q/W_m geometry contains no useful root or terminal signal.

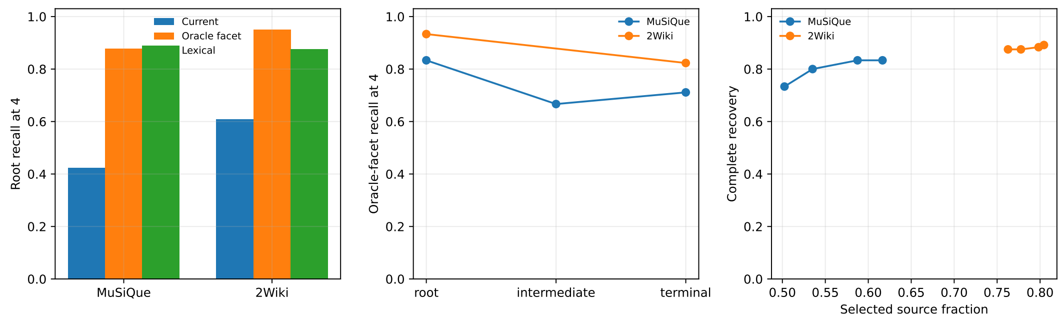

Figure 40: All-offset query audit. Left: current, oracle-facet, and lexical root R@4. Center: semantic visibility by evidence role. Right: the bounded executable union spends more source without realizing the oracle ceiling.

Taken together, the two gates define the remaining interface narrowly. For propagation, a future system should preserve contextual coarse discovery state while exposing fine payload identities; training search control around absent fine edges would address the wrong component. For entry, the oracle ceiling motivates bounded facet selection, but the final measurement gate below first tests whether inexpensive statistics already predict the winner. Learned selection is deferred to a separate study. At this chronological stage, physical native-K/V materialization and serving were still deferred; the later Gate-3 output study in the main text adds bounded materialization and generation but not a production serving benchmark.

Controlled chains separate the mechanism from Hotpot annotation structure. For task-edge depths $D \in \{1, 2, 3, 4, 8\}$, grouping one, two, or four adjacent chain nodes yields observed depth $\lceil (D+1)/g \rceil - 1$; all 15 search controls recover exactly at that depth. Chunking is therefore graph contraction, not evidence that the underlying reasoning became shorter.

B Final Measurement Gate and Experiment Freeze

The last gate adds no retrieval mechanism. It reads the frozen facet score matrices, graph rows, systems rows, and layer caches already reported above. Validation identities fit only two tiny diagnostics: the most-common winning scale and a ridge linear classifier of winning scale from question length, contextual hidden-state norms, and four lexical question-prefix flags. Target parent identity is absent from every selector signature. Parent-score margin and entropy inspect the unlabeled score distribution and are inexpensive runtime diagnostics, not query-only features. All target ranks are computed after facet selection on the example-disjoint held-out partition.

B.1 Oracle views are visible but not cheaply predictable

Width one supplies 73.3%/65.6% of held-out target-level root winners on MuSiQue/2Wiki; the global state supplies another 24.2%/30.6%. Winning spans are not confined to one position: 44.2%/55.6% are in the final third, but 37.5%/23.3% are in the first third. Table 37 evaluates each selected facet at the same example–seed level as the oracle ceiling. No inexpensive statistic approaches post-hoc target-aware selection. The best result is the unmodified global facet, .622 on MuSiQue and .692 on 2Wiki; the validation-fitted linear scale selector ties it only on 2Wiki and falls to .500 on MuSiQue. The result argues for a properly supervised bounded selector in later work, not activation of this tiny diagnostic.

Table 37: Held-out example-level root R@4. Oracle all-offset uses the target only after scoring. Every other single-facet row selects without target identity. The bounded union is the previously reported executable four-root competition, not a single-facet rank.

Entry diagnostic	MuSiQue	2Wiki
Deployed fixed facets	.422	.608
Bounded all-facet union	.411	.475
Global facet only	**.622**	**.692**
Earliest facet	.511	.667
Highest unlabeled parent margin	.556	.650
Lowest parent-score entropy	.500	.625
Validation-fitted linear scale	.500	**.692**
Oracle all-offset ceiling	.878	.950

The visibility boundary is role-dependent even though width one is the most frequent winner for every role. MuSiQue target-level oracle R@4 falls from .833 at roots to .667 at intermediate evidence and .711 at terminals; the lexical baseline falls from .792 to .417/.444. On 2Wiki, semantic root/terminal visibility is .933/.823 while lexical visibility is .806/.423. Query views therefore remain useful at the terminal boundary, especially semantically, but the decline through the path supports a typed interface $q_{entry} \rightarrow$ root $\rightarrow$ memory association $\rightarrow$ terminal $\rightarrow q_{terminal}$. It does not revive the failed terminal stopping rule.

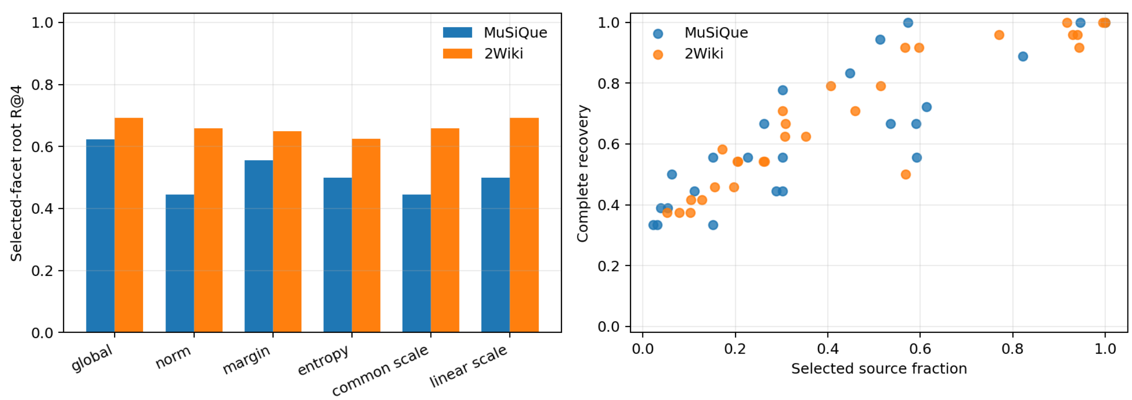

Figure 41: Left: held-out root R@4 for inexpensive selected-facet diagnostics. Right: the quality–selected-source Pareto points; high completion is available only at broad activation.

B.2 Context, layer, and chunk do not collapse to one control variable

The final correlation table repairs a subtle aggregation mismatch in the exploratory layer plot. For each graph metric, we now restrict contextualization to the same examples, aggregate by layer, and bootstrap example identities 1,000 times. Restricted-context displacement has Pearson correlation .470 with 2Wiki R@6 (95% bootstrap interval $[-.407, .828]$), .079 with MRR ($[-.432, .510]$), $-.013$ with MuSiQue depth contraction ($[-.460, .326]$), and $-.430$ with shortcut rate ($[-.725, .173]$). Only effective branching is stable: $r = -.785$ ($[-.807, -.754]$), although its Spearman estimate is a more modest $-.405$. With eight layer points these remain descriptive. They reject "more contextual means better graph" while supporting a narrower observation: accumulated context accompanies greater neighbor reuse.

The selected layer–chunk cross remains unchanged. At 32 tokens, early layer 0 gives 2Wiki R@6 .680 versus .480/.400 at layers 12/27. At 128, layer 12 leads .920 versus .880/.880. At 256, all reach .958–1.000 as coarse parents contract the graph. MuSiQue D=4 shows the same interaction without a monotonic layer law. At 128 tokens, completed-case $\Delta D = D - D^*_{native}$ is 2.2/2.5/3.0 at layers 0/12/27 with .833/1.000/1.000 completion. At 32 tokens, apparent contraction among survivors is not evidence of a better graph because completion is only .167/.333/.333. Layer-induced geometry and chunk coarse-graining are therefore separately reported and cannot be cleanly added.

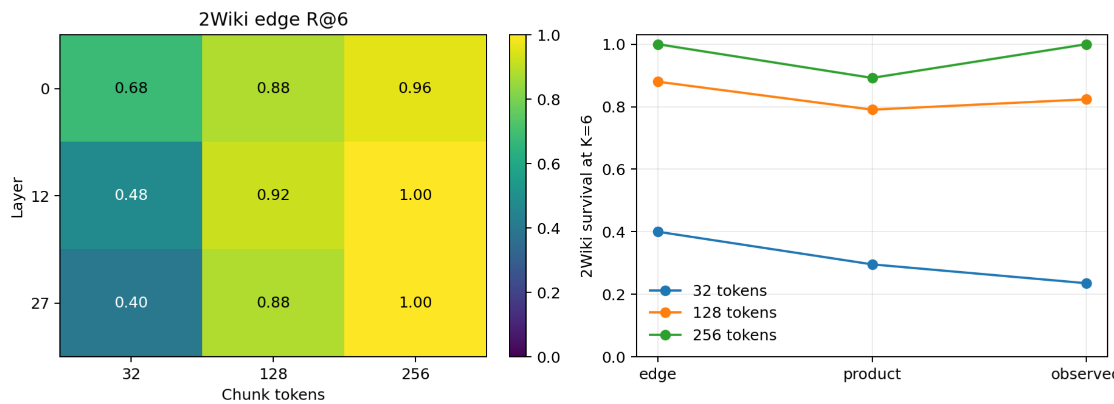

Figure 42: Left: the selected layer–granularity interaction. Right: 2Wiki K=6 edge recall, independent-edge product survival, and observed preserved-path survival by chunk size.

B.3 Missing local edges dominate the fine-graph failure

Table 38 replaces the earlier comparison between an edge product and whole-evidence recovery with a like-for-like preserved-path comparison. At K=6, missing local edges are the largest loss at 32, 64, and 128 tokens. Additional frontier/search competition is only .060/.029 at 32/64. At 128 and 256, observed survival exceeds the independent product by .033/.108, recorded as correlation or redundant-path gain rather than "negative search loss." The decomposition strengthens the conclusion that deeper search cannot repair an absent fine edge.

Table 38: Held-out 2Wiki edge–path decomposition at K=6. Extra is $\max(0, P_{product} - P_{observed})$; gain is the reverse positive part.

Tokens	Edge R@6	Product	Observed	Missing edge	Extra	Gain
32	.400	.296	.235	.600	.060	.000
64	.560	.441	.412	.440	.029	.000
128	.880	.791	.824	.120	.000	.033
256	1.000	.892	1.000	.000	.000	.108

B.4 Canonical quality–payload and cross-dataset view

The final Pareto artifact joins quality, selected source, evidence density, root breadth, K/B/H, expanded nodes, search time, adjacency time, and measured CUDA memory. Table 39 shows the frozen representatives. Search latency excludes adjacency construction. All rows in this chronological graph-only gate remain conceptual-routing measurements: materialized native-K/V tokens and bytes are exactly zero; "selected source" is the previously defined counterfactual whole-parent payload. The later Gate-3 output study materializes those identities and measures synchronized TTFT, TPOT, and total generation latency; neither stage measures concurrency or serving throughput.

Table 39: Held-out quality–payload representatives. C/B/H denote conservative, balanced, and high-recall labels. A combined label means the same non-dominated row serves both roles.

Dataset	Point	Tokens	K/B	Selected	Recall	Complete	Search ms
Hotpot	C	256	1/6	.552	.857	.714	1.05
Hotpot	B/H	256	4/6	.948	1.000	1.000	1.24
MuSiQue	C	16	6/16	.063	.819	.500	4.24
MuSiQue	B/H	128	8/16	.573	1.000	1.000	4.73
2Wiki	C	16	6/16	.172	.796	.583	4.51
2Wiki	B	128	4/6	.567	.958	.917	1.44
2Wiki	H	128	8/16	.918	1.000	1.000	3.06

Table 40: Canonical cross-dataset held-out summary. Dashes mean the all-offset oracle or an annotated transition graph was not measured; older fixed-window diagnostics are not relabeled.

Dataset	Role	Routed root R@4	Oracle facet R@4	Edge R@6	Complete	Selected
QASPER	direct control	.967	–	–	.800	.227
Hotpot	shallow/granularity	.971	–	–	.312	.501
2Wiki	annotated edge/path	.608	.950	.880	.958	.770
MuSiQue	depth/topology	.422	.878	.833	.944	.513

The cross-dataset table is intentionally not a leaderboard. QASPER is a small direct-retrieval control with no transition graph. Hotpot exposes granularity but its coarse completion spends half the source. 2Wiki provides the cleanest local-edge/path measurement, while MuSiQue provides the strongest depth and shortcut audit. The accompanying negative-results registry points to every closed gate: unreliable parent closure, harmful static reranking, failed dynamic Q/A reconstruction, uncalibrated terminal thresholds, false-terminal facet complementarity, inconsistent native head splitting, fine-edge collapse, weak contextualization–quality prediction, and the executable facet gap.

Freeze decision. The graph-only program stopped here. Its code and artifacts characterize the remaining gaps without fitting a large router, fusing layers, adapting K/B/thresholds, changing materialization, or adding another search architecture. The subsequent Gate-3 output study tests frozen bounded materialization; future work may test a supervised facet selector and adaptive early-local/middle-global topology, while optimized serving kernels, batched concurrency, and production-calibrated timing remain outside this paper.

C Extended Chronological Synthesis

C.1 What the experiments establish

The synthetic result and the integration tests establish three mechanical facts. First, a gist can serve as a new query and recover a node that the root cannot afford at matched B. Second, deduplication, beam pruning, and hard budgets prevent exponential materialization. Third, the closure can remain entirely in compact routing state until final identities map to full native K/V.

The natural-data results establish a different boundary. Projection-correct closure confirms that shared dimensionality did not imply shared query and memory semantics: respecting the asymmetric interface repairs much of QASPER's loss. The local experiment then finds positive bridge deltas and better exact identity/coverage, so parent aggregation does hide some useful semantic relations. Yet those stronger local edges do not improve chain completion. Direct iterative reuse of the inherited one-shot routing representation does not yet provide a dependable associative traversal graph on natural data.

Gate 3 then tests the transformer's own position-neutral local matching geometry after semantic narrowing. Its exact synthetic recovery proves that the routing path, grouped heads, and tokenwise reduction can represent a native bridge. The natural result is nevertheless negative under the predeclared success criterion: primary native QK does not beat local semantic closure or one-shot across matched budgets. The isolated HotpotQA Top-4 gain is offset by other budgets and by QASPER degradation. Initial relevance geometry and propagation geometry may differ, but pretrained layer-27 QK does not establish that specialization here.

This completes the staged diagnostic sequence: Stage 0 is one-shot retrieval; Stage 1 is chunk iterative closure; Stage 2 corrects the asymmetric projection interface; Stage 3 moves propagation to contextual local semantic gists; Stage 4 tests tokenwise native QK after semantic narrowing; Stage 5 makes root identity persistence monotonic and allocates the remaining budget through validation-selected bounded competition. Stage 6 preserves several contextual views of the question through root competition instead of forcing one relational summary before search. Stage 7 separates facet type and query-support scope, then asks whether static current-query facets can ground a bounded native successor set without mixing score families. Stage 8 gives the frozen transformer the known-correct first group and reconstructs the current state before repeating the same bounded ranking test. Stage 9 removes query filtering from intermediate traversal entirely, follows only native local association, and reserves current-query facets for a terminal goal predicate. Projection role, propagation scale, and native attention geometry have therefore been tested before introducing learned associative supervision. Oracle-conditioned rank now shows that most true transitions survive within Top-4, so a learned head is not yet the next controlled intervention. The displacement audit separates two failures that the aggregate metric had mixed: QASPER often starts from the right parent and can lose it to the fixed root/propagation slot split, whereas Hotpot usually fails to discover the first group. Stage 5 confirms the distinction: monotonic persistence reproduces QASPER one-shot quality, but bounded competition cannot propagate from a missing Hotpot entry. Stage 6 then recovers part of that missing-root gap with fixed, parameter-free contextual facets. Its success and the failure of native split-head composition localize the tested problem to semantic query-span dilution. Propagation still does not add chain completion beyond the improved root, so any later learned query decomposition or associative component remains new machinery rather than an automatic extension. Stage 7 sharpens that boundary: the native transition geometry places every held-out successor in Top-4, but static semantic facets worsen its position on average. Candidate availability and candidate ordering are therefore separate failures. Stage 8 shows that explicit contextual integration is not by itself enough: A-derived facets dominate the max score when permitted, but the validation-selected dynamic state still worsens held-out ordering. The unselected primary $Q\|A$ condition is mildly positive and motivates replication, not activation-policy tuning. A learned state update, explicit relational supervision, or another learned discriminator would be a new intervention and must beat the reported association-only and static-query Top-4 controls. Stage 9 changes that diagnosis in one important respect. Once intermediate memories no longer have to resemble the query, native traversal recovers every held-out oracle parent at $K = 4, B = 6$. The remaining frozen failure is terminal discrimination: max-over-facets identifies many related passages but cannot decide whether they complete the evidence chain. Thus propagation geometry and stopping geometry must be treated as separate model interfaces. The result supports native association as a candidate generator under oracle entry, but not executable iterative retrieval. Stage 10 resolves why that successful held-out traversal was only one edge deep. Parent granularity is a graph-construction parameter, not neutral preprocessing. Finer parents distribute the same annotations across more nodes and expose failures hidden by coarse contraction; coarser parents recover evidence more easily but include far more non-evidence tokens. Exact controlled chain contraction confirms the causal direction of this topology effect. The current native geometry is therefore adequate for a coarse evidence neighborhood, not yet for fine-grained deep discovery. Query-facet complementarity also fails as a completion discriminator, so neither larger breadth nor another frozen semantic threshold is the next justified intervention. Stage 11 is deliberately measurement-only. It shows that the winning query view is usually a single token or the global state, but position, norm, score margin, entropy, the validation-most- common scale, and a tiny validation-fitted linear scale model do not predict that view well enough to approach the oracle ceiling. It also aligns contextualization and graph correlations by example, corrects the edge/product comparison, and freezes conservative/balanced/high-recall payload points. This closes Paper 2.5 without converting a diagnostic into a production policy.

C.2 Relation to retrieval, agents, and recursive models

Memory Networks repeatedly read differentiable memory to support multi-hop tasks, while RAG, RETRO, and interleaved retrieval systems bring external evidence into generation [10, 3, 19]. Agentic systems can reformulate a text query after each model pass. Iterative PRA is narrower: it traverses already computed latent addresses inside one memory subsystem. It avoids repeated text serialization and full-model round trips, but it cannot replace an agent that must plan tools, verify evidence, or change the retrieval language.

Tiny Recursive Models repeatedly transform internal latent and answer state [9]. Controlled autoregressive work finds no reliable gain from the full autoregressive TRM architecture over simpler refinement baselines [13]. Iterative PRA does not apply a tiny neural network repeatedly. Its recurrence is *query* $\rightarrow$ *memory node* $\rightarrow$ *memory node*. This distinction motivates the parameter-free Level-1 test: retrieval recursion should earn its complexity before neural state recursion is introduced.

C.3 Materialization boundary

Papers 1–2 conservatively compress search but materialize every selected chunk in full. Iterative closure produces more structure before transfer: hop, parent edges, direct relevance, path score, confidence, branch entropy, and duplicate/overlap statistics. The versioned graph keeps these signals even when the final payload remains unchanged. This leaves a separate question: given a discovered closure, how much detail should each node receive? The natural-data chronology does not test partial materialization. Its granularity sweep makes that boundary quantitative. At a fixed six-parent conceptual budget, moving from 16 to 256-token parents raises evidence recall but increases the counterfactual selected payload fourteenfold and lowers evidence density more than threefold. A follow-up should therefore compare whole-parent, evidence-local, and progressively expanded materialization on the same discovered set rather than treating chunk size as a settled constant. The controlled study in this revision is self-contained and uses indivisible fact parents, making whole-parent and selected-chunk transport an exact identity control.

C.4 Controlled implementation map

Table 41 identifies the minimal implementation surface needed to reproduce the controlled experiment. Line numbers refer to the archived source revision shipped with this paper; function and field names are the stable anchors.

Table 41: Code-level map for the controlled LocalSA and evolved-state PRA experiment.

Source anchor	Contract
`src/pra_torch/masks.py`, line 8, `causal_attention_mask`	Returns a Boolean $[T, T]$ hidden-position mask. Finite W exposes exactly the current token and previous $W - 1$ positions; `None` is ordinary global causal attention.
`src/pra_torch/config.py`, lines 30 and 226	Declares and validates `self_attention_window`; line 512 normalizes exact `td_layered_pra` consumer IDs.
`src/pra_torch/model.py`, lines 353–366	Builds ordinary position-aware blocks except at explicitly requested PRA layers. Conversion copies layer normalization, Q/K/V/O, FFN, embeddings, and output head.
`src/pra_torch/model.py`, line 603, `forward_progressive_pra`	Runs blocks in decoder order, routes from each evolved final-token state, removes visited URI identities before the next PRA block, and returns logits plus a per-layer route/cost trace. Lines 610 and 666–685 restrict candidates to declared identities only for explicit oracle controls; ordinary routing leaves this argument unset.
`src/pra_torch/memory\_batching.py`, lines 229–351, `native_kv_attention`	Applies the one shared softmax over memory and causal native K/V. Lines 323–328 retain only the final query's $[H, M]$ memory weights when per-head metrics are enabled; the default path performs no per-position host copy.
`src/pra_torch/controlled\_local\_sa.py`, line 93	Generates fresh chains. Lines 152–174 force balanced terminal-relation decoys; lines 191–197 randomly interleave distractors while preserving reverse causal evidence order.
`experiments/paper2\_5\_iterative\_pra/run\_controlled\_local\_sa.py`, line 178	Trains all matched windows, selects checkpoints on validation only, evaluates a disjoint test split, and records tokens, approximate FLOPs, time, parameters, and convergence history.
Same file, line 378, `topology_rows`	Captures per-layer native Q/K edge ranks and contextualization metrics without PRA.
`experiments/paper2\_5\_iterative\_pra/run\_controlled\_pra.py`, lines 73 and 133	Encodes layer-native reference K/V once, reruns routing/materialization for every placement, and scores gold paths only after inference. Line 169 activates the no-replay progressive forward.
`experiments/paper2\_5\_iterative\_pra/run\_toy\_mechanistic.py`, lines 44–197	Defines the eight-label margin, exact evidence/distractor/native mass partition, payload-shuffle control, and unique matched oracle/wrong identity plans.
Same file, lines 199–381, `MechanisticCollector`	Hooks pre-residual, post-PRA, and post-block states; applies the frozen readout; records per-head attention, value norms, residual displacement, and local answer-direction alignment without feeding diagnostics back into inference.
`experiments/paper2\_5\_iterative\_pra/summarize\_outcome\_b.py`, lines 198–676	Joins one-shot and iterative rows by stable model/example identity, computes seed-level effects and example-level bootstrap diagnostics, constructs path strata and consumer profiles, and renders the Outcome-B figures. Lines 677–812 write the hypothesis and freeze audits only after metrics exist.

D Extended Limitations and Threats to Validity

The natural-data pool has 16 examples per dataset, repeated over five router seeds and then split by stable example identity for the final policy-selection gate. Seeds quantify projection sensitivity but do not create 80 independent questions. Hotpot evidence groups are source-contiguous spans, not annotated causal hop labels. QASPER's selected subset is yes/no and often one-group, so it tests direct retrieval more than multi-hop closure.

The protected-root result is an oracle-labeled upper bound, not an implementable policy. The calibration controls rerank only the fixed union of identities already exposed by frozen routing;

they cannot recover an unseen candidate and should not be read as a trained-router comparison. Their validation/test partitions are example-disjoint, but both remain small.

The adaptive-competition policy family is selected on nine HotpotQA and ten QASPER validation examples and reported on seven and six held-out examples, respectively; five router seeds measure projection sensitivity but do not enlarge those example counts. The validation objective weights Hotpot chain/recall, QASPER recall, and QASPER propagation cost explicitly. Different task weights could select another transparent rule, although the full selection audit is retained. The preservation winner degenerates to full Top-B locking on this sample and should be interpreted as a parity result, not evidence that $z > -0.5$ generalizes as a universal threshold.

The natural-data sample remains small. Projection correction, local hierarchy, native QK, and contextual query facets remove four specific confounds, but the experiment does not include adaptive zoom rules, evidence-edge training, a layer sweep, or a full agentic baseline. The local and native bridge controls are deliberately geometric and do not approximate natural language. Hotpot evidence groups are approximate source-order targets rather than annotated transition edges, so they can understate or misidentify the useful successor relation. At this stage, downstream F1/EM was measured only in a two-example execution smoke and was not used as evidence of quality; the later Gate-3 study supersedes that scope with a frozen 84-example generation cohort while preserving the smoke as a compatibility control. Finally, the scorer batches token pairs and heads but still launches small per-example routing operations; the reported wall time is not a production-kernel claim.

The facet selector is chosen on nine HotpotQA and ten QASPER validation identities and evaluated on only seven and six held-out identities. Five frozen projections expose sensitivity but are not independent questions. Max over more facets creates a multiple-comparisons effect even under one final budget; margins, false positives, precision, and Jaccard are reported, but a larger corpus is needed to estimate calibration. The native-head conditions necessarily change from the learned semantic representation to pre-RoPE QK because the semantic router has no head axis; they cannot identify a pure head effect relative to A/B. Finally, the 4-token winning windows can be grammatical fragments rather than explicit subquestions. The result demonstrates useful contextual localization, not interpretable decomposition or causal hop annotation.

The additive facet-type gate reuses the same small identity split, so its validation-selected 2-token width should not be interpreted as universal; the held-out width-4 local-only control is higher, illustrating the uncertainty directly. The stale-history prompt copies text from one non-oracle parent but is controlled rather than naturally distributed conversation. Latest-message support reduces comparisons and clean-prompt errors relative to full-prompt support, yet does not uniformly reduce stale-parent retrieval. The conditional grounding gate also grants oracle A and treats source-ordered evidence groups as approximate transitions. Its five projections expose router sensitivity but its seven held-out edges remain seven identities. Finally, exact native QK is exhaustive in this diagnostic; the millions of recorded dot products are not a production latency claim.

The dynamic gate inherits those seven held-out transitions and grants oracle A; it therefore tests successor discrimination, not an executable first-hop policy or end-to-end QA. Its validation tie-break selects $A\|Q$ question-only support, while the unselected $Q\|A$ Q+A row is better on held-out data. This disagreement is precisely why we enforce the frozen selection and why neither row establishes a general ordering. Re-encoding also decodes complete source-group chunks, not a minimal evidence sentence, and source-order evidence groups remain approximate causal hops. The measured cost is routing-only on one older GPU; no generated first token, concurrent request, or production native-KV transfer is present.

The semantic graph search also grants oracle roots and uses the same small identity split.

Its perfect held-out evidence recovery is therefore conditional, not end-to-end. A dense parent adjacency is precomputed for experimental control; production search would need a lazy or indexed successor implementation before the measured cold cost could be interpreted as serving latency. The manual false-terminal categories are qualitative judgments over 23 unique pairs, not new ground-truth labels. At the canonical 256-token parent granularity, the held-out annotations collapse to one-edge neighborhoods; the new fine-grained conditions require more oracle parents but are mostly unrecovered by $H = 4$. Thus this sample does not yet establish successful deeper natural traversal. Finally, the closed selected goal policy has perfect safety only by never stopping; routed roots are absent because the gate failed, not because routed search was evaluated and scored zero.

The chunk ladder retains only 16 Hotpot examples, of which 15 are bridge questions and one is a comparison question. Its bridge analysis is therefore primary and its comparison row descriptive. Supporting spans identify evidence locations, not a complete ordered causal graph; "minimum native-graph evidence-recovery depth" must not be read as reasoning depth. Zero overlap isolates granularity but does not test boundary-robust chunking. The 16-token condition also uses exact subdivision of the cached 32-token Q/K windows, while larger parents max-reduce those same atomic windows; this preserves native token scores but is not a learned multiscale index. Finally, selected K/V tokens are counterfactual because no payload is materialized. A subsequent natural benchmark should have more than two evidence hops, distributed document-level evidence, and an explicit or defensibly derived support graph.

The MuSiQue/2Wiki extension addresses annotation depth and path semantics but remains a diagnostic cohort: 18 and 24 held-out identities yield wide bootstrap intervals, and 2Wiki contributes only 25 preserved held-out transitions from 17 paths. The internal split is a hash partition of each official development set because public test labels are absent. Some 2Wiki edges without entity IDs use explicitly labelled normalized lexical joins. At 128 tokens, four selected transitions contract completely and eight partially overlap; none enters direct edge recall through an already-active source parent. The broad selected budget visits half of MuSiQue and three quarters of 2Wiki source tokens, so high completion is not a sparse-routing result. Finally, annotated task depth and native graph distance differ: $D = 4$ evidence is more distributed, but recovery saturates by $H = 2$. A larger held-out cohort, iterative-answer state, and stricter active-token budgets are required before claiming deep natural traversal.

The cross-dataset ladder reuses those 18/24 held-out identities across five deterministic granularities; rows are paired controls, not 210 new independent examples. Zero overlap isolates topology but does not test boundary averaging. The 256-token contextual encoding with 16/32-token search nodes is the requested context control, yet it is only one encoding scale and one layer. The independent-edge product assumes stepwise exchangeability and can be exceeded by contractions, alternate native paths, or several oracle roots; it is a diagnostic decomposition, not a causal probability model. Minimum H averages only recovered examples and must be read beside complete recovery. Counterfactual source fractions still assume whole-parent K/V materialization.

The all-offset audit performs many post-hoc comparisons. Its oracle ceiling explicitly uses the target to select a facet and is not executable. Five projections expose score sensitivity but do not multiply the number of questions. Width-one dominance may reflect entity tokens rather than compositional subquestions, and token Jaccard is only a lexical control rather than BM25 or a trained sparse retriever. The bounded union has no oracle channel, but its within-facet normalization and Top-1 nomination are one simple policy, not a learned-selector benchmark. Its failure closes unbounded facet aggregation, not the broader possibility of calibrated selection.

The final selector diagnostics fit only 42 validation identities and evaluate 42 held-out identities; five projections again measure sensitivity rather than independent sample size. The linear model predicts only scale and resolves offsets by hidden norm, so its failure is not a general impossibility result for learned selectors. Parent-margin and entropy diagnostics inspect all parent scores and are inexpensive but not query-only. Role visibility is target-level, while the headline oracle and selected-facet root ceilings are example-level; the paper labels both denominators. Layer correlations have only eight depth points. Their identity bootstrap measures cohort uncertainty but cannot make transformer depth an independent random sample. Finally, the Pareto frontier is conditional on oracle roots and counterfactual whole-parent payload. It records zero actual K/V materialization within that graph-only gate. Gate 3 later materializes frozen sets and measures synchronized per-example TTFT, TPOT, and total generation latency, but still supports no throughput, concurrency, or production-serving claim.

E Extended Experimental Synthesis

Bounded associative closure is exact when latent memory contains the right edges. Correcting the asymmetric frontier is necessary and materially reduces inherited degradation. Routing at a finer local scale can recover bridges erased by parent means. Tokenwise pre-RoPE native QK then recovers every designed native bridge, showing that semantic narrowing and a different propagation primitive can compose without changing materialization. None of these parameter-free *propagation* variants produces a dependable natural-data chain advantage at matched payload budget, and each finer search adds work. One-shot PRA therefore remains the default efficiency/quality reference even after the separate query-facet gain reported below.

The scientific lesson is narrower than "pretrained geometry has no associative edges." Direct iterative reuse is sensitive to projection interface, propagation scale, candidate width, and score reduction. Local semantic and native attention spaces contain designed bridges and occasional natural positives, but neither supplies reliable cross-parent transitions on this sample. The oracle-rank diagnosis narrows the next step further: useful transitions are commonly present below Top-1, while root evidence discovery remains weak. The final frozen-policy gate confirms both sides. Monotonic identity locks preserve direct QASPER evidence exactly, and adaptive breadth recovers controlled rank-2/rank-4 edges, but neither yields a held-out Hotpot chain advantage. In 52.9% of exploratory Hotpot runs the first evidence group is absent before locking begins. The next controlled question is therefore entry discovery through local query facets, not another transition geometry or a larger breadth sweep. Fixed contextual facets do admit more missing Hotpot roots: at 20%, entry presence reaches 0.457 from 0.343 and oracle recall reaches 0.295 from 0.205, while QASPER recall remains 0.883. The resulting Hotpot chain rate is 0.171 versus 0.143, but reconnecting bounded propagation adds no further chain gain and costs orders of magnitude more comparisons. Multi-span entry search is therefore the supported opt-in extension; one-shot remains the default. The broader facet ladder preserves that conclusion: validation freezes fine 2-token facets and latest-message support, but the held-out sample does not identify one universal width or uniformly reject stale memory matches. More importantly, static query reranking of native Top-4 successors changes conditional held-out R@1 from 0.286 to 0.257, despite perfect R@4 availability. A frozen dynamic pass over known-correct A then makes the intermediate memory visible but does not supply validation-stable composition: the selected $A\|Q$ row changes static-reranker held-out R@1/MRR from 0.257/0.500 to 0.229/0.479. The predeclared $Q\|A$ Q+A control reaches 0.314/0.538, yet its +0.057 R@1 gain is below the +0.10 gate and was not the validation-selected policy. We therefore stop before the F/K/B/ threshold surface and end-to-end dynamic closure. The evidence now points to learned current- state update, query decomposition, or relational composition rather than a broader sweep around an unvalidated frozen updater. Any such extension must retain the same payload, split, leakage, candidate-generation, and execution controls and improve beyond both association-only and static query-reranking baselines.

The terminal-query experiment adds a sharper final boundary. When query similarity is removed from intermediate admission, oracle-root native search reaches every held-out Hotpot and QASPER oracle parent at $K = 4, B = 6$. Yet the max-facet terminal score offers no acceptable operating point: q95 still produces .389 validation false goals for .022 correct Hotpot closure, while the safe endpoint never closes. We therefore stop before routed roots. The next iteration should improve the terminal evidence-completeness predicate, not force the original query back onto every bridge and not merely increase K or B.

The granularity control changes the interpretation of that success. At 256-token parents the held-out task is an exactly recovered one-edge evidence neighborhood, but refinement to 16–32 tokens distributes evidence over roughly five to seven oracle parents and defeats the same frozen search. The coarse result is real, yet it is partly purchased by graph contraction and a counterfactual active payload approaching the full source. Fine parents reverse that systems tradeoff: they triple evidence density and reduce selected tokens fourteenfold, but native association no longer finds the evidence. Iterative PRA therefore has two distinct open problems: fine-grained associative discovery and selective native-K/V materialization. Hotpot remains useful for exposing this boundary, but a deeper evidence-graph benchmark is required before claiming natural multi-step closure.

MuSiQue and 2Wiki make that final boundary empirical. MuSiQue's four-step examples occupy more oracle parents than its two-step examples, so the depth label survives memory construction. Nevertheless, the frozen native graph recovers nearly all evidence by $H = 2$ under $K = 6, B = 16$; later annotated steps do not survive at their nominal frontier depth. 2Wiki is more encouraging about local path fidelity: direct transition recall reaches .720 at $K = 4$, .880 at $K = 6$, and 1.000 at $K = 8$, with corresponding complete-path survival .588/.824/1.000. The cost is broad activation, and routed-root recall remains weak. The central thesis is therefore refined rather than declared solved: pretrained attention contains traversable local evidence geometry, but faithful deep closure still needs better entry routing, tighter successor geometry, or a dynamic reasoning state that preserves task order.

Layerwise extraction narrows "tighter successor geometry" further. The native graph is not one fixed object waiting to be read from a transformer: the same source and parent partition induce different rank, branching, and distance structure at different blocks. Measured accumulated contextualization explains some contraction but not local edge fidelity. More importantly, layer choice interacts with parent size: early Q/K is strongest in the fine 32-token 2Wiki control, middle Q/K is strongest at 128, and external contraction makes all three representatives nearly perfect at 256. The next graph representation should therefore be adaptive and multiscale, while charging every exposed scale and layer to one routing budget. This paper deliberately stops before multi-layer fusion, so the current late-layer implementation remains unchanged.

The final measurement gate turns those broad options into two concrete interfaces and closes this iteration. At fine 16-token search nodes, selected source falls to .063/.172 on MuSiQue/2Wiki, but preserved edge R@6 falls to .667/.400 and deeper search adds no recovery after $H = 2$. Missing local links, rather than frontier depth, are therefore the dominant fine-scale failure. A multiscale contextual index should retain coarse discovery state while pointing to fine payload identities. At entry, an existing query span supplies an oracle Top-4 root in .878/.950 of runs, whereas the best cheap held-out selectors reach only .622/.692. Confidence, position, norm, and a validation-fitted linear scale selector do not close that gap. The measured sparse frontier spans .063 selected source at .819 evidence recall for MuSiQue and .172 at .796 for 2Wiki, while their complete-recovery points require .573 and .918 respectively. We therefore freeze Paper 2.5 at this measured boundary. Supervised or adaptive facet selection belongs to a separate follow-up; production native-K/V serving latency and concurrency belong to a separate systems study. The Gate-3 output experiment in this paper is a bounded efficacy measurement, not that serving study.

F External Behavioral-Judge Audit

The frozen judge package contains 294 underlying comparisons, each presented in exact A/B and B/A orders. Seven comparison groups cover balanced/broad discovery against one-shot, balanced against no memory and oracle evidence, selected versus all-layer consumption, and identical/corrupted calibration anchors. The scorer validates opaque IDs, numeric ranges, reason lengths, and complete order-reversed pairs; restores the private condition orientation; and averages reversals before aggregation. Positive ΔQ always favors the named target after unblinding.

GPT-5.6 Sol covers all 294 pairs. Claude Sonnet 5 covers 152 complete pairs and no half-pair, so the partial-response mode reports 51.7% coverage and computes cross-judge metrics only on that intersection. Partial acceptance is explicit and opt-in; strict complete-ID validation remains the scorer default. Claude's failure on literal identity and deterministic corruption anchors precludes efficacy interpretation despite structurally valid rows. This is an instrument failure, not evidence for or against PRA.

Table 42: Frozen behavioral-judge artifacts. The truth file is retained privately; its hash is published so later unblinding remains auditable.

Artifact	SHA-256
Prompt	`65f810c43576ce9ac8eab640c0f8e0ec0acde3d6cfd73c98ccb065eeaf67b9e1`
Blind package	`3fa13a3473bc2aca8baa8980a2364069dadbf67d86384e44711c6cca1df933c9`
Private truth	`45c7c1cd57855ced306234c215085599d8020d6e5fd8a53db6dadced344df6a6`
GPT-5.6 Sol response	`8318ecb5ae0e65008d8a294eb1ff483e3249d601c90e15388a648a2904d7a9fd`
Claude Sonnet 5 partial	`8b7cb41a872889f39e0dd8d84940dc44028f386f389df0c6a2f943d0d422448b`
Aggregate metrics	`4c96f8f67dc74def9ec6650c81405f41bb667e3a0653a02f3b5fe7dbfa9eda82`

The exact judge-specific aggregates, calibration rows, order diagnostics, and shared-pair correlations are stored in `output_validation/behavioral_judge/behavioral_judge_external_metrics.json`. Blinded raw responses remain alongside that aggregate. No unblinded pair table or truth metadata is tracked.

G Implementation Map

The implementation is designed so another system can reproduce the mechanism without adopting the entire PRA-HF package. File and line references below refer to the accompanying source revision.

Layer-local packed index. `src/pra_hf/iterative.py:69--134` packs variable gist sets as `[chunks,max_gists,routing_width]` plus a validity mask. Records retain chunk identities and payload handles, but packing copies no native K/V.

Asymmetric companions and contextual parents. Lines 387–438 of `injection.py` project each contextual local gist with both W_m and W_q and projects the occupancy-weighted parent hidden-state mean separately. The full path is `src/pra_torch/hf/injection.py`. The cache stores compact companion vectors, not another native K/V payload.

Hierarchical index and parent deduplication. Lines 137–254 of `iterative.py` define aligned parent/local tensors and construct them from contextual segment gists. Lines 641–787 perform root-to-parent/local and local-to-local expansion, aggregates candidates by parent, enforces the unique-parent budget, and maps only final parents to payload handles.

Graph contract. Lines 269–342 of `iterative.py` define node, edge, graph, and result records. The current JSON Schema is `docs/papers/shared/schemas/retrieval_graph_v2.schema.json`. Nodes retain parent identity, local span, resolution, representation and projection types, hop, path score, and materialization state. Native-edge extensions additionally retain source/target spans, query and K/V head identities, native score, threshold, and semantic candidate rank.

Exact scoring and deterministic top-k. `iterative.py:376--397` normalizes frontier rows and computes `einsum("fd,cgd->fcg")` before max reduction across gists. A stable index tie-break is applied before `torch.topk`, producing deterministic identities on CPU and CUDA.

Traversal. `iterative.py:399--614` performs root scoring, frontier expansion, anchoring, deduplication, path pruning, Level-2 updates, stopping, and cost accounting. The code never accesses `chunk.token_kv`. Conversion at lines 617–638 returns selected payload handles only after the graph is final.

Host-model integration. `src/pra_hf/model.py:374--439` captures one root query with memory disabled, dispatches one-shot, chunk iterative, or opt-in `local_iterative` routing, maps final parent identities to consumption layers, enables fixed memory, and only then marks graph nodes materialized. Generation thereafter uses the unchanged Paper 2 native-K/V path.

Native QK feature capture. `precompute_native_qk_features.py:50--107` encodes each bounded 256-token parent once, reads Qwen's exact layer-27 capture, transposes canonical Q/K to token-major form, and slices aligned 32-token regions. It stores FP16 pre-RoPE Q/K with a validity mask for a short final window; it never independently re-encodes a local region. The capture itself is generated by Qwen's ordinary `q_proj`/`k_proj` and native Q/K norms before rotary position encoding. A test recomputes those projections from the captured hidden states and requires exact tensor equality.

GQA-compatible tokenwise scoring. Lines 122–211 of `src/pra_hf/native_closure.py` map query head h to K/V head $\lfloor h/2 \rfloor$, packs candidate keys accordingly, and computes all frontier-by-candidate token/head products with one `einsum`. Invalid padded pairs become $-\infty$ before max or Top-4 reduction. The scorer returns strongest token and compatible-head identities as well as the physical dot-product count; no Python loop traverses token pairs or heads.

Semantic narrowing and native propagation. `native_closure.py:215--480` uses root semantic scores for initial parents, applies $W_q h_A \to W_m h_B$ to rank a hard 10/20% candidate-parent pool, gathers all contextual local windows in that pool, and invokes native QK once. Proposals are merged by parent before the hard final parent budget. Threshold mode retains the same cap and fills rejected slots from root semantic ranking, preserving materialization parity. The graph cost record separates semantic comparisons, native dots, candidate parents/regions/tokens, accepted transitions, selected parents, final K/V tokens, and routing wall time.

Oracle and edge-rank diagnostic. Lines 66–116 of `run_oracle_convergence.py` map each annotation to overlapping parents and call the Paper-2 oracle constructor unchanged; disagreement with the cached evidence mask is fatal. Lines 133–185 define target rank using only frozen score vectors, with equal scores sharing a rank and the forced source group excluded. Parent/local/native score vectors are constructed at lines 188–245; target identity enters only the subsequent metric. Lines 252–292 fit the offline margin threshold on a hash-stable validation partition and evaluate it unchanged on test. The main loop beginning at line 588 wraps the existing Gate-2/3 evaluators, so convergence does not duplicate or subtly alter their selection policy.

Displacement and calibration diagnostic. Lines 81–159 of the additive displacement runner listed in the reproduction README compare one-shot oracle hits with the final identities returned by each frozen evaluator and retain replacement hop, score family, rank, and budget reason. Lines 162–174 implement the oracle-labeled protected-root counterfactual with the same final B; this function is used only by the offline runner. Lines 184–277 construct a fixed candidate-identity union, fit score-family statistics on validation examples only, and rerank held-out examples by raw, z-scored, or empirical-quantile score. The root/semantic and native score capture at lines 337–478 follows the exact transformations used by Gate 2 and Gate 3. The main loop at lines 696–1095 calls those frozen evaluators, verifies all 4,000 selections row-for-row against the prior convergence artifact, and writes additive rows without changing SDK routing.

Fixed-identity payload parity. `tests/test_hf_integration.py` sends the same selected chunk through router, oracle, and native-closure labels. All three paths produce byte-identical post-RoPE K, native V, and attention output. Gate 3 therefore ends at identities: it neither changes nor reconstructs the payload consumed by Qwen.

Monotonic adaptive competition. `src/pra_hf/adaptive_competition.py:44--154` defines immutable root/transition policies and score/result records without any oracle field. Lines 157–234 compute cross-seed agreement and choose a lock only after constructing full root Top-B. Lines 237–291 standardize raw pre-RoPE native scores per source, measure Top-1/Top-4 spread, normalized entropy and semantic/native rank agreement, then select bounded breadth 1, 2, or 4. Lines 294–323 merge parent proposals while asserting that every locked identity survives; root order fills any unused slots. The opt-in router at lines 326–404 invokes frozen transition providers only when $B - |L| > 0$, uses raw native rank rather than sigmoid magnitude, and returns separate root, semantic, native-dot, and scorer time costs. It owns no K/V tensor.

The additive runner constructs projection-correct semantic and native traces at lines 162–247 of `run_monotonic_adaptive_competition.py`. Lines 285–383 call the label-free router, then attach exact-oracle and evidence-group metrics to its returned identities. Validation-only selection is at lines 475–509; synthetic direct/rank-2/rank-4 controls and cost-normalized held-out effects are at lines 551–692. The main loop beginning at line 775 preserves the full policy matrices, freezes preservation and exploratory roles separately, and writes held-out rows without altering the model dispatch path or default routing mode.

Contextual query facets and real-head audit. `src/pra_hf/query_facets.py:101--183` forms overlapping windows only from states captured in one complete contextual prompt pass and retains the exact final-token global row. Lines 193–247 normalize and batch all facet–parent semantic products; lines 251–337 pool valid native K tokens by parent with `index_add_`, enforce the 16:8 GQA map, and compute the full facet–query-head–parent tensor. Lines 340–388 deduplicate per-head nominations before one deterministic final Top-B. The runner at `experiments/paper2_5_iterative_pra/run_query_entry_facets.py:673--958` reproduces the old root, selects window/head policies only on validation identities, records winning span/head provenance and cost, and runs the three exact synthetic controls. The capture at `precompute_query_entry_features.py:42--191` asserts bit-exact final-token parity and records zero independent window forwards. The confirmation at `run_query_entry_propagation.py:74--443` changes only root scores before invoking the previously frozen monotonic native-rank policy; its final parent budget remains unchanged.

Facet types and query-grounded propagation. `src/pra_hf/query_facets.py:128--307` clips query-support spans and constructs deterministic phrase, arbitrary-span, multiscale, token, and overlapping-window facets from one contextual state sequence. Provenance records global/local family, scale, and source-token boundaries. `src/pra_hf/grounded_propagation.py:26--213` contains no oracle field: lines 80–102 form a bounded association-ordered candidate set, lines 105–141 validate its identities in one [*facets*, *candidates*] product, and lines 144–213 apply association order, query reranking, rank conjunction, or a query-threshold AND gate. Rank conjunction combines ranks only; native and semantic raw scores are never added.

The Gate-A runner at `experiments/paper2_5_iterative_pra/run_grounded_facet_gate.py:106--478` constructs every predeclared family, verifies exact global/width-4 baselines, and freezes facet/support choices on the hash-stable validation partition. The controlled capture at `precompute_grounded_query_features.py:46--228` renders old user/tool history plus one explicit latest user message, records all character-to-token boundaries, and asserts that the copied stale parent is non-oracle. Gate B at `run_grounded_propagation_gate.py:91--523` first scores native proposals, then computes semantic products only for the source group and bounded Top-8 union. Selection at lines 209–239 uses validation rows; lines 242–281 implement exact relevance, drift-stop, and unresolved-bridge controls. Candidate JSON retains $A \to B$ rank, query rank/score, validating facet span/text, root-facet equality, and admission. The runner records zero K/V materialization and stops before end-to-end evaluation when its predeclared conditional gate fails.

Dynamic current-state reconstruction. `src/pra_hf/dynamic_query.py:47--75` constructs the three deterministic Q/A orderings and retains disjoint character provenance. Lines 78–129 apply the model's ordinary chat template, truncate to the model-safe 768-token bound, and map those character regions back to exact token spans. Lines 132–190 form contextual width-2 facets only inside the configured question or question-plus-A support; A can condition question states even when its positions cannot issue independent queries. No target identity is accepted by this API.

`experiments/paper2_5_iterative_pra/precompute_dynamic_query_features.py:49--108` decodes the oracle-conditioned source group A, runs one memory-disabled frozen layer-27 capture per ordering, and records contextual hidden/pre-query tensors, transfer bytes, elapsed time, and CUDA peaks. Lines 110–268 bind those captures to the unchanged 17 evidence-group transitions and write a hash-pinned, regenerable 96.4 MB feature cache. The feature cache is ignored; its manifest is tracked.

`experiments/paper2_5_iterative_pra/run_dynamic_query_gate.py:96--296` applies each frozen semantic projection, scores only native-QK proposals in a packed facet–candidate product, and records candidate/native/query ranks, exact facet span/text, rank movement, comparisons, routing memory, H2D bytes, and zero K/V materialization. Lines 420–692 generate the unchanged native candidate ladder, freeze the dynamic variant on validation, assert exact prior static and K=4 candidate baselines, and enforce the held-out stop rule before writing rows, summaries, diagnostics, and Figure 28.

Terminal-query semantic graph search. `src/pra_hf/semantic_graph_search.py:25--118` defines the independent edge/goal thresholds, resource bounds, per-proposal provenance, and routing-only result record. Lines 122–183 batch every local source against packed native keys and scatter-reduce local scores into parent adjacency; neither query facets nor labels enter that operation. Lines 223–271 perform stable native successor ordering and the separate any-facet terminal reduction. The traversal at lines 274–524 admits a proposal from its native edge score, deduplicates parent identities, applies the shared B/beam/hop limits, and only then reads the query-goal tensor. The public function accepts roots and score tensors but no oracle, target, evidence, or correctness channel.

The runner at `experiments/paper2_5_iterative_pra/run_semantic_graph_search.py:152--383` builds the frozen CUDA adjacency/goal caches, reproduces the exact 17-transition K ladder, and partitions identities before calibration. Lines 600–754 execute all strategies and select θ_{goal} on validation only; lines 757–825 select the bounded operating point; lines 827–908 define the five synthetic controls. The orchestration at lines 1068–1342 evaluates oracle roots first and calls routed roots only if both closure and false-goal gates pass. Calibration surfaces run from the identical CPU-resident score cache to avoid thousands of tiny CUDA synchronizations; the selected condition is rerun on CUDA for timing and memory. Finally, `experiments/paper2_5_iterative_pra/review_semantic_graph_false_goals.py:44--112` reconstructs the q95 validation terminals from source token spans and writes the review rows and counts.

Granularity and post-hoc evidence topology. `src/pra_hf/chunk_granularity.py:61--168` constructs deterministic half-open parent spans, maps each supporting span to overlapping parent IDs, selects the first-group root by exact token overlap, and records collisions and evidence-token containment. Lines 175–260 attach oracle recovery only after traversal and compute facet concentration, normalized path coverage, and incremental coverage. The search function receives none of those annotations. Line 263 defines the controlled chain-contraction depth used by the synthetic companion.

`experiments/paper2_5_iterative_pra/precompute_chunk_granularity_features.py:38--149` captures per-token layer-27 hidden states under the same frozen 256-token block encoder and checks every ID and source-token count against the canonical native-QK cache. The ignored 52,058,925-byte artifact has SHA-256

`a3c1ffb7a2e7a9213f44c33cb678cd084a5f7fb992677539bec31098e31906cf`.

The runner's lines 117–146 split cached tokenwise Q/K into exact 16-token atomic windows or retain the canonical 32-token windows; lines 207–427 pool semantic parents, construct native adjacency, and produce the five-seed facet matrix. Lines 430–511 execute the full K/H/B surface with an infinite goal threshold and apply evidence metrics afterward. Lines 514–738 validate controlled contraction and produce topology, efficiency, density, system-scaling, and separability tables; lines 740–854 generate the two paper figures. The orchestration at lines 857–966 asserts exact canonical held-out reproduction before writing the result manifest.

Natural annotated graphs. `src/pra_hf/natural_reasoning_graph.py:21--160` defines immutable annotation nodes, serializes MuSiQue paragraphs, and converts only explicit decomposition references into edges. Lines 194–280 serialize 2Wiki at sentence resolution, map relation triples to supporting facts, and keep entity-ID and derived lexical joins distinguishable. Lines 283–363 convert exact character offsets to tokenizer spans, map nodes to all overlapping parents, choose one maximum-overlap root representative, and classify preserved, contracted, and unmappable edges. No search score is accepted by this module.

`experiments/paper2_5_iterative_pra/precompute_natural_graph_features.py:48--107` captures token hidden states and padded 32-token pre-RoPE Q/K windows from independent frozen 256-token encoding blocks. Lines 110–226 bind exact annotation spans and one contextual query pass to 84 identity-frozen examples. The ignored 1,324,029,241-byte cache has SHA-256

`a03d5a978c7b5eeef21402d88d8c293827744a93cd4b0c1b5a605b0bf4ff19dc`.

`experiments/paper2_5_iterative_pra/run_natural_graph_depth.py:125--218` computes direct annotated-transition rank and bounded native shortest path from score matrices, excluding shared source parents from target recovery. Lines 243–416 construct exact adjacency and five-seed facet diagnostics; lines 419–523 run the oracle-free K/H/B search and attach node, edge, path, depth, payload, and systems metrics afterward. Lines 526–553 select one operating point from validation identities; lines 555–593 restore routed roots only after each dataset's oracle gate. The remaining aggregation writes 252 mappings, 372 transition rows, 12,096 oracle searches, 33,408 hop-survival rows, 1,160 facet rows, and 1,260 routed-root rows.

Layerwise native graph and contextualization. `src/pra_hf/layerwise_context.py:18--178` defines the per-layer state record and installs observation-only hooks at the decoder input, normalized attention input, projected attention output, post-attention residual, MLP output, and block output. Lines 138–166 assert both exact residual identities. Lines 182–207 construct a causal radius mask in the original $[1, 1, T, T]$ coordinates: tokens remain present, absolute position IDs remain unchanged, and only accessible K/V positions change. Lines 209–287 compute normalized displacement, directional rotation, branch magnitude, native query-head entropy/effective support, and local/distant/evidence attention fractions. The FFN branch is retained as a local-transformation control and is never called contextualization.

`experiments/paper2_5_iterative_pra/precompute_layerwise_graph_features.py:135--305` begins capture on all eight actual adapters before one shared full forward, streams exact pre-RoPE Q/K to CPU, checks the full-mask parity control, and reruns self/16/32-token visibility with the same logical positions. Lines 336–488 write one regenerable file per example rather than holding all layers in memory. The ignored cache is 7,832,055,097 bytes over 84 SHA-256-pinned entries; it covers 158,170 tokens, took 1,863.9 seconds on the GTX 950M, peaked at 1.394 GB allocated CUDA memory, and has zero maximum full-context reproduction error.

`src/pra_hf/layerwise_graph.py:11--109` implements deterministic Top-K neighbors, bounded native distance, weak components, reciprocity, destination reuse, and two-hop effective branching. Lines 111–161 provide Pearson/Spearman and example-level bootstrap summaries. The runner at `experiments/paper2_5_iterative_pra/run_layerwise_graph_exploration.py:190--735` rebuilds each layer's exact adjacency, attaches annotations only after search, computes transition/path/depth/ shortcut/lexical/system rows, joins them to measured contextualization, and refuses output unless layer 27 reproduces the historical curve. The gated cross at `experiments/paper2_5_iterative_pra/run_layerwise_granularity_cross.py:109--242` changes only the a-priori 0/12/27 layer and 32/128/256-token parent assignment; it neither tunes a best layer nor fuses layer scores.

Final measurement gate. The module `src/pra_hf/final_metrics.py:15--196` contains the measurement-only primitives: disjoint identity enforcement, deterministic facet-group ranking, label-free confidence features, a validation-fitted ridge selector, path-survival decomposition, Pareto marking, and exact keyed joins. None accepts an evidence target or changes candidate retrieval. The runner at `experiments/paper2_5_iterative_pra/run_final_metrics.py:152--448` reconstructs all-offset facets from the frozen natural-graph cache, fits scale selection on the 42 validation identities, and evaluates only disjoint held-out identities. Lines 449–720 aggregate root/intermediate/ terminal visibility and compute matched-example layer correlations with 1,000 identity bootstraps. Lines 721–988 separate missing-edge loss from post-edge search loss, assemble the MuSiQue shortcut and sparse quality–payload frontiers, and join the four canonical datasets. Lines 1013–1129 write the plots and refuse completion until every required row family exists. The runner performs no training, generation, graph search, or native-K/V materialization; it is a strict synthesis over already frozen rows and regenerable feature caches.

Reproduction. The experiment directory is `experiments/paper2_5_iterative_pra/`. Its Gate-1 runner isolates projection correction. Gate 2 captures one 256-token contextual parent pass and derives 32-token local means. Gate 3 uses `precompute_native_qk_features.py` followed by `run_gate3_native_qk_closure.py`. The 620,984,671-byte feature cache is regenerable and excluded from Git; its public manifest pins Qwen revision `c1899de289a04d12100db37 0d81485cdf75e47ca`. Its SHA-256 is

`edc589cc7886ba843ee598e89e6dab97e500084df8b7ee8dc8ab44ea75993ae8`.

The dynamic gate derives 51 Q1 captures from 17 transitions and three orderings. Its ignored 96,400,286-byte cache has SHA-256

`130a2bd5f1d43e4bf3bfbe48145d368144dae0a98f666559ccb1d131ea7dac5e`.

Exact commands are listed in the experiment `README.md`; raw rows, aggregate tables, plots, and graph JSONL are under the corresponding shared result tree.

H Validation Gates

The test suite checks $D = 1$ equivalence to Top-B, deterministic batches, exact synthetic two-hop recovery, deduplication and cycles, depth/beam/unique budgets, all path and frontier modes, zero limits, layer locality, projection-contract enforcement, exact local-bridge recovery, parent deduplication, GQA head mapping, max and Top-4 native reductions, semantic narrowing, threshold caps, exact pre-RoPE Q/K capture, CPU/CUDA identity parity, fixed-identity payload parity, and the no-materialization-before-closure boundary. Gate-3 generation validates every saved graph against the backward-compatible schema-v2 extension and records zero K/V materializations during closure. The feature manifest verifies the frozen model revision, layer 27, 16/8 grouped heads, native head width 128, and no memory-use adapter. The focused iterative/oracle/native/adaptive suite passes 69 tests; the query-facet/head module adds nine focused unit tests for contextual windows, one-pass derivation, exact baseline scoring, GQA mapping, provenance, deduplication, budget conservation, and oracle metrics. All 160 saved Gate-3 graphs and 5,600 result rows satisfy schema, parent-budget, and no-closure-materialization checks. The additive diagnostic preserves 4,000 convergence rows and 340 raw edge-score rows, then adds 2,488 root-hit classifications, 17,710 score-family observations, 2,619 fixed-union calibration candidates, and 780 held-out control rows. Tests cover canonical-oracle identity, annotation groups, displacement, protected-root budgeting, score summaries, ranking ties, validation leakage, deterministic policy fitting, and cost accounting. The monotonic gate adds 19,200 root-policy rows, 14,080 transition rows, and 1,040 held-out frozen-policy rows; every row conserves its parent budget and retains all locked identities. The entry gate adds 1,792 root rows and 450 four-method confirmation rows. Its manifest records 32 one-pass contextual query captures, zero independent window passes, and zero maximum final-token baseline error; all three synthetic controls pass. The additive facet gate adds 6,400 family rows, 960 clean-support rows, and 3,840 stale-history rows from 128 one-pass controlled captures. The conditional propagation gate adds 2,210 rows with complete candidate/facet provenance, zero diagnostic K/V materializations, and three passing synthetic controls; its end-to-end stop is recorded explicitly. Thirteen query-facet and six grounded propagation unit tests cover support clipping, phrase/multiscale/token facets, local-only sets, rank-scale invariance, residual facets, thresholds, provenance, and budgets. The dynamic gate adds 4,760 rows and 112 aggregate cells across 17 transitions, five projections, seven query states, and eight candidate widths. Its checks reproduce the prior static R@1/MRR and native K=4 candidate recall exactly, enforce zero diagnostic K/V materialization, and verify the predeclared Gate-1/Gate-2 stop. Six dynamic-query unit tests cover deterministic Q/A layouts, exact region provenance, restricted support, tokenizer-state restoration, and the absence of an oracle-target API channel.

The terminal-query iteration adds 53,760 oracle-surface rows, 3,360 goal-calibration rows, 640 hop rows, 320 selected-condition rows, and 1,705 selected-node provenance rows. Seventeen focused tests cover native adjacency reduction, all three deterministic search strategies, any- and different-facet goals, a low-query-similarity bridge, minimum depth, first-path stopping, deduplication, cycles, finite/shared and unbounded budgets, cost accounting, and the absence of a task-label API. All five synthetic graph controls pass. The complete repository suite now passes 487 tests with the same one upstream deprecation warning.

The granularity gate adds 80 evidence-topology rows, 4,800 oracle-free discovery rows, 3,230 five-seed facet–parent rows, 80 system profiles, and 15 passing chain-contraction controls. Focused tests cover deterministic exact boundaries, invalid overlap geometry, evidence remapping, group collisions, exact root containment, $H = 0/1/2$ behavior, post-hoc oracle recovery, minimum recovery depth, facet statistics and path complementarity, and contraction. The runner audits all canonical 256-token boundaries and asserts exact held-out $K = 4, B = 6, H = 4$ oracle and later-evidence recall of 1.000 before emitting artifacts. The ignored token-hidden cache manifest pins all 16 identities, 25,416 tokens, the Qwen revision, layer 27, absolute positions, and its content hash.

The natural-graph gate adds nine focused adapter/metric tests for explicit MuSiQue dependencies, 2Wiki entity-ID paths, unordered comparisons, exact character offsets, root representation, preserved/contracted/unmappable transitions, deterministic identity splitting, and native rank/shortest-path behavior. The full-dev audit records both licenses, archive hashes, split schemas, and unresolved 2Wiki mappings before model execution. The feature manifest pins 84 identities, 158,170 source tokens, the frozen Qwen revision, layer 27, and zero training or generation.

The layerwise extension adds seven focused tests for exact layer indexing, residual capture, normalized branch metrics, position-preserving context restriction, deterministic intervention, full-context reproduction, GQA attention statistics, deterministic Top-K graph construction, shortest distance, topology, tied-rank correlations, and example-level bootstrap behavior. Its tracked outputs contain 10,752 context rows, 992 transition rows, 672 graph rows, 3,360 hop rows, and 672 systems rows. The gated cross adds 756 example conditions and 1,116 transition rows. Both runners assert exact layer-27/128-token R@6 reproduction before writing their result JSON.

The final measurement gate adds 1,160 target-level facet rows, 188 selector summaries, five role rows, six hop rows, eight layer summaries, 25 matched-correlation rows, nine layer–chunk cells, 30 edge-decomposition rows, 140 quality–payload frontier rows, 15 memory-audit rows, and four cross-dataset rows. Eight focused tests cover deterministic facet ranking, label-free confidence features, validation/held-out identity separation, selector API leakage, fitted selection, path decomposition, Pareto membership, and exact joins. The runner asserts disjoint selector splits, uses 1,000 identity bootstraps, preserves blank values for unmeasured cross-dataset cells, and records zero materialized native K/V tokens rather than relabeling counterfactual source payload as serving memory.

The cross-dataset extension is implemented in `experiments/paper2_5_iterative_pra/run_natural_graph_depth.py:93--1568`. Lines 93–126 preserve 256-token contextual capture while assigning exact 16-token native Q/K halves to their own search parents. Lines 336–609 build the depth–granularity, transition, strict/conditional path, independent-edge product, system-cost, topology, and sparse-frontier tables. Lines 677–857 remap every annotation and construct exact dense adjacency without query or oracle labels. Lines 1450–1539 assert the canonical 2Wiki curve before writing artifacts. The final run contains 420 mapping rows, 620 transition rows, 31,500 oracle-root searches, 87,000 node/hop rows, and 420 system profiles across five granularities.

Reusable post-hoc calculations live in `src/pra_hf/cross_dataset_diagnostics.py:1--253`. Lines 49–92 count evidence/source tokens without overlap; 95–163 enumerate annotation paths and stride-one multiscale facets; 170–232 compute target-group semantic and lexical ranks; and 236–256 enforce one deduplicated shared root budget. None accepts an oracle label in its runtime candidate-selection signature. The all-offset runner is `experiments/paper2_5_iterative_pra/run_natural_multiscale_query_audit.py:1--670`. Lines 98–124 assign ordered root/intermediate/terminal roles, 142–318 aggregate held-out ceilings and validation-only gate decisions, and 342–644 construct scores, post-hoc ranks, distractor controls, and the bounded executable union. It emits 1,160 target-rank, 4,615 false-parent, 2,395 scale-score, and 1,680 practical-router rows. The 1.86-MB score cache is ignored but pinned by SHA-256 in the tracked result JSON. Seven additional diagnostics tests cover union-aware evidence density, deterministic annotation paths, independent-edge products, all valid stride-one offsets, decoded facet provenance, semantic/lexical group ranks, terminal roles, deduplicated shared budgets, and the absence of an oracle argument from executable multiscale selection.

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